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Bahir Dar University
Bahir Dar Institute of Technology

Faculty of Computing
Department of Artificial Intelligence and Data Science

Masters Thesis
on
End-to-End Continuous Ethiopia Sign Language Recognition

Author:
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for partial fulfillment of the requirements for the degree of master of science in artificial intelligence and data science.

[July 2024]

End-to-End Continuous Ethiopia Sign Language Recognition

By
Anteneh Yehalem

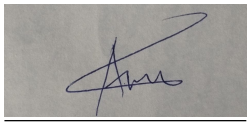
A Thesis Submitted to the School of Graduate Studies, Bahir Dar Institute of Technology, Bahir Dar University, in Partial Fulfillment of the Requirements for the Degree of Master of Science in Artificial Intelligence and Data Science.

Advisor
Million Meshesha (Phd)

[July 2024]

Declaration

This is to certify that the thesis titled “End-to-End Continuous Ethiopia Sign Language Recognition,” submitted as a partial fulfillment for the requirements of the Master of Science degree in Artificial Intelligence and Data Science, under the Faculty of Computing at Bahir Dar Institute of Technology, represents original research conducted by the undersigned. This work is entirely my own and has not been submitted to this or any other institution for any degree or certification.

Anteneh Yehalem Tegegne	18/06/2024	
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Bahir Dar University
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Approval of Thesis for defense

I hereby certify that I have supervised, read, and evaluated this thesis titled "End-to-End Continuous Ethiopia Sign Language Recognition" prepared by Anteneh Yehalem Tegegne under my guidance. I recommend the thesis to be submitted for oral defense.

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FACULITY OF COMPUTING

Approval of thesis for defense result

I hereby confirm that the changes required by the examiners have been carried out and incorporated in the final thesis.

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As members of the board of examiners, we examined this thesis entitled "End-to-End Continuous Ethiopia Sign Language Recognition" by Anteneh Yehalem. We hereby certify that the thesis is accepted for fulfilling the requirements for the award of the degree of Masters of science in " Artificial Intelligence and Data Science".

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Abstract

Sign language is a vital means of communication for the deaf communities, encompassing complex gestures and facial expressions to convey meaning. Recognizing sign language through automated systems presents a significant challenge due to its dynamic nature and the variability among individual signers. For effective communication, Ethiopian Sign Language (EthSL) relies on the accurate recognition of continuous sequences of signs. Previous research efforts have primarily concentrated on recognizing isolated signs at the character, word, and phrase levels. While these studies have been valuable, they fall short of capturing the full complexity and fluidity of natural sign language conversations, which occur at the continuous (sentence) level. Previous researchers have used traditional methods for EthSL recognition, typically decomposing the task into separate stages. The primary problem addressed in this study is the recognition of continuous EthSL in End-to-End approach, which involves understanding sequences of signs performed by multiple signers. This task is crucial for creating practical sign language recognition systems for real-world use, such as translation services and communication aids for the Deaf community.

To address the challenge of continuous EthSL recognition, we created the first-ever dataset for Continuous (Sentence-Level) EthSL. This dataset features recordings of 30 unique sentences performed by 22 different signers. Each signer performed each sentence twice, resulting in a total of 1,320 sentences. Our methodology uses an end-to-end approach that employs a 2D convolution to extract spatial features from individual video frames, followed by a 1D convolution to capture short-term motion details. These spatial and temporal features are then processed using a Bidirectional Long Short-Term Memory (BiLSTM) network to recognize long-term temporal patterns and the sequential nature of sign language. Finally, a classifier with Connectionist Temporal Classification (CTC) is used to align and recognize the continuous sign language sequences. This integrated approach effectively harnesses both spatial and temporal information to improve recognition accuracy.

The system evaluated on test dataset considering signer independence and unseen sentences split, achieving Word Error Rates (WER) of 8.02% and 47.02%, respectively. These results highlight the method's robustness in handling variability among different signers in unseen signers, showcasing its strength in signer independence. Achieving a WER of 47.02% for unseen sentences split is a notable result, but it also indicates a substantial area for improvement. To address this, future research should focus on expanding the dataset to include a wider variety of sentences, potentially through collaborative data collection efforts. Overall, the study presents a significant advancement in continuous EthSL recognition, with a robust approach to signer independence but also highlights the need for further efforts to enhance performance on unseen sentence data.

KeyWords: Ethiopian Sign Language; Sign Language Recognition; Deep Learning; Continuous SLR

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Abbreviations

ASL	American Sign Language
BILSTM	Bidirectional Long Short-Term Memory
BIGRU	Bidirectional Gated Recurrent Unit
CESLR	Continuous Ethiopia Sin Language Recognition
CPU	Central Processing Unit
CTC	Connectionist Temporal Classification
CNN	Convolutional Neural Network
EthSL	Ethiopian Sign Language
ENAD	Ethiopia National Association for Deaf
GPU	Graphics Processing Unit
GRU	Gated Recurrent Units
HMM	Hidden Markov Models
LSTM	Long Short-Term Memory
NN	Neural Network
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
RAM	Random Access Memory
SVM	Support Vector Machines
SLR	Sign Language Recognition
SLT	Sign Language Translation
WER	Word Error Rate
WHO	World Health Organization

Chapter One

Introduction

1.1 Background

Language plays a crucial role in facilitating intercultural dialogue, allowing individuals to articulate their thoughts and opinions within their societal context. Through language, people communicate and understand one another [1]–[3]. Unfortunately, not everyone has the privilege to do so, particularly those with disabilities, such as hearing impairments, which hinder their ability to hear and speak.

Hearing loss is referred to as deafness. Either one or both ears could be affected by the issue. In general, accidents or genetic conditions can lead to deafness. Based on the WHO Fact Sheet on Deafness and Hearing Loss, more than 5% of the global population suffers from hearing impairment. In Ethiopia, it's approximated that over 5 million individuals live with hearing loss out of a total population of 115 million[4].

The natural language of the deaf, sign language, is composed of signs, gestures, fingerprints, and facial expressions. Sign languages are typically classified according to the nation in which they are spoken, for example, Ethiopian Sign Language [5]. By its very nature, the Language is not universal. Regional variations in sign languages within the same nation may exist. Generally, sign languages are independent of spoken languages and do not use the same grammatical structures[6].

For deaf and hearing-impaired people, sign language has been developed as a language for daily interaction. While verbal languages are created using the lips, mouth, and vocal chords, processed by the ears, sign languages are created using the hands, head, face, and upper body. A sign language is made up of a set of signs, gestures, and syntax, as well as a mixture of movements, forms, and placements of the hands [7].

Sign language has its own rules, and grammar, like hand shape, hand movement, orientation, and non-manual features like facial gestures. Ethiopian National Association of the Deaf (ENAD) created the Ethiopian finger spelling in 1971. In 2008, ENAD also created a dictionary for Ethiopian Sign Language. The 33 "Geez Fidel" and the associated letters with vowels have separate finger-spelling gestures in Ethiopian sign language[8], [9].

There are two ways to conduct the signing process[8]. The first is the establishment of distinct

indicators for various objects. To indicate a father, or mother, a predetermined sign will be used. The other process spells the sign with their fingers. The translation of written alphabets into signs is called ”finger spelling”. Abebe and Meresha [1], [10] assert that another aspect of sign language is finger spelling, which is used to spell proper nouns, technical phrases, acronyms, and words from other sign languages. Only hearing people or deaf people with some hearing abilities can utilize finger spelling. Those who have been deaf since birth do not prefer using their fingers to spell[2].

Sign language recognition refers to the process of creating systems or technology capable of interpreting and understanding the gestures and movements used in sign languages. These systems often utilize various technologies, such as computer vision, machine learning, and sensors, to analyze and recognize the complex hand shapes, movements, and facial expressions inherent in sign language communication [8], [9], [11].The goal of sign language recognition is to bridge the communication gap between sign language users and non-signers, facilitating better accessibility and inclusion in various contexts, including education, employment, and everyday interactions. By accurately interpreting sign language, these systems can enable deaf and hard-of-hearing individuals to communicate more effectively with others who may not be fluent in sign language.

There are two categories of studies on the recognition of predetermined signs in sign language. The first one is the ability to recognize individual signs called isolated sign language recognition. The expectation in this kind of research is the identification of isolated (individual) signs that are neither preceded by other signs nor followed by them. In the second category, which involves the recognition of continuous signs, the signer will continually produce signs without a pause, and it is believed that these signs will be recognized in continuous sign language recognition research.[2], [12].

Numerous studies have been done in the field of sign language. These studies can be roughly split into three categories[9], [12]; Sign language recognition, Sign language translation and Sign language generation.

The process of analyzing signs created by a signer and producing corresponding text about the supplied sign is known as sign language recognition. Sign Language Recognition is a computational task that involves recognizing actions from sign languages. Different kinds of approaches are used to process the image or video that was collected and find a particular sign that the signer must display. This is important to solve especially in the digital world to bridge the communication gap that is faced by people with hearing impairments.[2], [3], [13]

1.2 Motivation

Seeing the challenges my nephew faces due to his hearing loss has deeply influenced my motivation for this research. At 18 years old, he grapples with difficulties in education, social interactions, and even simple family conversations, as no one else in our family understands sign language. His struggles extend to communicating with friends, often feeling isolated and misunderstood in social settings and school environments where few can know sign language. Witnessing his frustrations firsthand has ignited a personal drive to develop solutions that can alleviate these barriers to communication and foster greater inclusivity for individuals like him.

My nephew's experiences serve as a poignant reminder of the urgent need for continuous sign language recognition technology to improve the quality of life for those with hearing impairments, enabling them to fully engage and thrive in all aspects of society.

The compelling data on the widespread prevalence of hearing loss underscores the crucial demand for innovative solutions in communication and accessibility. According to WHO fact sheet 2024 With more than 430 million people currently in need of hearing rehabilitation, a significant portion being children, there's a clear imperative for effective tools and technologies to bridge communication barriers. Furthermore, projections suggesting that By 2050, nearly 2.5 billion people are projected to have some degree of hearing loss, and at least 700 million will require hearing rehabilitation, emphasize the urgent need for proactive measures to enhance inclusivity and quality of life for this expanding demographic. Within this context, the advancement of continuous sign language recognition represents not just a technological leap forward, but a vital means of empowering those with hearing impairments to fully participate in society. By harnessing state-of-the-art research and technology in this domain, we have the potential to profoundly impact millions of lives, fostering a more inclusive and just global community for everyone.

Some of the accessibility challenges faced by deaf and hard of hearing individuals include the following[3]–[5].

1. **Communication Barriers:** Deaf and hard of hearing individuals encounter significant challenges in communicating with others who do not understand sign language. Communication methods, such as spoken language or written text, may not always be effective or accessible for them. This communication barrier can lead to feelings of isolation, frustration, and exclusion from social interactions.

2. **Limited Access to Information:** Deaf and hard of hearing individuals often face barriers in accessing important information and services due to a lack of accommodations for their communication needs. Many public spaces, including government offices, healthcare facilities, and educational institutions, do not provide adequate support for sign language interpretation or alternative communication methods, making it difficult for deaf individuals to access critical services and participate fully in society.

3. **Educational Challenges:** In educational settings, deaf and hard of hearing students may struggle to access instructional materials, participate in classroom discussions, and communicate with teachers and peers. Without appropriate accommodations, such as sign language interpretation, captioning, or assistive devices, these students may experience delays in learning, academic underachievement, and social isolation.

4. **Employment Barriers:** Deaf and hard of hearing individuals face significant barriers to employment due to communication barriers and misconceptions about their abilities. Many employers are not equipped to accommodate the communication needs of deaf employees, leading to discrimination, limited job opportunities, and unequal access to career advancement opportunities. As a result, deaf individuals may experience higher rates of unemployment and underemployment compared to the general population.

5. **Healthcare Disparities:** Accessing healthcare services can be challenging for deaf and hard-of-hearing individuals due to communication barriers with healthcare providers. Without access to sign language interpretation or other communication accommodations, deaf

patients may struggle to understand medical information, communicate their healthcare needs effectively, and advocate for their own health. This lack of communication access can lead to disparities in healthcare outcomes and contribute to the underdiagnosis and undertreatment of health conditions among deaf individuals.

1.3 Statement of The Problem

Language is a system of communication used by a particular country or community. It is the method of human communication, and most of the language is represented by spoken words, but the major means of communication for hearing-impaired persons in their environment is sign language. However, the majority of hearing people are not familiar with sign language. As a result, the deaf and hearing-impaired group is unable to interact with hearing people on a daily basis. This makes it difficult for hearing people and the deaf community to communicate with one another.

There are only a few schools in Ethiopia that offer sign language to deaf and hard-of-hearing childern. As a result, deaf and hard-of-hearing did not have the same access to school as hearing students. Deaf students face numerous challenges within the school system, extending beyond the scarcity of sign language instructors. One pressing issue is the lack of specialized resources tailored to their needs, including assistive technology and trained staff. Without these resources, effective communication becomes a major hurdle, impeding both academic progress and social integration. Vital information presented verbally in classrooms often remains inaccessible to deaf students, undermining their comprehension and engagement. This communication barrier can exacerbate feelings of social isolation, hindering their ability to form connections and participate fully in school life.

Educational aids play a crucial role in facilitating learning for individuals with disabilities, such as using Braille to teach blind individuals. However, in Ethiopian deaf societies, widespread illiteracy persists, especially among those residing in rural areas, where they are often marginalized and perceived as lacking value throughout their lives. This systemic issue highlights the urgent need to address the educational disparities faced by the deaf community. Furthermore, while traditional educational approaches have made strides in improving literacy rates among other disability groups, such as the blind, there remains a significant gap in providing effective educational resources and support for the deaf population as result the deaf communities are more illiterate than other disabled individuals [4]. This gap not only perpetuates social and economic inequalities but also hinders the full participation and inclusion of deaf individuals in society.

Moreover, the lack of accessible educational materials and teaching methodologies tailored to the unique needs of deaf learners further exacerbates the problem. As a result, innovative approaches, particularly leveraging advancements in computer science, are increasingly recognized as a promising avenue for addressing the literacy challenges within the deaf community [15].

Deaf and hard-of-hearing individuals encounter a multitude of challenges when seeking medical care, particularly in emergency situations[8], [14]. Beyond the immediate need for communication with emergency physicians, there are persistent obstacles that impede their access to

quality healthcare. One significant issue is the limited availability of sign language interpreters, leading to difficulties in conveying critical information about diagnosis, treatment options, and post-treatment instructions. Consequently, misunderstandings and errors in healthcare delivery are more prevalent, exacerbating the risks associated with medical interventions. Moreover, concerns about privacy, coupled with a lack of awareness about healthcare rights, further compound the challenges faced by deaf patients. This can have a profound psychological impact, contributing to increased stress and mistrust in medical settings.

Much research has been conducted in different sign languages, but Ethiopian Sign Language (ETHSL) remains notably under-researched. For instance, studies on ETHSL recognition have been carried out by several researchers, contributing valuable insights. However, these studies possess limitations. The first limitation was, they only considered the recognition of character [16]. word [2], [6], [8], [10], [17], and phrase-level [3] Ethiopia Sign Language. They could not further extend to the work on continuous (sentence-level) recognition. In sign language, effective communication encompasses a multifaceted approach, extending beyond hand movements to encompass a spectrum of non-manual features. These include facial expressions, head movements, body posture, and other nuanced gestures. To fully understand sign language, it's essential to recognize signs at sentence level towards developing an applicable SLR system that considers the interplay between manual and non-manual components.

Previous approaches to Ethiopian Sign Language Recognition typically rely on traditional methodologies that decompose the recognition task into multiple independent stages, including segmentation, feature extraction, classification, and post-processing. Each stage requires specialized models and hand-crafted features, which complicates the system architecture and makes it challenging to develop, maintain, and optimize. This multi-stage pipeline also suffers from error propagation, where inaccuracies at any stage can cascade through the system, negatively impacting the final recognition accuracy. The reliance on hand-crafted features, which are often time-consuming to design and tune, further limits the system's ability to capture the full complexity and variability of Ethiopian Sign Language, hindering its generalizability to new or unseen data.

Moreover, traditional methods struggle with scalability as the number of signs and sentence complexity increase. The manual effort required to maintain and update hand-crafted features and models becomes impractical, particularly in the segmentation and feature extraction stages. Additionally, the sequential nature of these pipelines often results in slower processing times, making real-time recognition difficult to achieve. This delay is detrimental to interactive communication systems. Furthermore, traditional approaches often fail to adequately handle the unique phonological, syntactical, and grammatical nuances of Ethiopian Sign Language, leading to misinterpretations. The need for a more nuanced and context-aware approach is evident, as cultural contexts, idiomatic expressions, and specific gestures unique to EthSL are not effectively captured by these conventional methods.

So this research aims to work on continuous Ethiopian sign language recognition at the sentence level. To this end, this study attempts to explore and answer the following research questions:

1. Which deep learning algorithm (from LSTM, BiLSTM, GRU, and BiGRU) performs better in building an end-to-end model for continuous Ethiopian Sign Language recognition across signer-independent datasets and unseen sentence datasets?

2. What is the impact of integrating a correlation network on the performance of an end-to-end sign language recognition model?

1.4 OBJECTIVE OF THE STUDY

1.4.1 General Objective

The general objective of this study is to design an End-to-End Continuous Ethiopian Sign Language Recognition that simplifies the conversion of sentence level signs to computer understandable format.

1.4.2 Specific Objectives

To achieve the General Objective of the study the following specific objectives are formulated:

1. To conduct a comprehensive review of existing literature and related works to identify optimal methodologies and techniques applicable to the development of a system for recognizing Continuous Ethiopian Sign Language (CESL).
2. To gather and curate datasets comprising diverse signers to facilitate the training and validation of the proposed CESL recognition system.
3. To identify algorithms for image extraction and resizing that are computationally efficient and scalable, ensuring they effectively handle the dynamic nature of sign language gestures.
4. To propose an optimal model for the recognition of Continuous Ethiopian Sign Language on signer-indepedent and unseen sentence data,
5. To conduct thorough evaluation and testing of the proposed CESL recognition model using a designated test dataset, assessing key performance metrics such as Word Error Rate(WER) to validate its effectiveness.

1.5 Scope of the Study

The primary goal of this study is to recognize continuous Ethiopian Sign Language through hand gestures and facial expressions. The data is collected from Yekatit 23 Primary School and Fasilo Secondary School. The dataset is recorded in MP4 video format, primarily in outdoor environments. While we tried to control settings like background and lighting to some extent, we did not capture real-time environments.

The dataset has been compiled with a focus on topics around family, such as family introduction, job, family relationship and paying particular attention to social aspects. During the analysis of the data, factors such as gender and age distribution have been carefully considered.

Traditional Sign Language Recognition (SLR) approaches typically break down the task into several independent stages such as segmentation, feature extraction, classification, and post-processing, each requiring specialized models and hand-crafted features. In contrast, our research focuses on an end-to-end approach for continuous Ethiopian Sign Language Recognition, utilizing advanced deep learning techniques to create a unified model that directly maps raw video input to recognized sign language sentences. This methodology simplifies the system architecture, mitigates error propagation, enhances efficiency and scalability to address the unique linguistic challenges and practical needs of the Ethiopian deaf community.

1.6 Significance of the Study

Due to a lack of research,Ethiopia sign language development is still in its infancy. Numerous studies must be conducted in order to fulfill the unique demands of those who are hard of hearing. The effort put forth to address the issue of hearing-impaired individuals and hearing people communicating will increase as a result of this research.

The study will have a great contribution to:

1. **Provides a valuable dataset for future researchers in the field:**
   The dataset collected and prepared through this research is an invaluable resource for future studies in sign language recognition and related fields. Comprising recordings of 30 sentences performed by 22 different signers, each repeating each sentence twice for a total of 1,320 sentences, this dataset supports further advancements in technology, enabling continued innovation and improvement in communication aids for the deaf and hard of hearing. The dataset is available on Zenodo for future research: Zenodo Dataset.

2. **Provide end-to-end approach for Ethiopia Sign Language Recognition**
   Our research aims to revolutionize Ethiopian Sign Language Recognition by employing an end-to-end approach with advanced deep learning techniques. Unlike traditional methods that require multiple independent stages and hand-crafted features, our unified model directly maps raw video input to recognized sign language sentences. This simplifies system architecture, reduces error propagation, and enhances efficiency and scalability. Ultimately, this approach addresses the unique linguistic challenges and practical needs of the Ethiopian deaf community.

3. **Reduces the communication gap between hearing and hearing-impaired individuals:**
   This study aspires to bridge the communication gap between hearing and hearing-impaired individuals by developing a continuous sentence-level recognition system for Ethiopian Sign Language (ETHSL). Currently, no comprehensive research has been conducted on continuous Ethiopian Sign Language recognition, leaving a significant gap in the field. This pioneering study aims not only to address this gap but also to lay the groundwork for future research endeavors. By breaking new ground, this research hopes to inspire and encourage other scholars to explore and advance the field of ETHSL recognition.

4. **Enables individuals with hearing loss to complete daily tasks more effortlessly:**
   A robust sentence-level recognition system greatly assist individuals with hearing loss

in performing everyday tasks. Whether it's communicating with service providers, navigating public transportation, or simply interacting with others in their community, this technology enable them to manage these activities more independently and efficiently.

5. **Facilitates effective treatment of hearing-impaired patients by physicians in healthcare institutions:**
   In healthcare settings, clear communication between patients and healthcare providers is crucial. A sentence-level recognition system allow physicians to better understand and respond to the needs of hearing-impaired patients, ensuring they receive accurate diagnoses and appropriate treatments.

6. **Decreases feelings of discrimination among the hearing-impaired community:**
   The ability to engage in fluid and natural conversations without reliance on interpreters can significantly reduce feelings of isolation and discrimination among hearing-impaired individuals.

1.7 Methodology

For this particular research, we used an experimental research methodology. We can adjust various hyper-parameters and experimentation setups under this methodology to examine how they affect the model. This research approach was to implement different experimental sets and assess the performance of recognition algorithms employed in this research. In general this study follows the following steps.

1.7.1 Data Collection and Preparation

All the data required for the research was gathered using a variety of data collection techniques. The gathering process involved communicating with experts and educators actively engaged in Ethiopian Sign Language research and development in Yekatit 23 primary school and Fasilo secondary school and Translator at Amhara Mass Media. In addition to specialists and educators, native sign language speakers who regularly used sign language in their daily lives were included in the data collection process. Sign language recordings were obtained from 14 students, 6 teachers, and 2 other individuals who communicated through sign language and people used to communicate with family or friends at Bahir Dar Yekatit 23 and Fasilo Secondary School. After we collect the data we pass through different preprocessing techniques like frame extraction, and resizing images.

1.7.2 Implementation tools and algorithms

Vision-based continuous sign language recognition systems often learn to automatically output the gloss labels in the correct order from image sequences of signers' performances as input [18]. For the implementation, after the dataset is collected in the form of a video from which frames are extracted then some processing tasks are done after that that we use as a dataset that we feed to the model. We use, convolutional neural networks (CNNs) to extract spatial

characteristics from input images. The spatial information retrieved by the CNNs is then temporally modeled by Bidirectional Long Short Term Memory Layers (BLSTM). The networks may finally be trained with various length videos and label sequences thanks to a Connectionist Temporal Classification (CTC) Loss Layer[3], [11], [19], [20]. In order to design the model different supporting tools are required. These include Python programming language, Pytorch, and Pytorch packages like torchvision to get a pre-trained CNN model like ResNet. we also use Google Sheets for our dataset annotations.

1.7.3 Evaluation Method

The aforementioned methods have been tested using various experimental setups. Word Error Rate(WER) are performance indicator that was used to assess our sign language recognition models. WER is a metric used to evaluate the performance of models in tasks like continuous Ethiopia sign language recognition. WER measures the rate of inaccuracies in transcribing or recognizing words compared to a reference or ground truth. As WER decreases, it indicates that the model is performing better, as it's making fewer errors in recognizing signs and translating them into words or sentences. The algorithm that performed well throughout the model evaluation phase was chosen under the performance evaluation results.

1.8 Organization of The Study

The thesis is organized into five chapters. The first chapter introduces the study. It describes statement of the problem, objective, scope, significance and methodology of the study. The second chapter is a literature review that explores previous works on end-to-end CESLR systems and related studies by foreign and local scholars. The review summarizes the previous studies and their recommendations, which could inform the present study. The third chapter delves into end-to-end CESLR, including the model architecture of the end-to-end CESLR model. The fourth chapter presents the experiments conducted and the results obtained from these experiments. The chapter also includes an interpretation of the findings from the experiments. In the final chapter, chapter five, the study's conclusion and recommendations for future research are presented

Chapter Two

Literature review

2.1 Overview

The literature review section of this thesis explores existing research and advancements in the field of sign language recognition, with a specific focus on Ethiopia Sign Language (ESL). It begins by providing an overview of the Language, Deaf and Hard of Hearing, and sign languages specifically about Ethiopia Sign Language.

The review encompasses studies on various approach of sign language recognition, including data collection methods, feature extraction techniques, algorithms, and evaluation methodologies. Moreover, the literature review delves into the unique characteristics of Ethiopia Sign Language, examining linguistic properties, cultural aspects, and previous efforts in Ethiopia Sign Language recognition. Special attention is given to research relevant to continuous Ethiopia sign language recognition, as this represents a significant area of interest for the current thesis.

2.2 Deaf and Hard of Hearing

Based on the 2021 WHO Fact Sheet on Deafness and Hearing Loss, approximately 5% of the global population experiences some form of hearing impairment. In Ethiopia, it is estimated that over five million individuals, out of a total population of 115 million, live with hearing loss [4]

The community of individuals who are Deaf or hard of hearing is a diverse and active group whose lives are significantly impacted by their hearing impairments [3], [9]. This community encompasses individuals with differing degrees of hearing loss, and their experiences are greatly shaped by these impairments. Being part of this community isn't solely about facing a medical condition; it's about embracing a distinct identity that influences their interactions, perspectives, and connections with others who share similar experiences.

Deaf and hard-of-hearing individuals encounter a myriad of challenges in their daily lives, stemming from societal attitudes, communication barriers, and limited accessibility. In educational

settings, for example, many Deaf students face obstacles due to a lack of sign language interpreters, insufficient captioning, or inaccessible learning materials. In the workplace, communication barriers can impede job opportunities and career advancement, leading to feelings of isolation and exclusion. Furthermore, accessing healthcare services can be a daunting task for many Deaf and hard-of-hearing individuals, as healthcare providers may lack proficiency in sign language or fail to provide necessary accommodations. Social interactions, too, can be fraught with challenges, as Deaf and hard-of-hearing individuals navigate environments that are often not designed with their needs in mind.

Despite these challenges, technological advancements offer promising solutions to enhance accessibility and communication for the Deaf and hard-of-hearing community. Sign language recognition technology, for instance, has the potential to revolutionize communication by enabling real-time translation between sign language and spoken or written language.

According to the 2007 Ethiopian Population and Housing Census, disabilities fall into the following categories:

1. **Deaf:** Persons who were not able to hear from both ears but can speak were grouped under this category.
2. **Hearing Difficulty:** Persons who totally lost hearing in one ear and partially hear by one or those who were able to hear partially by both ears were considered as persons with hearing difficulty.
3. **Dumb:** Persons who were not able to speak at all but can hear were considered as dumb.
4. **Speaking Difficulty:** Persons who were not able to speak as a healthy person could speak or persons whose speech couldn't be understood clearly to others were considered as having speaking difficulties.
5. **Deaf Mute:** Persons who were not able to speak and hear at all were categorized as deaf mute.

2.3 Sign Language

Language is a system of communication used by humans, typically consisting of a set of sounds, gestures, symbols, or written symbols that convey meaning and allow individuals to express thoughts, ideas, emotions, and intentions. It's a fundamental aspect of human cognition and culture, serving as a tool for both social interaction and individual expression[2], [8]. Language can be communicated through speech or writing. When spoken, language is expressed verbally and created through the intricate interaction of articulatory organs and air pressure from the lungs. The resulting speech signal, composed of a sequence of sound units, conveys the speaker's message to the listener [21].

Languages vary greatly across different cultures and regions, with thousands of languages spoken worldwide. Each language has its own unique grammar, vocabulary, syntax, and rules for communication. These linguistic elements enable individuals to understand and interpret the messages conveyed by others within the same linguistic community.

Language is vital across many facets of human existence, spanning education, literature, science, business, politics, and beyond. It functions as a tool for safeguarding and passing on cultural heritage, while also enabling cooperation, collaboration, and the sharing of ideas among individuals and communities [3], [9].

Language plays an indispensable role in every dimension of human experience, encompassing education, literature, science, business, politics, and beyond. Not only does it serve as a mechanism for preserving and transmitting cultural heritage, but it also fosters cooperation, collaboration, and the exchange of ideas among individuals and communities

spoken language relies on the speaker's articulate use of their articulatory organs, the suitable choice of communication medium, and the listener's well-functioning auditory system. However, for deaf individuals who do not meet these conditions, sign language becomes their primary avenue for communication, providing them with essential opportunities[8], [17].

Sign language, a multifaceted and intricate form of communication, serves as the primary mode of interaction for deaf and hard-of-hearing individuals across diverse cultures worldwide [4], [22]. Rooted in visual-manual modality, sign languages harness a rich array of gestures, hand shapes, facial expressions, and body movements to convey nuanced meanings, emotions, and concepts.

Beyond serving as a means of communication, sign languages also play a crucial role in fostering cultural identity and community cohesion within the Deaf community. They serve as mediums for artistic expression, storytelling, and cultural transmission, enriching the cultural landscape and promoting inclusivity and diversity. Moreover, sign languages facilitate access to education, employment, and various societal domains, empowering individuals with hearing impairments to fully participate in social, economic, and cultural life.

Despite the inherent complexities and richness of sign languages, they have often faced marginalization and lack of recognition in mainstream society. Efforts towards promoting awareness, acceptance, and accessibility are essential to ensuring the rights and inclusion of deaf individuals. Through ongoing advocacy, education, and technological advancements, sign languages continue to evolve and thrive as vibrant linguistic systems, embodying the resilience and creativity of the Deaf community worldwide.

There exists a common misconception regarding Sign Languages (SLs) as merely manual representations of spoken languages, disregarding their status as independent languages[4]. Some also wrongly perceive SLs as primarily consisting of pantomime and gestures, lacking the expressive capabilities of spoken languages. However, despite these misconceptions, linguists have identified an increasing number of natural Sign Languages. Ethnologue (https://www.ethnologue.com/), a prominent online database on world languages, documents in 2024 over 150 distinct Sign Languages utilized by various Deaf communities. Additionally, the World Federation for the Deaf (WFD) asserts the existence of over 300 Sign Languages, which are used by more than 70 million Deaf individuals globally [4].

2.4 Ethiopian Sign Language

Ethiopia Sign Language (EthSL) is the primary sign language used by the Deaf community in Ethiopia. It has its own distinct features and regional variations. Despite facing challenges such as limited formal recognition and educational resources, efforts are being made to preserve and promote EthSL through education, advocacy, and linguistic research.

In 2021, the World Health Organization (WHO) revealed that Ethiopia is home to 5 million deaf individuals, emphasizing Ethiopian Sign Language as the exclusive avenue of communication for this sizable segment of the population [4].

Ethiopia Sign Language is a primary means of communication for deaf and hard-hearing people in Ethiopia. Sign language was first introduced to Ethiopia in the 1960s by American and Nordic missionaries who established deaf schools in the country[12]. They introduced the sign language used in their own countries, which became integrated into Ethiopian deaf culture over 50 years.

Certain local signs initially originated and were utilized within specific schools for the hearing impaired in Ethiopia. Over time, these signs were integrated into Ethiopian Sign Language. Examples include words like "እንጀራ", "ሀበሻ", "ሬሳ" among others. Initially introduced by American missionaries, Ethiopian Sign Language was initially influenced by American Sign Language (ASL) and signed English. While it has been adapted to align with Ethiopian culture, there still retains some similarities with ASL. As Ethiopia developed its own sign language, it incorporated elements such as finger spelling, notation, structure, and signing style from the local hearing-impaired community. [1], [9], [16]

Ethiopian Sign Language (EthSL) evolved over time, incorporating finger spelling and signs devised by the Ethiopian deaf community. While initially influenced by American Sign Language (ASL) and other Nordic countries like Finnish Sign Language.

The hearing-impaired community, in collaboration with The Ethiopian National Association of the Deaf and other relevant organizations, have been dedicated to establishing Ethiopian Sign Language as an official communication tool. Through their concerted efforts, they have successfully cultivated a robust sign language system, now employed as a primary instructional medium in schools and utilized by the media for effective communication. [5], [16]

2.4.1 Component of Sign Language

In sign language, effective communication encompasses a multifaceted approach, extending beyond mere hand movements to encompass a spectrum of non-manual features. These include but are not limited to facial expressions, head movements, body posture, and other nuanced gestures. To fully understand sign language, it's essential to recognize and appreciate the intricate interplay between manual and non-manual components. Here, we delve into a comprehensive breakdown of these elements, highlighting their significance in facilitating clear and expressive communication within the deaf and hard-of-hearing community [3], [22].

Some of the manual features of sign language include the following[2], [6], [9]:

Handshape: The hand form is the initial setup for making a symbol. The orientation of the hand and fingers is specific to the signer. A right-handed signer uses their right hand to perform

signs, while a left-handed signer uses their left hand. Both right and left-handed signers can convey signs for the same letter.[3], [16], [23]

Movement: In sign language, meaningful expressions are conveyed through gestures like arcs, parallel lines, and curved shapes. Timing is crucial for interpreting these movements. Along with hand action, eye motions and body movement (movement across the chest) can be used. Movements can vary in speed, direction, and duration, and they contribute to the meaning of signs.[1], [3], [16], [21]. The hand movement causes variations in the letters to be communicated by a given sign. Ethiopian sign language employs seven alphabet orders, each with similar signs distinguished by hand shape movement[16], [23].

Location: The specific place in the signing space where signs are produced relative to the signer's body or other reference points. Different locations can indicate different meanings or grammatical features. Signs that are similar to others often differ only in their location. For instance, in Ethiopia Sign Language, the signs for "father" and "mother" share the same hand shape but vary in location. "Father" is signed with the hand placed on the forehead, while "Mother" is signed with the hand at the chin.[16], [23]

Orientation: The direction in which the palm or fingers of the hand are facing while producing signs. Orientation can change the meaning of signs and is important for clarity in communication. Orientation in signing refers to the direction of signs, which is crucial as it can alter their meanings. A sign may convey one meaning when executed in one direction and a different meaning when executed in the opposite direction. For instance, the sign for "give" requires the motion of giving from oneself to another person, while the sign for "take" involves the same gesture but in the reverse direction, indicating the action of receiving from another person.[9], [23]

In addition to manual features, the following are the non-manual features of sign language:

Facial Expressions: Facial expressions play a crucial role in sign language grammar and convey various aspects of meaning, including emotion, intensity, and grammatical information. For example, raised eyebrows can indicate a question, while a furrowed brow can convey negation[1], [3].

Eye Gaze: Eye gaze is used to establish and maintain visual contact between signers. It can also indicate topics, participants, and the direction of action within a conversation.

Head Movements: Head movements can provide important grammatical and syntactic information, such as indicating yes/no questions, wh-questions, or indicating the subject or object of a sentence.

Body Posture: Body posture can convey information about the speaker's attitude, mood, or emphasis. It can also be used to distinguish between different types of discourse, such as narrative versus dialogue.

These manual and non-manual features work together to create a rich and nuanced communication system in sign languages. They are essential for conveying meaning, grammatical structure, and emotion and contribute to sign language communication's complexity and expressiveness.

2.4.2 Signing Hand

Sign language is a dynamic visual language that relies on the expressive capabilities of the body. Beyond hand shapes, orientation, movement, and facial expressions, it encompasses a rich array of gestures and body movements to convey nuanced meanings. The use of space, including the placement and trajectory of signs, also plays a significant role in sign language communication, allowing for spatial grammar and emphasis. Furthermore, the rhythm and pacing of signing contribute to the overall fluency and clarity of expression in sign language conversations.

The distinction between dominant and non-dominant hands is significant in sign language, varying according to the signer's handedness. Typically, a right-handed individual uses their right hand as the dominant hand, while a left-handed individual utilizes their left hand in this role[1], [3], [16], [23]. This classification influences the execution of signs, where one hand takes precedence over the other depending on its dominance. In sign language, signs are categorized into three types based on how the hands are utilized[12]s.

- **Single-handed signs:** These signs are performed using either the dominant right or left hand exclusively. For example, in signing the Amharic alphabet letters, only one hand is used.
- **Symmetrical two-handed signs:** In these signs, both the dominant and non-dominant hands are engaged, and they move in tandem, mirroring each other's movements. This creates symmetry in the signing gestures.
- **Asymmetrical two-handed signs:** Similar to symmetrical two-handed signs, asymmetrical signs involve both dominant and non-dominant hands. However, in this case, only the dominant hand moves while the non-dominant hand remains stationary, providing a contrast in movement between the two hands.

2.4.3 Ethiopian Finger Spelling

The Ethiopian National Association for the Deaf (ENAD) introduced Ethiopian Finger Spelling in 1971, which was later officially recognized by the Ministry of Education, referred to as Ethiopian Manual Alphabets.As shown in Fig 2.1 Ethiopia sign language consist of 33 base alphabet signs, with each base alphabet having six variations either sharing the same hand shape or incorporating unique hand movements [2], [16], [22].

Ethiopia finger spelling stands apart from its counterparts in American, and British Sign Languages. Unlike American Sign Language(ASL) finger spellings, which typically involve minimal motion and are executed with a single dominant hand for most letters, Ethiopia finger spelling varies in its approach. While ASL and Ethiopian Sign Language predominantly utilize a single hand shape to represent letter signs, with only 'J' and 'Z' requiring movement in ASL, British finger spelling diverges by incorporating both hands for most letters, although 'C' is an exception and can be formed with a single hand. This contrast in finger spelling techniques highlights the unique characteristics of each sign language system.

In Ethiopia, the initial set of Amharic alphabets, known as "Ge'ez," are static (see Figure 2.1)and do not necessitate any hand movement to represent them. Conversely, the subsequent orders of

Amharic alphabets require motion to convey their respective signs[3], [16]. Similar to American Finger Spelling, Ethiopian Finger Spelling employs a single dominant hand to depict all alphabet signs.

In many cases like names of persons, countries, cities and some other common words, their signs are first letter based. The sign for the name **"አበበ"** for instance, is the sign for **"አ"** and touching part of our face. First alphabet-based signing is not limited to names only however the most common words like **"እንጀራ"** is also described by using the sign of **"እ"** and adding some movement that shows the method of baking **"እንጀራ"** [16], [23].

Signing numbers is different from the normal signing that most Amharic speakers use. All numbers below one thousand are signed using a single hand, and the second hand is only used to designate that a number is in the thousands or millions [12].



Figure 2.1: Ethiopia Manual Sign Language Alphabet[8]

2.4.4 Signed Ethiopian Sign Language

Signed Amharic is a sign language dialect which matches each spoken Amharic word. All aspects of Sign are signed by following Amharic grammar [2], [12]. People generally speak at the same time when using Signed Amharic to give hearing impaired children the benefit of both signed and spoken Amharic. Example, Amharic phrase **"አበበ ወደ ቤት ሄደ"** (Abebe went home) can be signed as direct word by word **"አበበ + ወደ +ቤት + ሄደ"** (Abebe + went + home)

Communication barriers exist between deaf and hearing individuals due to the inability of deaf people to hear spoken language and the challenge hearing individuals face in understanding

sign language. Sign-spoken language, employing signs and finger spelling, serves to convey the syntax of spoken language to deaf individuals, primarily utilized in educational and TV program contexts to facilitate understanding.

Signed Amharic serves as a bridge between spoken and signed languages. It's a methodical approach aimed at fostering communication between hearing and deaf communities, especially facilitated by Ethiopian Television news and programs for disseminating crucial information to the deaf populace in Ethiopia[1], [3], [5], [6], [8], [10], [15], [17]. This form of communication diligently mirrors spoken Amharic grammar, ensuring that every word is represented in sign language, thereby catering to the linguistic needs of both hearing-impaired individuals and the wider Amharic-speaking society. Moreover, in the context of Signed Amharic, speakers often accompany their signing with simultaneous spoken articulation, providing a comprehensive linguistic experience that nurtures the language development of deaf individuals, particularly children.

2.4.5 Ethiopian Sign Language Grammar

In Ethiopia Sign Language, the structure of sentences differs from that of spoken Amharic, following a distinctive arrangement known as "(Time) (Topic) (Comment)". This format diverges from spoken language's traditional "Subject + Object + Verb" syntax [2], [16].

The inclusion of an optional "Time" element in Ethiopia Sign Language serves to convey the tense of the sentence, whether it pertains to the present, past, or future. The "Topic" component assumes a role akin to that of the subject in spoken language, representing the primary focus or subject of discussion. Finally, the "Comment" aspect plays a pivotal role by furnishing a description or supplying supplementary information about the topic at hand.

In spoken Amharic, the sentence "The man is reading a book" is structured as "**ሰውየው መጽሐፍ ያነባል**" (Subject + Object + Verb). However, in Ethiopian Sign Language, this same sentence is conveyed through a distinctive sequence: Firstly, a sign indicating the present tense, such as tapping the wrist with the index finger, may be employed as the "Time" component (although this is optional).

Next, the "Topic" is established using a sign representing "man," which might involve holding the hand near the forehead and moving it forward to signify the individual being discussed.

Finally, the "Comment" is conveyed through a sign depicting "reading," which could entail holding hands in front of the face and mimicking the act of turning pages. By integrating these components, the sentence in Ethiopia Sign Language effectively communicates the same message as its spoken counterpart, albeit with a distinct structural arrangement that accentuates the topic and enriches the discourse with additional contextual information.

2.5 Sign language recognition

Sign language recognition refers to the process of developing and implementing technological systems capable of understanding and interpreting manual gestures, hand movements, and facial expressions used in sign languages. Sign language recognition systems serve the purpose

of transforming sign language into either written text or spoken language, facilitating communication between individuals who are unfamiliar with sign gestures and those who rely on them for communication.[5]

These systems typically utilize various sensors, cameras, and algorithms to capture and analyze visual information, converting it into textual or auditory output for communication purposes. Sign language recognition technology plays a crucial role in bridging the communication gap between individuals who are deaf or hard of hearing and those who do not understand sign language. It enables real-time translation of sign language into spoken language or text, facilitating interactions in educational, professional, and social settings. Sign language recognition systems vary in complexity, from simple gesture recognition applications to advanced machine learning-based models capable of accurately interpreting complex sign language grammar and nuances. As the field continues to advance, researchers focus on improving the accuracy, speed, and accessibility of sign language recognition systems to enhance communication inclusivity and empower individuals with hearing impairments.

Steps in Sign Language Recognition

1. **Data collection:** Data collection is a foundational step in developing a Sign Language (EthSL) recognition system. Data collection for Sign Language (EthSL) recognition involves both device-based and vision-based approaches[3], [12], [24]. Device-based recognition uses sensors such as accelerometers, gyroscopes, and flex sensors attached to various parts of the signer's body to capture detailed movements. These sensors record parameters like finger flexion, hand orientation, and acceleration, which are processed using pattern recognition algorithms. While data gloves and motion capture devices provide precise tracking, they can be cumbersome and intrusive, limiting the signer's natural movements and making them less practical for everyday use. Despite these challenges, the detailed data obtained from these sensors are valuable for accurately identifying signs.

   In contrast, vision-based recognition systems rely on cameras, including web, stereo, and 3D cameras, to capture video footage of sign language gestures. These systems use computer vision techniques to extract features such as hand shape, movement, and facial expressions from the video data. The collected videos are annotated with glosses, temporal boundaries. This annotated data is then used to train machine learning models, such as deep learning algorithms, to interpret and classify the gestures. Vision-based approaches are less intrusive and can capture the natural signing process, making them suitable for real-time applications and bridging communication gaps between sign language users and non-users.

2. **Preprocessing:** Preprocessing involves transforming the raw video data into a format suitable for feature extraction and model training. This step is crucial to ensure the data is consistent, manageable, and informative for the subsequent phases of the sign language recognition system.

   Frame Extraction:

   Choose an appropriate frame rate that balances capturing sufficient detail with managing data size. A higher frame rate ensures finer detail and smoother motion capture, but it increases the data volume and computational load. A typical frame rate for sign language recognition ranges from 25 to 30 frames per second (fps), depending on the complexity of the signs and the speed of the signer's movements.

Frame Extraction: Extract frames from the video at the selected frame rate. This involves segmenting the continuous video stream into individual frames, each representing a discrete moment in time. This step ensures that the temporal dynamics of the sign language are preserved while creating a manageable dataset. The extracted frames should maintain high visual quality to capture the subtle details of hand shapes, orientations, and movements essential for accurate recognition.

Resizing: Resize frames to a standard resolution to maintain consistency across the dataset. A common resolution might be 224x224 pixels, which balances detail with computational efficiency. Resizing ensures that all frames have the same dimensions, which is necessary for feeding them into machine learning models. In image processing, various resizing methods are employed to alter the dimensions of images while maintaining quality. One prevalent approach is the interpolation resizing technique, which utilizes known pixel values to estimate unknown values during the resizing process. Image interpolation functions in two primary ways: it aims to find the best approximation of pixel intensity based on the values of surrounding pixels in one direction and simultaneously in the perpendicular direction. This dual approach ensures a more accurate representation of the image.

Interpolation algorithms are broadly classified into adaptive and non-adaptive categories[3], [13]. Non-adaptive methods, such as nearest neighbor, bilinear, bicubic, spline, and Lanczos, treat all pixels uniformly without considering the content of the image. These algorithms apply a fixed mathematical formula across the entire image. For example, the nearest neighbor method assigns the value of the nearest pixel to the new pixel, leading to a blocky appearance, while bilinear and bicubic methods use linear and cubic polynomial functions, respectively, to smooth the image, resulting in better quality but higher computational costs.

On the other hand, adaptive interpolation algorithms adjust their calculations based on the image data they are processing. These methods analyze the image content and adaptively modify the interpolation process to preserve edges and finer details, thereby improving the visual quality, especially in images with high-frequency details like sharp edges and textures.

Brightness Adjustment: Normalize brightness levels across frames to mitigate the effects of varying lighting conditions. This involves adjusting the brightness and contrast to ensure uniform illumination, which helps in reducing the impact of shadows and highlights that can obscure hand and facial features.

3. **feature extraction:** Feature extraction is a crucial step in digital image processing, aimed at increasing recognition rates by converting raw pixel data into a compact feature vector representing the image's essential characteristics such as color, shape, and texture. This process reduces computational complexity and enhances recognition outcomes, particularly in applications like sign language recognition. Key features include color, which is captured using techniques like color histograms; texture, which reflects intensity variations and material surface roughness; and shape, described through region-based or contour-based methods. Handcrafted techniques and deep learning approaches are used in feature extraction[3], [8], particularly Convolutional Neural Networks (CNNs), are employed for feature extraction[25]. CNNs, consisting of layers such as convolutional, pooling, activation, and fully-connected layers, automatically identify and extract complex features from images, making them a state-of-the-art method for image recognition.

4. **Pattern recognition** Pattern recognition, also known as classification, is the foundational

process of recognizing recurring structures within complex information and assigning labels or categories to new instances based on their features. It aims to assign a new data to a specific class using a trained model, where the training set comprises standard examples of various classes. Classification methods can be divided into supervised and unsupervised approaches. Supervised classification uses pre-labeled samples to guide the classification of new data, while unsupervised classification groups unknown data based on inherent patterns. Several algorithms facilitate sign language recognition, such as Support Vector Machines (SVM), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). CNN, pivotal in deep learning, perform both feature extraction and classification by transforming input images through convolutional layers and culminating in a fully connected layer for class probabilities. RNNs, designed for sequential data, leverage intrinsic memory to retain contextual information, making them ideal for tasks requiring understanding of sequences. Each algorithm optimizes classification performance through various techniques, enhancing accuracy and reducing computational complexity, thereby playing a critical role in applications such as face recognition, text classification, and sign language recognition.

Challenges of sign language recognition The development of sign language recognition and translation tools is significantly hindered by two key challenges: resource scarcity and lack of standardization in available datasets[26], [27]. The scarcity of large, annotated datasets suitable for training machine learning models is a major obstacle. Collecting, annotating, and curating comprehensive datasets that cover the breadth of sign language vocabulary and usage is extremely difficult due to factors like limited access to signers, lack of linguistic expertise, and funding constraints. Sign languages have rich vocabularies and nuanced expressions, making it challenging to capture the full spectrum of signs and their variations in datasets. Collecting data that adequately represents the diverse lexicon and subtle nuances of sign language requires extensive efforts and collaboration across linguistic, cultural, and technological domains

Sign language recognition faces numerous challenges, primarily stemming from the variability, scarcity, and complexity inherent in sign language communication.

Variability in Signs:

1. **Signer Variability**: Just as spoken languages exhibit individual differences in accent, rhythm, and tone, sign languages showcase variability among signers. Factors such as signing speed and style contribute to the challenge of developing models that generalize effectively across diverse signing behaviors. Addressing this variability requires robust recognition algorithms capable of adapting to individual signing styles[25]–[27].
2. **Contextual Variation**: The meaning of a sign is not fixed and can vary depending on the context in which it is used. Factors such as topic, audience, and emotional expression influence sign interpretation, adding complexity to recognition tasks. Developing context-aware recognition systems that can dynamically adapt to contextual cues is crucial for accurate and nuanced sign language understanding[26], [27].

Data Scarcity:

Limited Datasets: The scarcity of large, annotated datasets suitable for training sign language recognition models hinders progress in the field. Acquiring, annotating, and curating comprehensive datasets that encompass the breadth and depth of sign language vocabulary and usage is

a daunting task, often constrained by factors such as access to signers, linguistic expertise, and funding[26]–[28].

Diverse Signs: Sign languages have rich vocabularies and nuanced expressions, posing challenges in capturing the full spectrum of signs and their variations. Collecting data that adequately represents the diverse lexicon and subtle nuances of sign language requires extensive efforts and collaboration across linguistic, cultural, and technological domains[11], [25].

Hardware Restriction: Another technical gap that is worth mentioning is the limits of hardware for conventional deep learning architectures. The model architecture that has multiple layers with millions of parameters is expected to be heavy in size and might require high resources and computational power [11], [29].

Environmental Challenges Sign-language recognition poses significant challenges due to various environmental and technical factors, particularly those related to sophisticated background and light conditions. These factors can greatly affect the accuracy of recognition [3].

Background Complexity: Complex or cluttered backgrounds can make it challenging to distinguish hand movements and positions from other objects and patterns in the scene. This visual noise can confuse the recognition algorithms, leading to errors in detecting and interpreting signs.

Dynamic Environments: In environments where there are moving objects or people in the background, tracking the hands accurately becomes even more difficult. The motion in the background can be misinterpreted as part of the signing activity.

Inconsistent Lighting: Uneven lighting, where different parts of the scene are lit differently, can cause parts of the hands or face to be improperly illuminated, complicating the detection process

2.6 Types of Sign Language Recognition Model

Sign language recognition, a crucial aspect of technology aimed at facilitating communication and accessibility for the deaf and hard of hearing community, can be classified categories based on the approach to signer variability: signer independent and signer dependent recognition systems. These classifications address the fundamental challenge of accommodating the diverse range of signing styles, speeds, and individual characteristics present among sign language users[3], [5] .

2.6.1 Types of SLR based on signer variability

There are two main categories of sign language recognition based on signer variability, such as signer dependent and signer independent.

Signer-dependent sign language recognition refers to systems that are specifically trained to recognize sign language gestures performed by individual signers. In this approach, each signer undergoes a personalized training phase where the system learns the signer's unique signing style,

handshapes, and gestures. The training data used for these systems typically includes samples of the signer performing a variety of signs and gestures. By tailoring the recognition model to the characteristics of each individual signer, signer-dependent SLR systems can achieve high accuracy for that particular signer. However, they may require a significant amount of user-specific data for training, making them less practical for scenarios involving multiple signers. Signer-dependent SLR systems are often used in applications where the signer's identity is known in advance and personalized recognition is critical for effective communication, such as personalized assistive technologies or individualized sign language tutoring systems.

Advantages of Signer-Dependent SLR

- High Recognition Accuracy: Signer-dependent SLR systems can achieve high accuracy for the trained signer, as the recognition model is specifically tailored to their unique signing style and gestures.
- Personalized Training: These systems allow for personalized training, where the signer's individual characteristics are taken into account, resulting in a recognition model optimized for their signing patterns.
- Effective Communication: Signer-dependent SLR can lead to more effective communication for the trained signer, as the system can better understand and interpret their signs with greater accuracy.

Disadvantages of Signer-Dependent SLR

- Data Dependency: Signer-dependent SLR systems require a significant amount of user-specific data for training, which may not always be readily available, especially for applications involving multiple signers.
- Limited Generalization: These systems may struggle to generalize to new or unknown signers, as the recognition model is optimized for a specific individual and may not perform well for others.
- Lack of Scalability: Signer-dependent SLR may not be scalable for applications with a large number of users, as personalized training for each individual signer can be time-consuming and resource-intensive.

Signer-independent sign language recognition

Signer-independent sign language recognition refers to systems designed to interpret sign language gestures from any signer without requiring personalized training data. These systems generalize across signers, allowing them to recognize a wide range of signing styles and variations. Signer-independent SLR systems do not rely on individual-specific characteristics but rather aim to capture common patterns and features present across different signers. This approach is particularly useful in applications where the signer's identity is unknown or changes frequently. Signer-independent SLR systems provide a versatile solution for various communication aids, interactive systems, and assistive technologies aimed at serving diverse user populations.

Advantages of Signer-Independent SLR

- Generalization: Signer-independent SLR systems generalize across signers, allowing them to recognize a wide range of signing styles and variations without requiring personalized training data.
- Versatility: These systems are versatile and can be applied in various scenarios where the signer's identity is unknown or changes frequently, making them suitable for diverse applications.
- Accessibility: Signer-independent SLR enhances accessibility by providing a solution that accommodates different signers, facilitating communication and interaction for a broader user population.

Disadvantages of Signer-Independent SLR

- Reduced Personalization: Signer-independent SLR systems may lack the level of personalization achieved by signer-dependent systems, potentially leading to lower recognition accuracy for individual signers.
- Limited Adaptability: These systems may struggle to adapt to the unique characteristics and variations in signing styles of specific signers, particularly in scenarios where personalized recognition is critical.
- Dependency on Training Data: Signer-independent SLR systems rely heavily on the availability and quality of training data, which may limit their performance in recognizing less common or culturally specific signs.

2.6.2 Types of SLR based on temporal dynamics

Another important distinction lies in how signs are perceived and interpreted over time. Sign language recognition can be categorized into three main types: static, isolated, and Continuous Sign Language Recognition. These classifications delineate the temporal aspect of sign language processing, addressing the varying durations and complexities of sign sequences [3], [5], [16].

Static Sign Language Recognition Static sign language recognition focuses on analyzing and interpreting individual sign gestures captured at a single moment in time. It involves identifying the static visual features of handshapes, orientations, and positions of the hands and other relevant body parts within a single frame or image. Static recognition systems are typically used for recognizing simple sign gestures or isolated signs where temporal information is not crucial for understanding.

Advantages static sign language recognition

- Simplicity: Static sign language recognition systems are relatively straightforward and easy to implement.
- Fast Processing: Analyzing single frames or images enables quick processing, suitable for real-time applications.

- Robustness: Static recognition can be more robust to factors like motion blur or rapid movements.
- Suitable for Isolated Signs: Well-suited for recognizing individual gestures or isolated signs.

Disadvantages static sign language recognition

- Limited Context: Lack of temporal context makes it challenging to interpret signs relying on temporal dynamics.
- Difficulty with Continuous Signing: Struggles to handle continuous signing sequences seamlessly.
- Reduced Expressiveness: Does not capture dynamic aspects like motion or facial expressions fully.
- Limited Vocabulary: Difficulty recognizing signs with complex handshapes or movements, limiting vocabulary coverage and accuracy.

Isolated sign language recognition

isolated sign language recognition focuses on recognizing individual signs presented in isolation, with clear boundaries between each sign[12]. In this approach, signs are typically performed one at a time, allowing for easier segmentation and recognition compared to continuous signing. Isolated sign language recognition involves identifying and interpreting individual signs or gestures performed independently, without any preceding or subsequent signs [3]. This technology analyzes hand movements, facial expressions, and body postures to recognize and understand the meaning of each isolated sign, without considering its context within a larger sequence.

Advantages Isolated Sign Language Recognition

- Flexibility: Isolated sign language recognition systems can recognize individual signs within a sequence, allowing for versatile applications such as sign language translation or educational tools.
- Tailored Feedback: Enables personalized feedback for sign language learners by identifying and correcting errors in individual signs.
- Scalability: Systems can be designed to accommodate a wide range of sign vocabularies, making them suitable for diverse user needs and applications.

Disadvantages Sign Language Recognition

- Context Limitations: May struggle to interpret signs accurately without considering the broader context of the signing sequence.
- Transition Challenges: Difficulty in handling transitions between signs smoothly, leading to potential errors or misinterpretations.

- Reduced Naturalness: Isolated recognition may not capture the natural flow and rhythm of sign language communication as effectively as continuous recognition systems.
- Training Data Requirements: Requires large and diverse datasets for training to ensure accurate recognition of a wide range of sign gestures.

Continuous Sign Language Recognition

Continuous Sign Language Recognition involves the real-time interpretation of sign language gestures within a continuous flow of signing. In this approach, signs are not presented individually with clear boundaries but rather as a continuous stream of movements and expressions [3], [12]. The system must accurately segment and interpret this stream, identifying the boundaries between signs and recognizing each sign as it occurs. This process is particularly challenging due to variations in signing speed, pauses between signs, and co-articulation (overlapping movements between adjacent signs). Continuous Sign Language Recognition systems typically employ advanced algorithms such as dynamic time warping, hidden Markov models, or deep learning architectures to handle these complexities and accurately recognize signs within the continuous stream of signing.

Advantages Continuous Sign Language Recognition

- Natural Interaction: Continuous Sign Language Recognition enables more natural and fluid interaction between signers and non-signers, resembling face-to-face communication.
- Real-time Translation: Allows for real-time translation of sign language into spoken language or text, facilitating communication between deaf and hearing individuals.
- Contextual Understanding: Captures the contextual information and nuances present in continuous signing sequences, enhancing the accuracy and richness of interpretation.
- Enhanced Expressiveness: Recognizes dynamic aspects such as motion, rhythm, and facial expressions, capturing the full expressiveness and naturalness of sign language communication.

Disadvantages Continuous Sign Language Recognition

- Computational Complexity: Continuous Sign Language Recognition systems are computationally intensive and may require advanced algorithms and processing techniques.
- Training Data Requirements: Requires extensive and diverse training datasets to effectively model the complex variations and nuances present in continuous signing sequences.
- Error Propagation: Errors in recognizing individual signs or transitions may propagate throughout the sequence, leading to inaccuracies in interpretation.
- Handling Variability: Must accommodate variations in signing speed, style, and individual characteristics among different signers, posing challenges for robust recognition.

These classifications provide a framework for understanding the temporal dynamics and processing requirements of sign language recognition systems. Each type of recognition addresses different aspects of sign language communication, from individual static gestures to fluent, continuous signing, contributing to the development of more effective and inclusive technologies for the deaf and hard-of-hearing community.

Overall, while static sign language recognition offers simplicity and speed, its limitations in handling temporal context and continuous signing sequences make it less suitable for capturing the natural flow and complexity of sign language communication compared to systems that incorporate temporal analysis.

2.6.3 Types of SLR Based on data collection

Data Collected on Controlled Environment

SLR model may be developed for the recognition of data collected in a controlled or uncontrolled environment.

Controlled Environment SLR refers to sign language recognition systems that operate within carefully regulated settings where environmental factors such as lighting, background noise, and camera angles are tightly controlled. Typically, these systems are deployed in well-equipped studios or laboratory setups specifically designed for data collection. In controlled environments, signers perform gestures in front of high-quality cameras under consistent lighting conditions, ensuring clear and reliable data capture. The advantage of controlled environments lies in their ability to provide clean and standardized data for training and testing SLR models. Researchers often use controlled environment SLR setups for conducting experiments, benchmarking algorithms, and developing robust recognition models. These systems serve as a foundation for understanding sign language gestures and developing accurate recognition algorithms before transitioning to real-world applications.

Advantages of Controlled Environment SLR:

1. **Standardized Conditions:** Controlled environments ensure consistent lighting, background noise levels, and camera angles, providing clean and reliable data for training and testing SLR models.
2. **Precise Data Collection:** Researchers can precisely regulate experimental conditions in controlled environments, allowing for accurate data capture and analysis of sign language gestures.
3. **Benchmarking and Comparison:** Controlled environment SLR setups facilitate fair comparisons between different algorithms and approaches, enabling researchers to benchmark performance and identify areas for improvement.

Disadvantages of Controlled Environment SLR:

1. **Limited Realism:** Controlled environments may lack the realism of natural settings, potentially leading to models that perform well in the lab but struggle to generalize to real-world scenarios.

2. **Resource Intensive:** Establishing and maintaining controlled environments can be resource-intensive, requiring specialized equipment, dedicated facilities, and careful calibration.

3. **Restricted Application Scope:** While beneficial for research and development, controlled environment SLR setups may not fully capture the complexities of real-world sign language communication, limiting their applicability in practical settings.

Uncontrolled (Real-Time) Environment SLR

Real-Time (Uncontrolled) Environment SLR refers to sign language recognition systems that operate in natural, uncontrolled settings where environmental conditions are less predictable. In these environments, such as public spaces, classrooms, or everyday communication settings, sign language gestures are captured as they occur naturally without constraints on factors like lighting, background noise, or camera angles. Real-time SLR systems face challenges such as variations in lighting, background clutter, occlusions, and unpredictable signer behavior. Despite these challenges, they are designed to provide practical solutions for real-world applications, such as assistive technologies, communication aids, and interactive systems. These systems aim to accurately interpret sign language gestures in real-time to facilitate communication and enhance accessibility for individuals with hearing impairments, making them essential tools for inclusive communication environments.

Advantages of Real-Time (Uncontrolled) Environment SLR

1. **Realistic Data:** Real-time SLR systems operate in natural environments, providing access to real-world sign language data with authentic variations in lighting, background noise, and signer behavior. This realism enhances the robustness and adaptability of recognition models trained on such data.

2. **Practical Applications:** Real-time SLR systems are well-suited for practical applications, including assistive technologies, communication aids, and interactive systems, where users require immediate and accurate interpretation of sign language gestures in everyday settings.

3. **Increased Accessibility:** By operating in uncontrolled environments, real-time SLR systems help bridge communication barriers for individuals with hearing impairments, enabling them to interact more effectively with the surrounding world and facilitating inclusive communication experiences.

Disadvantages of Real-Time (Uncontrolled) Environment SLR

1. **Environmental Variability:** Uncontrolled environments introduce variability in factors such as lighting conditions, background noise, and camera perspectives, which can pose challenges for accurate sign language recognition. This variability may lead to decreased recognition accuracy and reliability.

2. **Complex Signal Processing:** Real-time SLR systems must contend with complex signal processing tasks, including noise reduction, background subtraction, and motion tracking, to extract meaningful features from the input data. These tasks become more challenging in uncontrolled environments with unpredictable conditions.

3. **Increased Computational Requirements:** Processing sign language gestures in real-time within uncontrolled environments requires significant computational resources. This includes processing large amounts of data in real-time, running complex algorithms for feature extraction and recognition, and handling the inherent complexities of uncontrolled environmental conditions.

Despite these challenges, real-time SLR systems play a crucial role in enhancing accessibility and fostering inclusive communication environments for individuals with hearing impairments. Advances in technology and algorithm development continue to improve the robustness and reliability of real-time SLR systems, expanding their applicability in diverse real-world scenarios.

By classifying sign language recognition based on the setting of data collection, researchers and developers can tailor their approaches and algorithms to address the specific challenges posed by each setting. Techniques for handling noise, variability, and adaptability may differ between controlled and uncontrolled settings to achieve accurate and robust sign language recognition.

2.7 Approach of Sign language recognition

Sign language recognition systems typically involve the use of sensors, cameras, and machine learning algorithms to interpret and translate gestures into textual or spoken language. These systems can be categorized into two main approaches: vision-based and Device-based recognition [3], [12].

Device-based Recognition

Device-based SLR systems rely upon microcontrollers and specific sensors, such as data gloves, power gloves, digital cameras, accelerometers, depth cameras, Kinect, leap motion controllers, dexterous master gloves, and so on to capture hand movements and gestures directly. These devices measure parameters such as finger flexion, hand orientation, and acceleration, which are then processed using pattern recognition algorithms to identify signs. Data gloves offer precise tracking of hand movements but may be less practical for everyday use due to their intrusive nature. These approaches are difficult for signers because they need cumbersome devices to be worn by the signer and also limit their movement[9], [24], [30].

The Device-based approach to sign language recognition involves the use of wearable sensors or motion capture devices to directly capture the movements of the signer's hands and body. These sensors could include accelerometers, gyroscopes, or flex sensors attached to various body parts involved in signing. The data collected from these sensors are then processed to extract relevant features related to hand and body movements, which are subsequently used for recognition purposes.

Advantages Device-based Recognition

- **Precise Movement Capture:** Device-based approaches provide precise data on hand and body movements, allowing for accurate recognition of sign language gestures.

- **Real-time Processing:** Since sensor data is captured directly, it can enable real-time recognition, making it suitable for interactive applications.
- **Less Sensitive to Environmental Factors:** Unlike vision-based approaches, Device-based methods are less affected by variations in lighting conditions or background clutter.

Disadvantages Device-based Recognition

- **Limited to Capturing Physical Movements:** Device-based approaches may not capture facial expressions or subtle hand shapes, limiting their ability to interpret all aspects of sign language.
- **Intrusive Setup:** Wearable sensors require the signer to wear additional equipment, which may be perceived as intrusive and uncomfortable.
- **Cost and Maintenance:** Acquiring and maintaining sensor equipment can be costly, especially for systems requiring multiple sensors or sophisticated motion capture devices.

Vision-Based Recognition

Vision-based SLR systems utilize camera inputs such as web cameras, stereo cameras, and 3D cameras to capture video of a person's hand movements (gestures) and then use computer vision techniques to interpret and understand sign language gestures performed by a person.

Features such as hand shape, movement, and facial expressions are extracted from the visual data, and machine learning algorithms are trained on these features for recognition[3], [9], [24], [30].

This technology often employs machine learning algorithms, such as deep learning models, to classify and interpret the gestures. These algorithms are trained on large datasets of sign language videos to learn the relationship between hand movements and the corresponding sign language symbols or words.

Vision-based sign language recognition has the potential to bridge communication gaps between people who use sign language and those who do not, by translating sign language gestures into spoken or written language in real-time. It can be used in various applications, including communication aids for the deaf and hard of hearing, educational tools for learning sign language, and interfaces for controlling devices through gestures.

Advantages Vision-Based Recognition

- **Rich Information**: Vision-based approaches can capture rich information including hand shapes, movements, and facial expressions, enabling more comprehensive recognition.
- **Non-intrusive**: Since vision-based systems do not require wearable sensors, they are non-intrusive and more comfortable for users.

- **Flexibility**: Vision-based systems can potentially recognize sign language gestures from a distance, offering flexibility in deployment scenarios.

Disadvantages Vision-Based Recognition

- **Sensitive to Environmental Factors**: Vision-based approaches are sensitive to lighting conditions, background clutter, and occlusions, which can degrade recognition accuracy.
- **Complexity**: Analyzing visual data and extracting meaningful features can be complex and computationally intensive.
- **Limited Capture Range**: Vision-based systems may have limitations in capturing fine-grained hand and finger movements, especially at a distance.

Vision-based sign language recognition is a field of research that aims to enable computers to interpret sign language gestures through visual inputs that get form a camera. Several methods have been used for sign language recognition, some of which are discussed in the following subsections

2.7.1 Machine learning

Machine learning is a subset of artificial intelligence (AI) that involves the development of algorithms and statistical models that enable computers to learn and improve performance on a specific task without being explicitly programmed. ML algorithms learn from data, identify patterns, and make decisions or predictions based on that data. Sign language recognition involves teaching a computer to understand and interpret sign language gestures performed by humans. The are many algorithms that are used for the recognition of sign language some of them are discussed below:

Hidden Markov Model (HMM)

Hidden Markov Models (HMMs) are sophisticated statistical tools employed to unravel the intricacies of sequential data, where both observable and hidden states interplay. In these models, observable states manifest as discernible symbols or data points, while hidden states orchestrate the transitions between these observable states, probabilistically emitting the symbols observed. Their versatility finds application across diverse domains like speech recognition, natural language processing, bioinformatics, and beyond, primarily due to their adeptness at discerning hidden patterns within sequential datasets.

In sign language recognition, a similar approach utilizes a statistical model known as a hidden Markov model (HMM). this method analyzes the sequence of observable symbols in sign language gestures, such as hand shapes, movements, and facial expressions. The HMM is then used to predict the most likely sequence of sign language gestures. Hidden Markov Models (HMMs) are statistical models that excel in modeling sequences of observations, even when the underlying states generating these observations are not directly observable. They have been successfully applied to various pattern recognition and machine learning tasks, including sign language recognition.

Support Vector Machine (SVM)

Support Vector Machines (SVMs) are powerful supervised learning models used for classification and regression tasks. They work by finding the optimal hyperplane that best separates different classes in the feature space. SVMs aim to maximize the margin, which is the distance between the hyperplane and the nearest data points (support vectors) from each class. This results in a robust classifier that generalizes well to unseen data

In sign language recognition, Support Vector Machines (SVMs) can be employed as classifiers to distinguish between different sign gestures. Each sign gesture is represented as a set of features extracted from the hand movements, facial expressions, and other relevant characteristics. These features serve as input to the SVM, which learns to classify each gesture into its corresponding category or meaning.

A support vector machine was developed for binary classification, consists of projecting input vectors into a high-dimensional feature space and then looking for a linear decision boundary that maximizes the minimum distance between two class classes.[8]

Support Vector Machines (SVMs) can also handle multi-class problems using different methods like ”one-against-one” and ”one-against-all”. In the ”one-against-one” approach, SVMs are built for each pair of classes, while in the ”one-against-all” approach, one SVM is created for each class. In the latter strategy, each SVM is trained to distinguish samples of one class from all other classes. This means that multi-class labels are broken down into several two-class labels, and SVMs are trained to solve these individual classification problems. Finally, the solutions from these individual classifiers are combined to solve the overall multi-class problem.

For inappropriate (non-linear separable) data, the SVM first maps the original data to a higher dimensional space using a kernel function K, such as Radial Basis Function (RBF) and polynomial kernels and then separates them linearly [8], [16].

Naïve Bayes Classifier

A Naive Bayes classifier is a type of probabilistic classifier based on Bayes' theorem with an assumption of independence between the features. It is commonly used for classification tasks, where the goal is to assign a class label to a given input based on the features of that input [8], [31].

When classifying patterns using Bayesian classification, two key probabilities are involved: the prior probability and the posterior probability[32]. The prior probability indicates the likelihood that a pattern belongs to a certain class based on existing evidence or knowledge. In contrast, the posterior probability $P(C_i|x)$ represents the final probability that a pattern x belongs to class C_i. In a binary classification scenario with classes C_1 and C_2, where the prior probabilities are denoted as $P(C_1)$ and $P(C_2)$ respectively, the posterior probability $P(C_i|x)$ is calculated using the formula [31]:

$$P(C_i|x) = \frac{P(x|C_i) \cdot P(C_i)}{P(x)} \tag{2.1}$$

According to Bayes theory, the probability of the classification error can be minimized by the following rule:

x is classified to C1, if $P(C1|x) > P(C2|x)$; otherwise

x is classified to C2, if $P(C2|x) > P(C1|x)$

In the context of sign language recognition, a Naive Bayes classifier could be used to identify the meaning or intent of a sign based on features extracted from the sign itself. These features might include the hand shape, movement, orientation, and position, among others. The classifier would learn the statistical relationships between these features and the corresponding sign meanings from a training dataset and then use that knowledge to classify new signs it encounters.

In‘ building a sign language recognition system, you would collect data on various signs along with their corresponding features. You would then train a Naive Bayes classifier on this data, which would learn the likelihood of certain features given a particular sign. When presented with a new sign, the classifier would use this learned information to predict the most likely meaning or intent of the sign based on its features

2.7.2 Deep Learning

Deep learning, a branch of machine learning, employs neural networks to extract insights from data. as shown in the figure 2.2 these networks consist of interconnected nodes that handle information. Deep learning specifically utilizes deep neural networks, which comprise multiple layers of nodes. Each layer in such networks specializes in extracting distinct features from input data, and this process is iterated across all layers. The culmination of these layers' outputs yields the final result of the network.

There are many different architectures and variations of deep neural networks, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs). CNNs are commonly used for image and video analysis, while RNNs are well-suited for sequential data, such as speech and natural language processing. GANs generate new data that resembles a given training set, such as realistic images or music.

Training a deep neural network involves feeding it a large dataset of labeled examples and adjusting the weights and biases of the neurons to minimize the difference between the network's predictions and the true labels. This process, known as backpropagation, involves computing the gradient of a loss function concerning the network's weights and biases, and using this gradient to update the weights and biases in the direction of the steepest descent.

One of the key strengths of deep learning is its ability to learn from large, complex datasets without the need for manual feature engineering. In traditional machine learning, engineers must design and select features that are relevant to the task at hand, but in deep learning, the network learns the relevant features automatically from the data. This makes deep learning particularly useful in domains where the underlying patterns are difficult to express in a concise set of features, such as image or speech recognition.

In summary, deep learning stands as a formidable domain within machine learning, employing artificial neural networks comprising numerous layers to scrutinize and glean insights from data. Its adeptness at discerning intricate patterns from vast datasets renders it indispensable across various domains, spanning from image and speech recognition to cutting-edge scientific

endeavors. The continual evolution of novel architectures and methodologies promises to further enhance deep learning's potential and broaden its scope of applications in the years ahead [8], [31], [33].

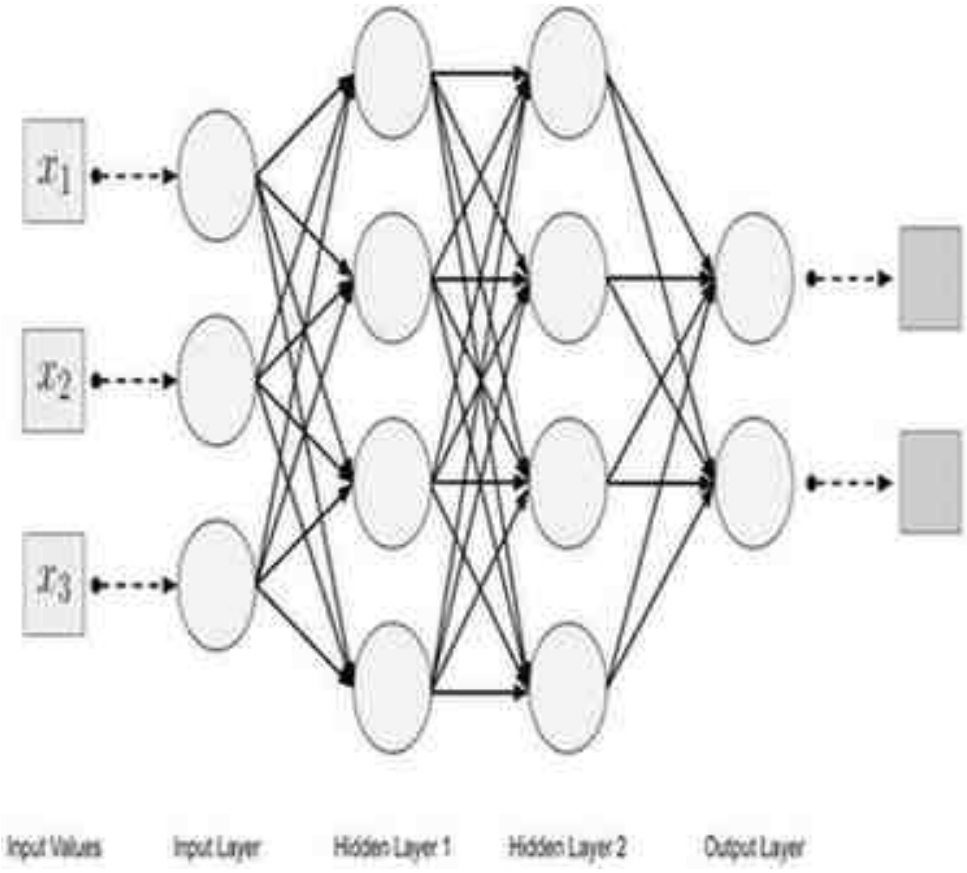


Figure 2.2: Deep Learning Architecture

2.7.3 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are a type of deep learning architecture specifically tailored for processing visual data. They have revolutionized the field of computer vision due to their ability to automatically learn hierarchical representations of features directly from raw input data, such as images or video frames.

In end-to-end sign language recognition systems that utilize Convolutional Neural Networks (CNNs), the CNN performs feature extraction directly from the input data, such as video frames of sign language gestures. This approach eliminates the necessity for using distinct methods specifically designed for feature extraction, resulting in a more comprehensive and streamlined solution. By incorporating feature extraction within the CNN architecture, the system becomes more efficient and effective at recognizing sign language gestures, as it automatically learns to extract relevant features during the training process. This integration of feature extraction within the CNN contributes to a more cohesive and robust sign language recognition system.

Architectures in Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) is a deep learning model specifically designed for processing and analyzing visual data, such as images or videos. It consists of multiple layers, including convolutional layers, pooling layers, and fully connected layers[33]. Architectures in Convolutional Neural Networks (CNN) essentially refer to the different configurations and arrangements of layers within CNN models. These architectures are tailored for specific tasks and datasets, with variations in the number of layers, the type of layers used (convolutional, pooling, fully connected), and other architectural choices. Each architecture may have its strengths and weaknesses, and researchers often explore and develop new architectures to improve performance on various visual recognition tasks.

LeNet:

LeNet, developed by Yann LeCun et al. in the 1998, is one of the pioneering Convolutional Neural Network (CNN) architectures. It comprises several layers, including convolutional layers followed by subsampling layers (pooling), and concludes with fully connected layers. LeNet was originally designed for handwritten digit recognition tasks, making it a foundational architecture in the field of computer vision. Its simplicity and effectiveness laid the groundwork for subsequent advancements in CNN architectures.

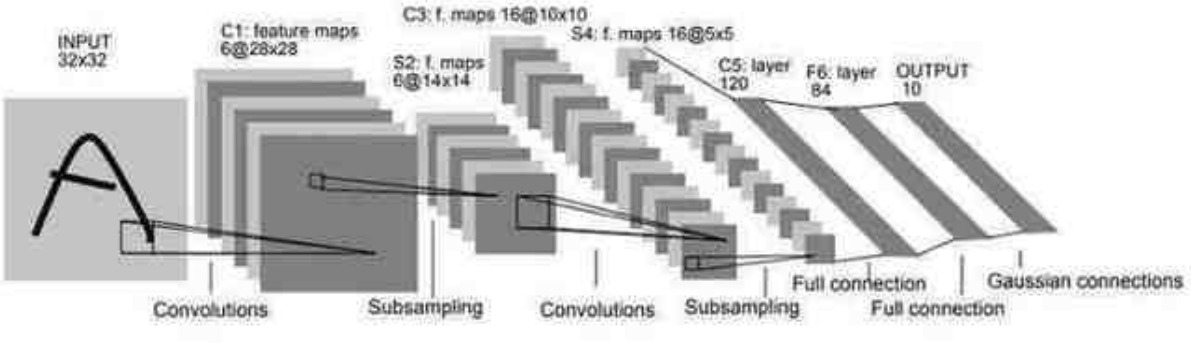


Figure 2.3: LeNet Architecture [34]

AlexNet:

AlexNet, introduced by Alex Krizhevsky et al. in 2012, marked a significant breakthrough in the domain of deep learning. It consists of several convolutional layers with rectified linear unit (ReLU) activation functions, interspersed with max-pooling layers to reduce spatial dimensions. AlexNet's deep architecture, compared to previous models like LeNet, enabled it to learn intricate features from images, leading to superior performance in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2012. AlexNet's success popularized the adoption of deep CNNs for various computer vision tasks.

ResNet:

ResNet, short for Residual Network, was introduced by Microsoft Research in 2015 to address the challenges of training very deep neural networks. ResNet leverages residual connections, also known as skip connections, to facilitate the flow of gradients during training, thereby mitigating the vanishing gradient problem. These skip connections enable the network to learn residual mappings, making it easier to train extremely deep architectures with hundreds of layers. ResNet architectures have been instrumental in achieving state-of-the-art performance in image classification, object detection, and other visual recognition tasks.

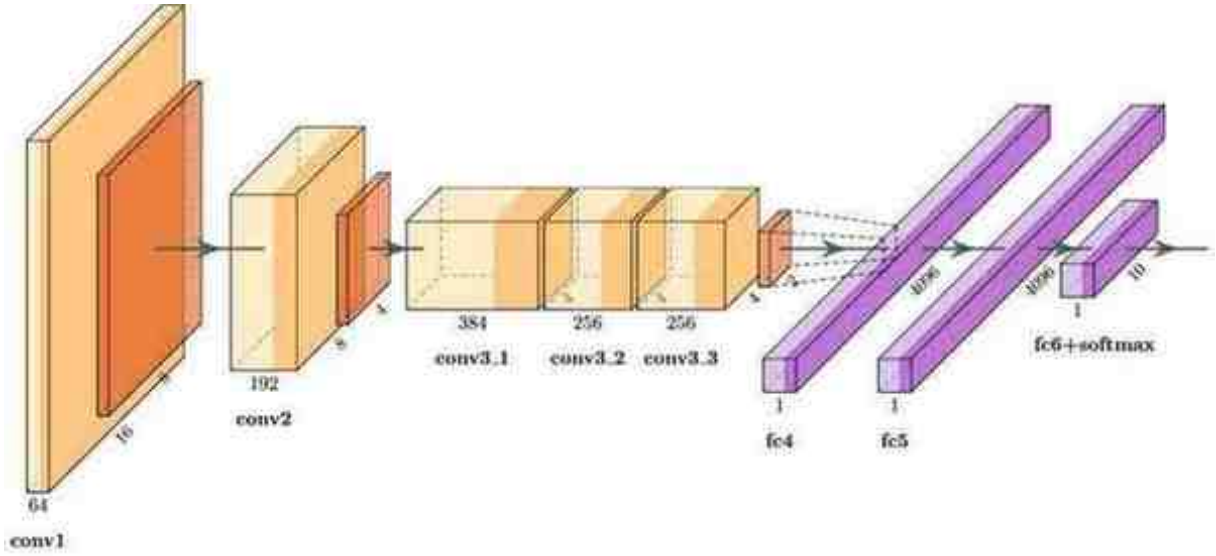


Figure 2.4: AlexNet Architecture [35]

In the context of sign language recognition, ResNet has been used in various ways. For example, in [36], ResNet-50 was used as one of the five deep learning models to recognize American Sign Language (ASL) alphabet gestures. The study found that ResNet-50 achieved the highest accuracy rate of 99.98% among all models, demonstrating its effectiveness in recognizing ASL gestures

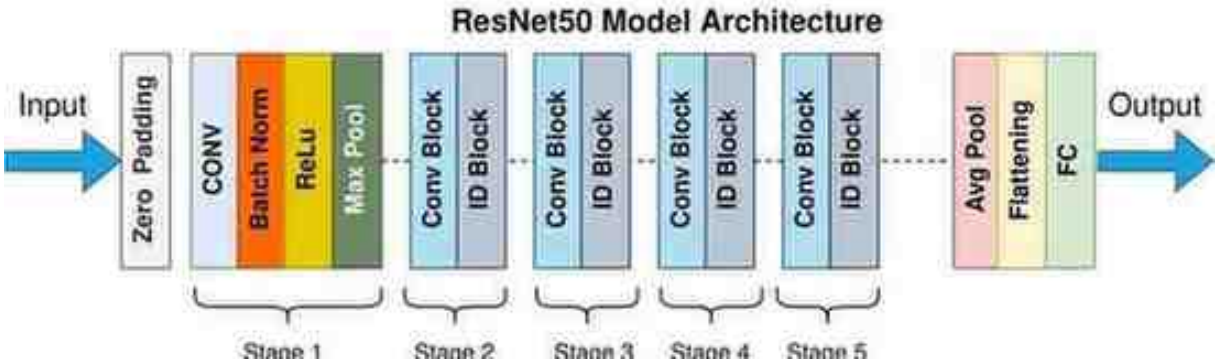


Figure 2.5: ResNet Architecture [37]

MobileNet:

MobileNet is a lightweight CNN architecture designed by Google to meet the computational constraints of mobile and embedded devices. Introduced in 2017, MobileNet employs depth-wise separable convolutions to reduce computational complexity and model size while maintaining high performance. Unlike traditional convolutional layers, depth-wise separable convolutions decompose the standard convolution operation into separate depth-wise and point-wise convolutions, significantly reducing the number of parameters and computational costs. The MobileNet architecture is characterized by its speed, low latency, and low power consumption, making it ideal for various use cases such as image classification, object detection, embeddings, and segmentation. By reducing the number of parameters required for processing, MobileNet optimizes resource utilization without compromising performance, enabling efficient image recognition on mobile platforms.

In summary, MobileNet is a versatile and efficient deep learning model that leverages depth-wise separable convolutions to achieve high accuracy in image classification tasks while being optimized for deployment on mobile devices with limited computational capabilities.

2.7.4 Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNNs) are a type of artificial neural network designed to efficiently process sequential data by maintaining an internal memory or state. Unlike traditional feedforward neural networks, which process data inputs in a fixed order, RNNs can incorporate information from previous inputs to make predictions or decisions about the current input. This makes them particularly useful for tasks such as natural language processing, speech recognition, time series analysis, and more. RNNs have connections that form directed cycles, allowing them to exhibit dynamic temporal behavior. This architecture enables them to handle inputs of varying lengths and to capture dependencies over time in sequential data.

One of the most common types of recurrent layer used in RNNs is the Long Short-Term Memory (LSTM) layer. LSTMs were introduced in 1997 by Hochreiter and Schmidhuber, and have since become one of the most popular types of recurrent layer. The key innovation of the LSTM is its ability to remember or forget information from previous time steps selectively[38]. This is accomplished through the use of three ”gates”: the input gate, the forget gate, and the output gate. The input gate controls how much new information is allowed into the hidden state, the forget gate controls how much previous information is forgotten, and the output gate controls how much information is output at the current time step.

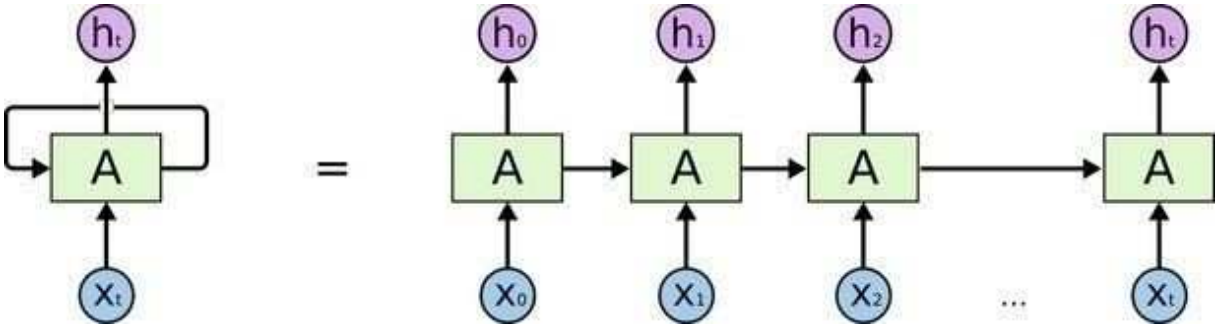


Figure 2.6: Architecture of RNN [39]

Another popular type of recurrent layer is the Gated Recurrent Unit (GRU) layer. GRUs were introduced in 2014 by Cho et al., and are similar to LSTMs but have a simpler architecture. GRUs have two gates: the reset gate and the update gate. The reset gate controls how much information from previous time steps is reset, while the update gate controls how much information is updated with the current input

In conclusion, RNNs are a powerful class of neural networks that are well-suited to processing sequential data. They have proven to be highly effective in a wide range of applications, and continue to be an active area of research in the field of deep learning. With the increasing availability of large datasets and powerful computing resources, we can expect RNNs to continue to advance and find new applications in the years to come.

Long Short-Term Memory (LSTM):

Long Short-Term Memory (LSTM) networks emerge as a breakthrough solution to the vanishing gradient problem in vanilla RNNs. LSTM architectures incorporate specialized memory cells and gating mechanisms—input, forget, and output gates to selectively retain or discard information over multiple time steps. This enables LSTMs to capture long-range dependencies more effectively, making them indispensable for tasks requiring a nuanced understanding of sequential data, such as natural language processing and time series prediction.

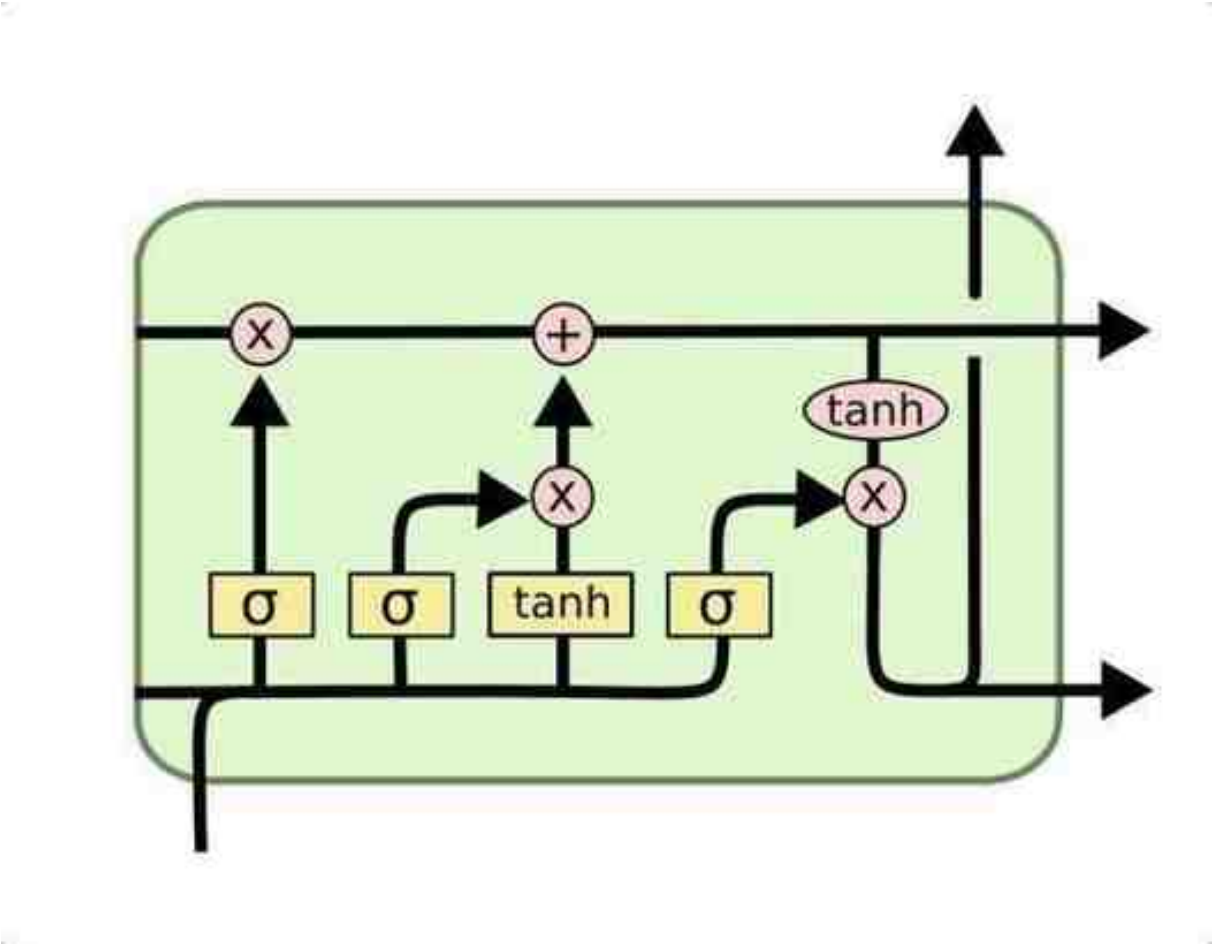


Figure 2.7: Architecture Of LSTM [38]

Bidirectional LSTM (BiLSTM):

Bidirectional LSTM (BiLSTM) networks elevate the capabilities of traditional LSTMs by processing input sequences in both forward and backward directions simultaneously. This bidirectional processing enables BiLSTMs to capture contextual information from past and future time steps, enhancing their ability to model complex temporal patterns. By incorporating bidirectional context, BiLSTMs excel in tasks where a comprehensive understanding of sequence context is essential, including sentiment analysis, machine translation, and named entity recognition.

The basic architecture of a BiLSTM cell consists of two LSTM layers, one processing the sequence in the forward direction and the other processing it in the backward direction. The output of each layer is concatenated to form the final output of the BiLSTM cell. During training, the weights of both LSTM layers are learned through backpropagation.

BiLSTMs are utilized to analyze and recognize the sequential nature of sign language gestures, enabling the model to understand the temporal dynamics and context of the gestures for accurate recognition[40]. The BiLSTM model is particularly effective in scenarios where understanding the sequence of events is crucial, such as in recognizing the dynamic and continuous nature of sign language gestures.

Gated Recurrent Unit (GRU):

Gated Recurrent Unit (GRU) networks represent a streamlined alternative to LSTMs, featuring a simplified architecture with fewer parameters. GRUs also incorporate gating mechanisms—update and reset gates—that regulate the flow of information within the network, facilitating efficient learning and memory management. Despite their simplicity, GRUs exhibit comparable performance to LSTMs in many tasks while requiring less memory and computational resources, making them an attractive choice for sequence modeling applications.

The architecture of GRU is similar to that of RNN, but it has an additional gating mechanism that allows it to capture long-term dependencies effectively. The GRU unit consists of three gates that control the flow of information. The Update Gate determines how much of the past information should be passed along to the future. The Reset Gate controls which parts of the past information are important for predicting the current output. There is a Candidate Activation that Computes the new candidate activations based on the input and the reset gate. Finally, hidden State represents the memory of the network, updated based on the update gate and the candidate activation.

The Gated Recurrent Unit (GRU) comprises several key components that enable it to process sequential data effectively. These components include the update gate, reset gate, and candidate activation.

The update gate is a neural network that takes as input the previous hidden state (h_{t-1}) and the current input (x_t). It outputs a value between 0 and 1, indicating the extent to which the gate should be open or closed. This gate is pivotal in determining how much information from the previous hidden state should be preserved for the current time step.

Similarly, the reset gate is another neural network that operates on the previous hidden state (h_{t-1}) and the current input (x_t). It also produces a value between 0 and 1, indicating the degree to which the gate should forget or retain information from the previous hidden state. The reset gate plays a crucial role in deciding which parts of the previous hidden state are relevant for predicting the current output.

The candidate activation is calculated by combining information from both the previous hidden state and the current input. First, the previous hidden state (h_{t-1}) is multiplied by the reset gate (r_t), effectively filtering out irrelevant information. This filtered hidden state is then concatenated with the current input (x_t) and passed through a neural network to generate a new candidate activation ($\tilde{h}_t$).

Finally, the update gate determines how much of the previous hidden state to retain and how much of the new candidate activation to incorporate into the current hidden state. By balancing the preservation of relevant information from the past and the integration of new information from the current input, the GRU effectively manages long-range dependencies and adapts to the dynamics of sequential data.

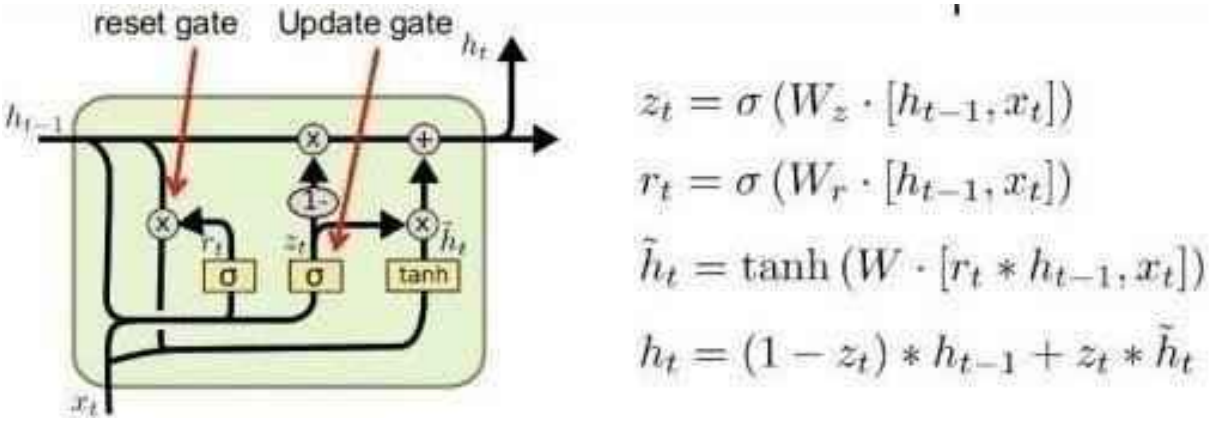


Figure 2.8: Architecture of GRU [41]

Bidirectional GRU (BiGRU):

Bidirectional GRU (BiGRU) networks combine the bidirectional processing capability of BiLSTMs with the efficiency and simplicity of GRUs. By processing input sequences in both forward and backward directions simultaneously, BiGRUs capture bidirectional context while maintaining computational efficiency. This makes BiGRUs particularly well-suited for tasks where memory and computational resources are limited, such as mobile and embedded systems. With their balanced performance and resource efficiency, BiGRUs have gained widespread adoption across diverse domains, from speech recognition to gesture recognition.

The BiGRU architecture extends the capabilities of the standard GRU by processing the input sequence in both forward and backward directions simultaneously. Instead of a single GRU layer, BiGRU incorporates two separate GRU layers: one for the forward direction and another for the backward direction. These two layers operate independently, each capturing different aspects of the input sequence.

The outputs of the forward and backward GRU layers are then concatenated to form the final output. This allows BiGRU to capture dependencies in the input sequence from both directions, providing a more comprehensive understanding of the context. In natural language processing (NLP) tasks, this enhanced capability often leads to better performance, as it enables the model to capture nuanced relationships and semantic dependencies present in the text. s

2.7.5 Connectionist Temporal Modeling (CTC)

Connectionist Temporal Classification (CTC) is a machine learning technique employed for tasks involving sequence labeling. Developed by Alex Graves and Jürgen Schmidhuber in 2006, it addresses the challenge of identifying sequential data of varying lengths with no predefined alignment[42]. CTC finds extensive application across diverse domains including speech recognition, handwriting recognition, and machine translation.

In CTC, the objective is to learn a mapping from input sequences to output sequences without requiring explicit alignment between them during training. This is achieved by introducing a special blank label and allowing the model to output this label to indicate ”no output” for a

particular time step. This enables the model to implicitly learn the alignment between input and output sequences

The CTC algorithm is alignment-free and doesn't require an alignment between the input and the output. However, to get the probability of an output given an input, CTC works by summing over the probability of all possible alignments between the two. We need to understand what these alignments are in order to understand how the loss function is ultimately calculated.

Connectionist Temporal Classification (CTC) has been instrumental in advancing the field of continuous sign language recognition, particularly in the development of end-to-end systems. These systems aim to directly translate video sequences of sign language gestures into corresponding textual representations without requiring explicit alignment information. One notable application of CTC in this context is its utilization within neural network architectures, such as long short-term memory (LSTM) networks or transformer models.

These architectures process the input video sequences and produce output sequences of linguistic units, such as words or characters, through training with the CTC objective function. The CTC loss function allows the neural network to learn the mapping between input video sequences and output textual representations without the need for manual alignment, thereby streamlining the recognition process.

During training, the network is optimized to maximize the likelihood of generating the correct textual representation for each input video sequence, ensuring robust performance. In inference, the trained model utilizes decoding techniques like beam search or greedy decoding to predict the most likely sequence of linguistic units for a given input video sequence. This approach enables the development of accurate and efficient continuous sign language recognition systems, with broad applications in communication accessibility for the deaf and hard-of-hearing community

CTC (Connectionist Temporal Classification) in end-to-end Continuous Sign Language Recognition refers to a loss function used to bind the recognition and translation problems into a single unified architecture. This approach does not require ground-truth timing information and simultaneously solves two interdependent sequence-to-sequence learning problems, leading to significant performance gains in sign language recognition and translation tasks[11]. It allows for joint learning of Continuous Sign Language Recognition and Translation in an end-to-end manner, enhancing the overall performance of the system without the need for explicit timing information.

2.8 End to End Continuous Sign Language Recognition

End-to-end Continuous Sign Language Recognition refers to the process of automatically recognizing and interpreting sign language gestures or sequences of gestures without interruption or breaks in the flow of communication.

Traditionally, sign language recognition systems have been composed of multiple stages, involving hand gesture detection, tracking, feature extraction, and classification. End-to-end systems, however, aim to streamline this process by directly mapping the raw input (e.g., video sequences of sign language gestures) to the desired output (e.g., text or spoken language translations) without relying on intermediate steps.

End-to-end Continuous Sign Language Recognition systems have the potential to improve accessibility for deaf and hard-of-hearing individuals by providing real-time translation of sign language into spoken or written language, facilitating communication with hearing individuals. However, they also pose significant technical challenges due to the complexity and variability of sign language gestures. Ongoing research in this field aims to address these challenges and improve the accuracy and usability of such systems.

Continuous Ethiopia Sign Language (CESL) is a complex and expressive visual language used by the deaf community in Ethiopia. Recognizing and understanding CESL gestures have significant implications for the development of assistive technologies, educational tools, and communication aids tailored for the Ethiopian deaf community. This section explores the current state of sign language recognition techniques, challenges specific to CESL recognition, and potential avenues for future research.

These systems typically employ deep learning techniques, such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs), to learn directly from the raw input data. They require large amounts of annotated sign language data for training, as well as sophisticated architectures to handle the temporal nature of sign language gestures.

2.9 Related Works

End-to-end Continuous Sign Language Recognition is a technique where the whole pipeline from the image stream to text output is modeled using a single neural network, without the need for any intermediate feature extraction or other steps.

2.9.1 Related Work For Foreign Sign Language

The paper[25] introduces a novel approach to weakly supervised learning in the video domain. Through the integration of CNN-LSTM models into a multi-stream HMM framework, the authors propose a comprehensive solution for jointly learning from multiple modalities. Their approach addresses the challenges of weak labeling by introducing intermediate synchronization constraints and multiple loss functions, eliminating the need for preprocessing steps like tracking and cropping.

The authors contextualize their work within the broader field of weakly supervised learning for videos, highlighting the importance of parallelism in addressing sequence learning tasks. They specifically focus on sign language recognition, discussing the limitations of previous approaches in handling simultaneous sub-problems like hand motion and mouth gestures. Building upon prior research in lip-reading and articulated hand shape recognition, the paper presents a significant advancement in weakly supervised learning, offering a comprehensive solution that integrates multiple modalities and eliminates preprocessing steps.

The paper presents compelling results obtained through the application of the proposed approach across three diverse datasets. Firstly, on the PHOENIX 2014 dataset, which encompasses a broad vocabulary with over 1000 sign language classes, the method demonstrates a notable

reduction in the word error rate (WER) compared to state-of-the-art methods. This suggests the effectiveness of the approach in accurately recognizing sign language sequences.

The experimental evaluation showcases the effectiveness of the proposed approach on challenging datasets, demonstrating superior performance in continuous sign language recognition, mouth shape detection, and hand shape classification. The clear presentation and thorough evaluation make this paper a valuable contribution to the field of sequence learning in the video domain, offering promising results and potential applications beyond sign language recognition.

One potential drawback of the paper is the lack of detailed discussion on the computational complexity and scalability of the proposed approach. While the method shows promising results, its efficiency on large-scale datasets or real-time applications remains uncertain. Additionally, the reliance on weak labels for training may introduce noise and uncertainty, which could impact the robustness of the learned classifiers, particularly in scenarios with limited or noisy training data.

Challenges in the paper include the need for further exploration and validation of the proposed approach across diverse datasets and domains beyond sign language recognition. The generalizability of the method to other sequence learning tasks and its ability to handle varying levels of complexity in real-world scenarios require thorough investigation. Furthermore, the interpretability of the learned features and the extent to which they capture meaningful representations of the input data remain areas for future research and improvement.

The papers[43] address the challenge of overfitting in Vision-based Continuous Sign Language Recognition (CSLR) training by proposing a Visual Alignment Constraint (VAC) to enhance the feature extractor with alignment supervision. The VAC comprises two auxiliary losses, focusing on visual features and enforcing prediction alignment between the feature extractor and the alignment module. Experimental results on two challenging CSLR datasets demonstrate that the proposed VAC enables end-to-end trainable networks and achieves competitive performance.

The introduction provides a comprehensive overview of CSLR, emphasizing its significance in bridging communication gaps between Deaf and hearing individuals. It outlines the weakly supervised nature of CSLR, with network architectures composed of a feature extractor and alignment module. The paper highlights the limitations of end-to-end training, citing previous studies that leverage iterative training schemes to enhance the feature extractor. It introduces the Visual Alignment Constraint (VAC) as a solution to efficiently train CSLR models, addressing the overfitting of the alignment module and improving the generalization ability of the trained model.

The paper reports experimental results on two challenging CSLR datasets: PHOENIX14 and Chinese Sign Language (CSL). The proposed Visual Alignment Constraint (VAC) enables end-to-end trainable networks and achieves competitive performance on these datasets. Specifically, the method achieves significant improvements in CSLR training by addressing the overfitting issue, leading to enhanced generalization ability. Additionally, two metrics, Word Deterioration Rate (WDR) and Word Amelioration Rate (WAR), are introduced to evaluate the contributions of the feature extractor and alignment module, serving as indicators of overfitting. Overall, the experimental results validate the effectiveness of the proposed VAC approach in improving CSLR performance on real-world datasets.

The paper[11] introduces a novel transformer-based architecture for jointly learning Continuous Sign Language Recognition (CSLR) and Translation in an end-to-end manner. This approach is

facilitated by employing a Connectionist Temporal Classification (CTC) loss, which binds the recognition and translation problems into a unified architecture. The joint approach eliminates the need for ground-truth timing information and significantly improves performance in both recognition and translation tasks. Evaluation on the challenging RWTH-PHOENIX-Weather-2014T (PHOENIX14T) dataset demonstrates state-of-the-art results, surpassing previous models in both sign language recognition and translation. Notably, the translation networks outperform existing models by a considerable margin, showcasing the effectiveness of the proposed Sign Language Transformers.

The introduction provides a comprehensive overview of the challenges and complexities involved in sign language translation. It highlights the importance of recognizing the sequence of sign glosses (CSLR) and emphasizes the distinction between CSLR and Sign Language Translation (SLT) due to differences in grammar and linguistic structure. The paper addresses the unsolved sub-tasks in sign language translation, particularly sign segmentation, which is crucial for realizing continuous sign language translation. While previous studies have focused on CSLR, the proposed approach tackles the spatio-temporal machine translation task by jointly addressing CSLR and translation, thereby advancing the state-of-the-art in sign language translation.

2.9.2 Related Works For Ethiopia Sign Language

The study conducted by Nigus and Menore concentrates on the recognition of Amharic alphabet signs and their conversion into corresponding text using a machine learning approach [16]. Ten students actively participated in the data collection process facilitated by the researchers. The video footage, lasting 3 to 4 seconds each, was recorded in the third-generation partnership (3GP) format. The researchers propose an architecture comprising six distinct processes. Initially, the video footage of sign language is converted into frame images. Subsequently, the frame images undergo preprocessing to eliminate any noise, followed by the separation of the background from the sign characters in the third process. The fourth step involves feature extraction utilizing Fourier descriptors, motion descriptors, and color difference measures. These extracted features are then utilized to train respective models for learning machines such as Neural Networks (NN) and Support Vector Machines (SVM). During the testing phase, these trained models are utilized to identify sign language and convert it into printed text. The performance of the NN and SVM models is evaluated, employing a ten-fold cross-validation testing approach. This assessment also scrutinizes the classification efficiency of the NN and SVM models. Ultimately, employing NN and SVM classifiers, the device achieves sign identification with precision rates of 57.82% and 74.06%, respectively. Consequently, the Multi-Class SVM classifier demonstrates superior classification efficiency compared to the NN classifier. Nonetheless, a notable limitation of this study is the relatively modest accuracy of the model. Moreover, the research fails to provide comprehensive translation functionality as it solely detects alphabet signs. Furthermore, both the accuracy of the model and the dataset size are deemed inadequate. Therefore, there is a clear imperative to develop a model with improved accuracy, incorporating a more extensive dataset that encompasses not only alphabets but also words, numbers, and sentences.

The study conducted by Biruk Menegiste centers on advancing Ethiopia Sign Language recognition at the word level, employing deep learning techniques to convert Amharic word signs

into their corresponding textual representations [6]. The system takes video frames of Ethiopia Sign words as input and produces Amharic text as output, with its architecture comprising preprocessing, feature extraction, and classification stages. Preprocessing involves cropping and converting RGB to grayscale, while feature extraction utilizes a deep residual network (ResNet-34), storing features in .csv format. Classification is performed using the same ResNet-34 architecture. The system is trained and evaluated using a specifically curated dataset for this thesis, encompassing all Ethiopia Sign words. Evaluation metrics such as precision, recall, F1 score, and accuracy assess its performance, which achieves a 95% accuracy in classifying 60 sign words. However, a notable limitation is the small dataset size, lacking coverage of numbers and alphabets.

In another work, Tefera & Yaregal proposed the recognition of Isolated Signs in Ethiopian Sign language [2]. Three signers were involved in the data collection. The input video was shot using a laptop webcam camera with 30 frames per second frame rate. The camera was placed in front of the signer and set to a static location. The signers wore long-sleeved shirts to assist with skin identification and segmentation of the hands and heads. Twenty-five signs were selected to participate in the study. Twenty of the signals were performed with one hand, while the remaining five are performed with two hands. The researchers chose the word because it is commonly used in the community. The researchers used HIS model skin color detection used for the hand and the head segment. The selected features extracted for each region which are segmented as body parts. The Hidden Markov Model was used by the researchers for training and recognition (HMM). However, the gap of this work is the researcher gives attention only to words without including numbers and the Amharic alphabet, so that the study couldn't provide full translation functionality. Besides, the researchers used a static background when capturing videos of the signers. Moreover, the signers wore long-sleeved clothes even if it is difficult to find the signers who wore similar long-sleeved clothes in the real world. Furthermore, the accuracy of the model is not enough, and the dataset that is used by the researcher is not sufficient. So, it is better to develop a letter level and numbers to narrow the communication gap between deaf and healthy people.

In another study Teferi Girma and Belete Bizen's study tackles the communication barrier between hearing-impaired individuals and the hearing community in Ethiopia by developing a model that converts Ethiopian Amharic phrase signs into text [3]. Their approach combines Convolutional Neural Networks (CNN) with Long Short-Term Memory (LSTM) networks in a hybrid architecture to enhance performance, with CNN handling feature extraction and LSTM managing classification. Leveraging a dataset of 10,500 samples, the model achieves an impressive testing accuracy of 96%. Evaluation metrics such as accuracy, precision, recall, f1-score, and confusion matrix demonstrate the model's effectiveness. Overall, their study represents a significant step towards improving communication accessibility for the hearing-impaired in Ethiopia, offering promising results in Amharic phrase sign recognition and bridging the communication gap with the wider population.

Previous research on Ethiopian Sign Language (EthSL) recognition has primarily focused on isolated signs such as alphabet, word and phrase, which do not accurately reflect the fluid nature of real-world communication where signs are produced in a continuous, uninterrupted manner. These studies have been constrained by small, insufficiently datasets. Moreover, previous researcher used traditional approaches involve multiple preprocessing steps such as feature extraction, background segmentation, and hand tracking, which add complexity and require significant human intervention. These steps can introduce errors and inconsistencies, ultimately

reducing the overall accuracy and efficiency of the recognition system.

In contrast, our proposed end-to-end architecture for continuous EthSL recognition, addressing these limitations by eliminating the need for intermediate preprocessing steps. An end-to-end system directly processes raw video input, which reduces the time and effort required for data preparation and minimizes potential sources of error. This approach allows the model to learn more comprehensive and contextually rich representations of sign language, leading to improved accuracy and robustness. By leveraging a larger and more diverse dataset that includes continuous signing, the proposed system seeks to achieve higher performance and better generalization. This shift from isolated to continuous sign language recognition will significantly enhance communication for the deaf and hard-of-hearing community, providing a more practical and scalable solution for real-world applications in EthSL communication.

Author	Approach	Method	Data & Result	Limitation
Nigus and Menor	Amharic SLR based on Amharic Alphabet Signs	ANN and SVM	• They prepare Alphabet dataset and collect 1,710 total videos • 57.82% and 74.06% by ANN & SVM classifiers respectively.	• only consider Alphabet of Ethiopia Sign Language. • could not extend the work to word, phrase & sentence level.
Tefera and Yargal	Recognition of isolated sign in Ethiopia Sign Language	Hidden Markov Model	• They prepare word-level sign language dataset • a total of 1500 videos for all 25 signs words. • Achieved 86.9%	• Not Consider Ethiopia continuous sign Language • Requires manual frame-level alignment • Only word sign is consider • only 3 signer is used.
Biruk and Getachew	Word level Amharic SLR using Deep Learning algorithms	Deep learning (ResNet-34)	• They collect 2400 Amharic sign words for 60 distinctive classes. • accuracy 95%	• small words • does not consider phrase and sentence of sign language
Teferi and Belete	Amharic Phrase Level Sign Language Recognition Using Deep Learning	Hybrid Network (CNN and LSTM)	• They Prepare Phrase level dataset of 10500 in total • Achieved 96%	• Smaller dataset only collect for 10 phrases • Not Consider Ethiopia continuous sign Language

Table 2.1: Related Work on Amharic Sign Language Recognition

Chapter Three

Methods

3.1 Overview

This chapter presents an approach to the development of an End-to-End Continuous Ethiopia Sign Language Recognition model. This chapter is structured into several key sections, each contributing to the comprehensive methodology. Firstly, the Data Preparation section outlines the meticulous process of gathering and organizing the necessary data, encompassing Data Collection and Data Annotation to ensure the availability of high-quality, annotated datasets. Following this, the Data Processing section describes the preprocessing steps undertaken to prepare the data for model training, including Frame Extraction and Image Resizing to optimize data format and quality. The methodology's core lies in the Model Development section, which discusses both the Model Architecture and its Implementation. Here, the design choices behind the model's structure and the technical aspects of translating this architecture into a functional implementation are elucidated. By delineating these key steps, this chapter provides a clear road map of the research process, ensuring transparency and reproducibility, and serving as a foundation for the subsequent evaluation and discussion of the developed Sign Language Recognition model.

3.2 Proposed Architecture

The architecture depicted in figure 3.1 is designed for Continuous Ethiopia sign language recognition and it is based on an end-to-end approach that combines spatio-temporal modeling, temporal modeling, and classifier components.

Overview of Model Architecture

1. Spatio-temporal modeling: Spatio-temporal modeling involves capturing both spatial and temporal information from data that varies over both space and time. In sign language recognition, spatio-temporal modeling aims to understand how sign language gestures evolve and how they are spatially distributed within the video frames.

In the spatiotemporal module for sign language recognition, we utilize a customized ResNet architecture with 3D convolutional layers to extract spatial and temporal features from video frames. Correlation layers enhance motion information capture by computing spatio-temporal correlations across frames. Temporal convolution layers further abstract temporal features using 1D convolutions and max-pooling to capture short-temporal features, preparing the data for recurrent layers. This integrated approach allows the model to effectively learn and recognize complex spatiotemporal patterns in continuous sign language sequences.

2. Temporal Modeling: Temporal modeling specifically addresses the temporal dynamics present in sign language sequences. This component involves techniques such as recurrent neural networks (RNN) long short-term memory (LSTM) networks, to capture sequential dependencies and temporal patterns in the sign language data. By analyzing the temporal evolution of gestures. Temporal modeling helps in recognizing the sequence of signs and understanding the context within the sign language utterance.

In our approach to end-to-end continuous sign language recognition, we employ Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal modeling. BiLSTM enhances the capabilities of traditional recurrent neural networks (RNNs) by maintaining an internal state that effectively models transitions in sequential input data. This bidirectional nature allows the network to process information from past and future contexts simultaneously, leading to a more comprehensive understanding of the temporal dynamics inherent in sign language sequences.

3. Classifier The classifier component is responsible for predicting the sign language gestures based on the learned representations from the spatio-temporal and temporal modeling stages. Our classifier component utilizes Connectionist Temporal Classification (CTC) as the primary method for predicting sign language gestures. CTC is a popular technique in sequence labeling tasks, particularly suited for problems where the alignment between input and output sequences is not explicitly known. In the context of continuous Ethiopia sign language recognition, CTC learns to directly map input sign language sequences to output labels without requiring precise frame-by-frame alignment between gestures and labels.

By leveraging CTC, the architecture learns to recognize continuous sign language gestures by jointly optimizing possible alignments and label predictions. This allows for end-to-end training of the model, enabling it to learn discriminative features and temporal dynamics directly from the input data.

3.3 Data Preparation

In this section, we explain how we prepare a dataset for continuous Ethiopian sign language recognition experiments.

3.3.1 Data Collection

Dataset Collected from Television Program and News

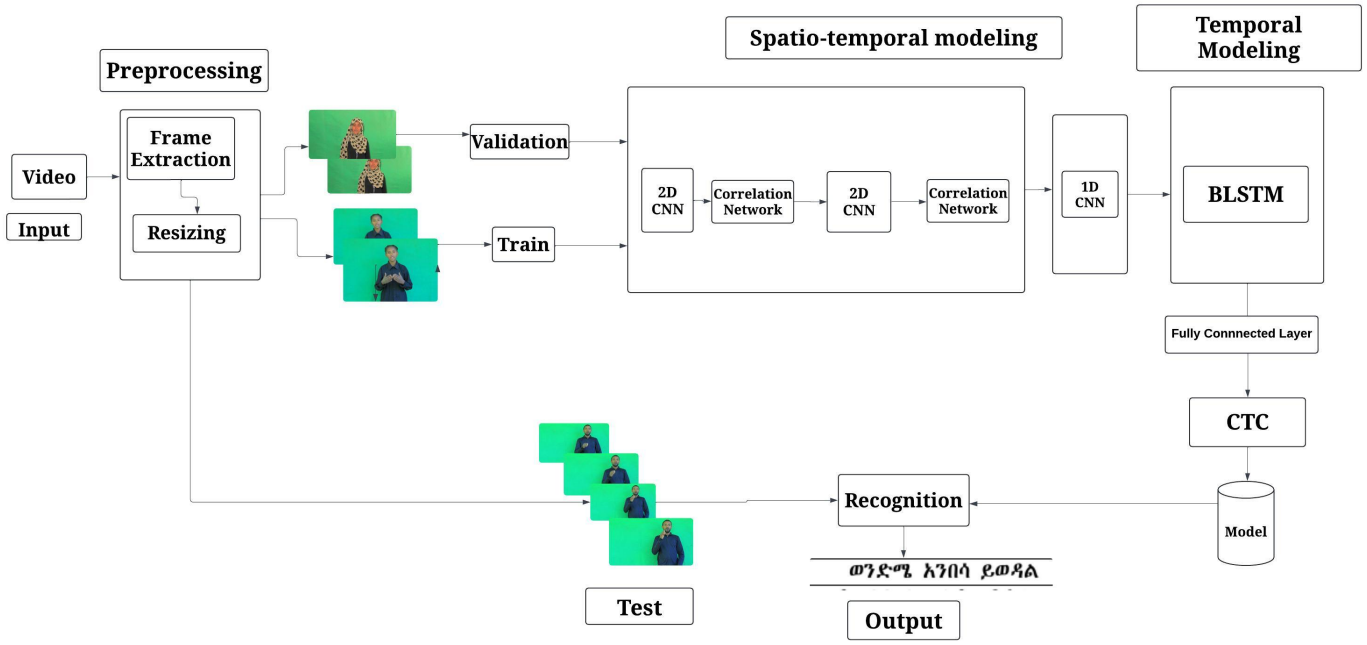


Figure 3.1: Proposed Architecture for End-to-End Continuous Ethiopia sign language.

Since there is no ready-made dataset for continuous Ethiopian Sign Language, we plan to collect the dataset from various television programs that broadcast Ethiopian Sign Language to the deaf and hard-of-hearing communities. We have selected three television channels for this purpose: Amhara Mass Media Agency, Ethiopian Television, and Mahber Kidusan Television Program, as they regularly transmit programs and news to these communities.

During our study, we diligently collected a total of 300 sign language video programs and news from those channels. We then segmented the video into individual sentences.

Once we've downloaded the sign language video, we intend to segment the video sentences into durations ranging from 5 to 20 seconds. However, our initial attempt to collect a comprehensive dataset of Ethiopian Sign Language (EthSL) from television programs faced significant challenges. One major issue was the high signing speed of the interpreters, resulting in continuous signing with no clear pauses between sentences. This made it extremely difficult to segment the videos into individual sentences of the desired duration, which was between 5 to 20 seconds. Consequently, many sentences extended well beyond this range, with some lasting over 2 minutes, making them impractical and computationally intensive for machine processing.

We've also encountered a significant issue regarding the lack of standardized vocabulary among translators. Because these translators are hearing individuals, there are inconsistencies in how they sign the same words. For instance, the word "enat" (meaning mother) was signed differently by various translators, leading to at least two distinct signs for the same word. This variability complicates our dataset, making it challenging to ensure consistency and reliability in sign language recognition. Consequently, our initial strategy of collecting sign language data from television programs hasn't been as effective as we hoped, prompting us to develop our dataset independently.

Preparing the sentence

Our dataset was prepared with the assistance of two experienced sign language teachers from Yekatit 23 Primary School in Bahir Dar, who have been teaching sign language to deaf and hard-of-hearing groups for many years. They carefully curated the 40 most commonly used sign language sentences from a book titled ”Sign Language for Beginners,” published by the Sign Language community. These sentences focus on introducing family members’ occupations, favorite foods, and other family-related topics that are frequently used in sign language communication.

Drawing upon their extensive experience, embarked on a comprehensive journey to curate a robust collection of sign language sentences. Their endeavor involved a thorough examination of various linguistic resources, with particular emphasis on the esteemed publication ”Sign Language for Beginners.” This authoritative text, endorsed by the Sign Language community, served as a cornerstone in their selection process.

Once we had selected 40 sentences at Yekatit 23 School, we went to Fasilo Secondary School to meet with a translator from the Amhara Mass Media Agency, as well as a translator from the school itself. We presented the 40 sentences chosen by the teacher at Yekatit 23 Primary School. After a thorough review, we prioritized the sentences from 1 to 40 based on their complexity and the frequency of words related to family topics.

In the end, we selected the top 30 unique sentences, which contain 71 unique words. These sentences were unanimously agreed upon by three sign language teachers. They were ready to select the signer and record the signing of these sentences.

As Shown in Table 3.1 the selected 30 sentences contain simple to complex words and follow the signed Ethiopia sign language structure as shown in the table below.

Selection of Signer

We enlisted signers from two schools, Yekatit 23 Primary School and Fasilo Secondary School in Bahir Dar, Ethiopia, both of which have sign language teachers and students.

Initially, we sought assistance from a teacher at Yekatit 23 Primary School, who has partial hearing ability, to aid in data collection and sentence preparation. They then collaborated with another teacher to select the sentences. Once the sentences were chosen, we enlisted numerous students and teachers to sign each selected sentence twice.

Ultimately, we enlisted a total of 14 signers from Yekatit 23 Primary School, as detailed in Table 3.2. Among these signers, there were 8 men and 6 women, with 11 being native signers and 3 non-native signers. This diverse composition contributed to the quality and diversity of our dataset, making it suitable for recognition tasks.

At Fasilo Secondary School, we collaborated with a teacher and journalist from the Amhara Mass Media Agency, who also serves as a translator at the school. We presented the 40 selected sentences from Yekatit 23 Primary School for review and signer selection. After reviewing and prioritizing the sentences based on the complexity and frequency of family-related words, the teacher engaged 8 students to sign each selected sentence twice, as indicated in Table 3.2. Among these signers, there were 5 women and 3 men, with 6 being native signers and 2 non-native signers.

We finalized our selection of the first 30 sentences unanimously agreed upon by the three teachers, which were then signed by 22 individuals from two schools, including both teachers and

Table 3.1: Selected Sentence for End-to-End Continuous Ethiopia Sign Language Recognition

	Sentence
1	አባቴ ዘመድ ይወዳል
2	ሆቴሉ ጠባብ ነው
3	እኔ ዳቦ በላሁ
4	ወንድሜ ዳቦ በማር በላ
5	እህቴ ጽሎት ታደርጋለች
6	እናቴ ማር ትወዳለች
7	ወንድሜ ወፍራም ነው
8	እኔ እንጀራ በላሁ
9	ወንድሜ ከኔ ይበልጣል
10	ዛሬ ሆቴል እሄዳለሁ
11	እህቴ ንጽህ ናት
12	እናቴ የት ናት
13	ሆስፒታሉ ሰፊ ነው
14	ወንድሜ አንበሳ ይወዳል
15	እህቴን በጣም እወዳታለሁ
16	አባቴን አመሰግናለሁ
17	እህቴን ጸሎት ማድረግ ትወዳለች
18	የእናቴን ሻሽ እወድዋለሁ
19	እኔ ጓደኛ እወዳለሁ
20	አንበሳ ማየት እወዳለሁ
21	ማን ነው
22	የኢትዮጵያ ባንዲራ አረንጓዴ ቢጫ ቀይ ነው
23	እናቴ ቀጭን ናት
24	ቤቱ የእኛ ነው
25	ወንድሜ ሀኪም ነው
26	እኔ ዳቦ እወዳለሁ
27	ቤተሰብ ክብር ነው
28	ልጆች መዝናናት ይወዳሉ
29	እኔ ደህና ነኝ
30	ልጆች ፍቅር ይፈልጋሉ

Table 3.2: Statistics of Continuous Ethiopia Sign Language Dataset (CESL)

	Yekatit 23 Primary School	Fasilo Secondary School
Number of Signer	14	8
Men	8	3
Women	6	5
Native signer	11	6
Non-Native Signer	3	2

students. These signers spanned ages from 11 to 60, providing a diverse range of ages within our dataset. This diversity adds value to our dataset as it introduces variations in signing fluency and speed across different age groups. Younger signers tend to sign at a faster pace, while older signers exhibit a slower signing speed, contributing to the dataset's richness and quality through increased variability.

Table 3.3: Statistics of the CESL-v1 dataset.

Feature	Value
Video resolution	1920 x 1080
Number of Signers	22
Repetitions	2
Number of sentences	30
Video duration (seconds)	2-15
Body joints	Upper-body involved
Vocabulary size	65
Total number of videos	1320

Record the video

We used a green background and had the signer wear black clothing to ensure clear image quality when recording videos for the selected sentences, as depicted in Figure 3.2.In our controlled recording environment, we encountered variations in the intensity of the green light, resulting in bright, dim, and normal green backgrounds, as illustrated in the 3.2 and these give an opportunity for developing robust algorithms and tools that can handle diverse video qualities, lighting conditions.

To accurately capture each signer's upper body movements, we strategically positioned a Canon Mark IV camera at different distances in front of them. This adjustment was crucial due to the diverse heights of signers, as sign language gestures primarily occur above the upper body, as depicted in the visual reference provided in Figure 3.3 .

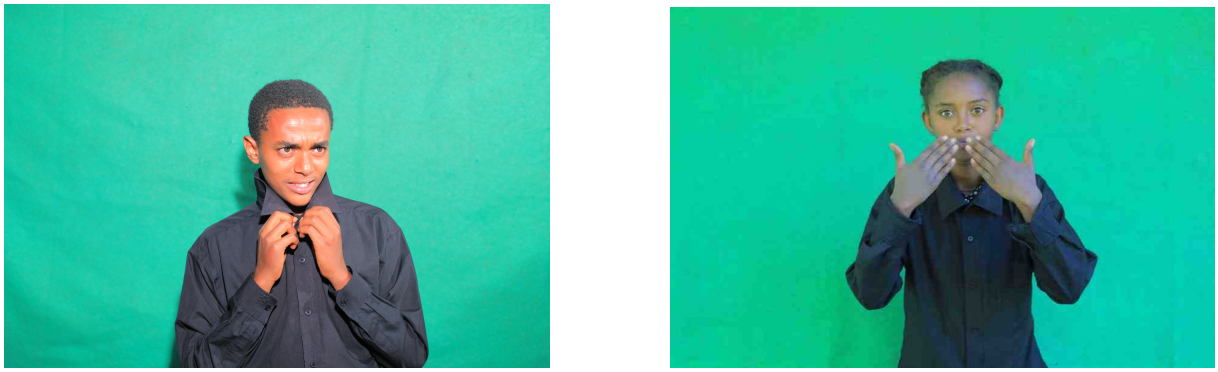


Figure 3.2: Background and Signer's Position

The camera settings were configured to record in full HD resolution (1920x1080 pixels) at 25 frames per second, ensuring smooth capture of signer movements. Each sentence is signed twice by each signer. The dataset's overall statistics are summarized in Table 3.2.

3.3.2 Data Annotation

The dataset annotation process began with a panel of three experts selecting sign language sentences. These sentences were transcribed in detail, capturing nuances like sign sequences, hand



Figure 3.3: Camera Position

configurations, facial expressions, and other non-manual markers seen in the sign videos.

Consistency and quality were maintained throughout the annotation process via multiple review rounds and verification steps, including agreement checks and expert consultations to resolve any discrepancies. The resulting dataset comprises a structured collection of sign language sentences, each accompanied by its transcription and detailed annotations, which facilitates subsequent research tasks.

The annotation file, as depicted in table 3.4 includes fields such as ID, folder, signer, and annotation text. The annotation text represents the video transcription, and each ID is unique, containing information about the folder and the signer who signs the sentence.

This annotation file serves as ground truth during the training of our deep learning model, enabling it to learn the relationship between the stream of images and corresponding text.

Table 3.4: Annotation File

id	folder	signer	annotation
01day_2016_sentence_0	01day_2016_sentence_0/1/*.png	Signer01	አባቱ ዘመድ ይወዳል
01day_2016_sentence_1	01day_2016_sentence_1/1/*.png	Signer01	አባቱ ዘመድ ይወዳል
01day_2016_sentence_2	01day_2016_sentence_2/1/*.png	Signer02	አባቱ ዘመድ ይወዳል
01day_2016_sentence_3	01day_2016_sentence_3/1/*.png	Signer02	አባቱ ዘመድ ይወዳል
02day_2016_sentence-0	02day_2016_sentence-0/1/*.png	Signer01	እኔ እንጀራ በላሁ
02day_2016_sentence-1	02day_2016_sentence-1/1/*.png	Signer01	እኔ እንጀራ በላሁ
02day_2016_sentence-2	02day_2016_sentence-2/1/*.png	Signer02	እኔ እንጀራ በላሁ
02day_2016_sentence-3	02day_2016_sentence-3/1/*.png	Signer02	እኔ እንጀራ በላሁ

3.4 Frame Extraction and Resizing

Frame extraction in video processing is a fundamental technique involving the capture of individual frames or images from a video sequence. Each frame represents a frozen moment in time during video playback. This process is essential for a range of video-related tasks, such as analyzing videos, detecting and tracking objects, estimating motion, and compressing videos.

In the context of continuous Ethiopian Sign Language recognition, frame extraction serves as a critical first step after collecting the video data. This process involves breaking down the continuous video into discrete frames, enabling detailed analysis and recognition of sign language gestures. Below are detailed steps explaining how frame extraction is performed.

Steps in Frame Extraction

1. Input Video: The step Starts with the recorded video of the sign language gestures or sentences you want to recognize. (see Section 3.3.1). This video will serve as the input for the frame extraction phase.

2. Video Loading: To bring a video file into the processing pipeline, we rely on video processing libraries or tools. One such widely adopted library is OpenCV, known for its extensive features in handling video files alongside other functionalities. Utilizing OpenCV in Python, we seamlessly load video files into memory, enabling a range of operations including frame extraction, object detection, and video manipulation.

3. Frame Rate Determination: Understanding the frame rate of the input video is crucial for several tasks like extracting frames at consistent intervals or identifying keyframes. This essential information is programmatically retrieved through video processing libraries, allowing us to precisely manage frame extraction and maintain synchronization with the video's temporal dynamics.

When it comes to choosing a frame extraction method, we have two primary options to consider:

1. Uniform Sampling: This method involves selecting frames at regular intervals from the video. For instance, if a video is captured at 30 frames per second and we decide to extract one frame per second, we would extract every 30th frame.

2. Keyframe Extraction: Alternatively, keyframe extraction involves identifying frames based on specific criteria such as notable changes in motion, scene transitions, or predefined time intervals.

For this thesis, we opt for the Uniform Sampling technique. This approach entails iterating through the video frames at the specified interval, ensuring a consistent and evenly distributed selection of frames for further analysis and processing.

Reasons for Choosing Uniform Sampling in Frame Extraction

Consistency Uniform sampling ensures consistent temporal distribution of frames throughout the video, simplifying subsequent processing steps and aiding model training.

Preservation of Temporal Information By evenly representing frames across the video's duration, uniform sampling preserves crucial temporal information, capturing the dynamics and motion patterns inherent in sign language gestures.

Simplicity Uniform sampling is straightforward to implement and understand, making it suitable for rapid prototyping and experimentation.

Robustness Uniform sampling may be more robust to variations in video quality, lighting conditions, and background clutter compared to key frame extraction methods, enhancing its suitability for real-world applications.

Computational Efficiency Avoiding the need for intensive processing to identify key frames, uniform sampling can be computationally efficient, particularly for long videos or large datasets.

Reduced Sensitivity to Noise Uniform sampling is less sensitive to noise or outliers in the video, mitigating the risk of selecting non-representative frames for sign language recognition.

For uniform sampling, iterate through the frames in the video file according to the specified interval.

4 Frame Extraction: Extract the image data from each chosen frame and store it for subsequent processing. These extracted frames serve as inputs for various processing techniques and will be utilized in sign language recognition algorithms during later stages of the workflow.

The code for the steps for Frame Extraction is stated below. The provided code exemplifies the Frame Extraction process using OpenCV in Python. Initially, the video file is loaded into memory with ‘cv2.VideoCapture‘, followed by extraction of crucial video details like frame rate, width, height, and total frames from the video file. Through a loop, each frame of the video is sequentially accessed with ‘cap.read()‘. Subsequently, frames are saved as individual image files, ensuring their accessibility and organization for subsequent data processing or model input.

Finally, system resources are efficiently managed by releasing the video capture object with ‘cap.release()‘. Overall, this code streamlines the extraction of frames from a video.

```
def extract_frames(video_path, output_folder):
    cap = cv2.VideoCapture(video_path)
    fps = int(cap.get(cv2.CAP_PROP_FPS))
    width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
    total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

    print(f"Processing {video_path}:\n  Video FPS: {fps}\n  Video Dimensions)

    frame_number = 0
    while True:
        ret, frame = cap.read()
        if not ret: break
        frame_filename = f"{output_folder}/{os.path.basename(video_path)}.png"
        cv2.imwrite(frame_filename, frame)
        frame_number += 1

    cap.release()
    print(f"  Frames saved to: {output_folder}\n")
```

Resizing image

Image resizing involves adjusting the dimensions of an image, usually by modifying its width and height. This modification can either enlarge or reduce the image's size, depending on the intended purpose.

Resizing is a crucial step in image processing and computer vision, enabling images to be tailored to specific needs and applications. It ensures that images are appropriately scaled for display, analysis, or other tasks, maintaining their visual quality and relevance to the context in which they are used.

In continuous Ethiopian Sign Language recognition, preparing image data is crucial for subsequent analysis and model training. A key step in this process is resizing the images while maintaining their original aspect ratio.

Steps in Resizing Image

1 Input Image: Start with an image that you want to resize. This image can be in various formats such as JPEG, PNG, or BMP, and it may have different dimensions depending on how it was captured or generated.

2 Desired Dimensions: Determine the desired dimensions for the resized image. This typically involves specifying the target width and height in pixels. Resizing can be performed to make images smaller for efficient storage and processing, or larger for better visualization or analysis.

3 Interpolation Method: Interpolation is used to estimate the values of pixels in the resized image based on the values of neighboring pixels in the original image. Common interpolation methods include nearest neighbor, bilinear, bicubic, and Lanczos interpolation. Each method has its characteristics in terms of quality and computational complexity.

4 Resizing Algorithm: Apply the chosen interpolation method to adjust the size of the image. The resizing algorithm calculates the new pixel values based on the original image and the desired dimensions. It may involve averaging pixel values or applying mathematical transformations to achieve the desired size.

5 Aspect Ratio: Consider the aspect ratio of the original image when resizing. The aspect ratio is the ratio of the width to the height of the image. To maintain the aspect ratio, you may need to crop or pad the image after resizing to ensure that it fits within the specified dimensions without distortion.

6 Implementation: Resize the image using image processing libraries or tools. Popular libraries like OpenCV, PIL (Python Imaging Library), and scikit-image provide functions or methods for resizing images with various interpolation options.

7 Quality Considerations: Pay attention to the quality of the resized image, especially when decreasing its size. Resizing can introduce loss of detail and sharpness, so it's important to evaluate the visual quality of the resized image and adjust the resizing parameters or interpolation method as needed.

8 Output: The resized image is generated as the result of the resizing process. It can be saved in the desired format and used for further analysis, or as input for our model for recognition.

The code for Resizing Image is stated as Below:

```
    def resize_img(img_path, dsize='256x256px'):
    dsize = tuple(int(res) for res in re.findall("\d+", dsize))
    img = cv2.imread(img_path)
    img = cv2.resize(img, dsize, interpolation=cv2.INTER_LANCZOS4)
    return img


def resize_dataset(video_idx, dsize, info_dict):
    info = info_dict[video_idx]
    img_list = glob.glob(f"{info_dict['prefix']}/{info['folder']}")
    for img_path in img_list:
        rs_img = resize_img(img_path, dsize=dsize)
        rs_img_path = img_path.replace("210x260px", dsize)
        rs_img_dir = os.path.dirname(rs_img_path)
        if not os.path.exists(rs_img_dir):
            os.makedirs(rs_img_dir)
            cv2.imwrite(rs_img_path, rs_img)
        else:
            cv2.imwrite(rs_img_path, rs_img)
```

The above function ‘resize_img‘ function handles the job for image resizing. It takes two things: the path to the original image and the desired size, which defaults to ’256x256px’. we use the Interpolation Method ‘cv2.INTER_LANCZOS4‘ to make sure the image still looks good after resizing. This means that even though the image is smaller or bigger, it still keeps the same proportions as the original. After resizing the image is done, ready to be used for recognizing sign language. So, by using the ‘resize_img‘ function, we can make sure images are just the right size without messing up how they look, for later sign language recognition system to work better.

3.5 Model Development For End-to-End Continuous Ethiopia Sign Language Recognition

After the data was collected and processed, it was then fed to the deep learning algorithm to build the desired Continuous Ethiopia Sign Language recognition model. The model was trained on the sign language video corpus, learning to recognize the various features of Ethiopian Sign Language gestures and relate them to the corresponding annotation.

In this study an end-to-end continuous sign language recognition is designed, integrating spatio-temporal and temporal modeling finally we employ a classifier that contains Connection Temporal Classification (CTC) loss.

3.5.1 Spatio-Temporal Modeling

The spatio-temporal module in the context of sign language recognition is essential for capturing the dynamic nature of sign language gestures over both space and time. Here's a detailed description of each component within this module:

1. Spatial Representation Module:

ResNet (3D Convolutional Layers) The customized ResNet architecture used by the model includes several 3D convolutional layers that serve the purpose of extracting useful features from the input video frames. The input video frames are a 3D representation of the sign language sequence, where the time domain is mapped to one dimension, and the spatial dimensions (height and width) are mapped to the other two dimensions. The initial convolutional layer, **conv1**, applies a set of learnable filters to the input video frames, producing a set of output feature maps that capture local spatio-temporal patterns and structures within the video. This layer uses a kernel size of $1 \times 7 \times 7$ and a stride of $1 \times 2 \times 2$, which results in a reduction of the spatial dimensions while preserving the temporal dimension. After the convolution operation, batch normalization normalizes the output of each batch to reduce internal covariate shifts, resulting in more stable and faster training. ReLU activation introduces non-linearity to the output, allowing the network to learn more complex and nonlinear features.

The subsequent layers consist of several `BasicBlock` modules, which are residual blocks designed to learn residual functions. Each block contains two 3D convolutional layers with a kernel size of $1 \times 3 \times 3$ and padding to maintain spatial dimensions. **Layer1** applies 3D convolutions with 64 filters and maintains the spatial resolution. **Layer2** increases the number of filters to 128 and applies downsampling via convolutions with stride $1 \times 2 \times 2$, reducing the spatial dimensions. **Layer3** further increases the number of filters to 256 and applies another downsampling. **Layer4** increases the number of filters to 512 and applies final downsampling. By applying these convolutional layers, the model can automatically learn relevant features from the input video frames without the need for manual feature engineering. This is a key advantage of using a 3D CNN architecture for video processing, as it allows the model to adapt to a wide range of input data and capture important patterns that may not be immediately apparent to a human expert.

Correlation Layers (*corr1, corr2, corr3*) The correlation layers compute spatio-temporal correlations between features, which are crucial for capturing motion information across frames. This is particularly important for tasks like continuous sign language recognition, where temporal dynamics play a key role. We use this correlation Layer in between Layers as shown in figure 3.1 to get motion trajectories of body parts across consecutive video frames from each spatial feature.

we use a correlation network that is designed to capture and leverage the motion trajectories of body parts across consecutive video frames. By computing and aggregating correlation maps between frames, the correlation emphasizes significant spatial and temporal patterns, such as the movement of hands and face, which are critical for accurate continuous sign language recognition. The correlation network is integrated into our framework by placing it after each stage of the Layer, and its outputs are combined with the original features. This integration allows the model to better utilize temporal and spatial information, significantly improving the accuracy of continuous sign language recognition.

Correlation Layer 1 (corr1) includes **down_conv**, which reduces the number of channels to a smaller dimension (8 channels). This helps in reducing the computational load for subsequent operations. The **spatial_aggregation** stage aggregates spatial features using a series of convolutions with increasing dilation rates (dilation=(1, 1, 1), (1, 2, 2), and (1, 3, 3)). This multi-scale approach allows the layer to capture correlations at different spatial scales. Finally, **conv_back** restores the original number of channels (128 channels).

Correlation Layer 2 (corr2) follows a similar structure with **down_conv** reducing the number of channels to 16. The **spatial_aggregation** stage again uses a multi-scale approach with dilation rates (dilation=(1, 1, 1), (1, 2, 2), and (1, 3, 3)). **Conv_back** restores the original number of channels (256 channels).

Correlation Layer 3 (corr3) reduces the number of channels to 32 using **down_conv**. The **spatial_aggregation** uses dilation rates (dilation=(1, 1, 1), (1, 2, 2), and (1, 3, 3)), and **conv_back** restores the original number of channels (512 channels). The purpose of these correlation layers is to enhance the representation of temporal dependencies and motion patterns, which are crucial for understanding continuous sign language sequences. The sample code to develop the spatial and correlation module is presented below.

We use a 3D convolutional layer (nn.Conv3d). While it is technically a 3D convolution due to its use of the Conv3d class, the specific choice of the kernel size (1,7,7) and corresponding strides and padding essentially make it operate similarly to a 2D convolution on slices of the input tensor.

Using a 3D convolutional layer with a kernel size designed to behave like a 2D convolution might seem redundant, but there are practical reasons for this design choice, By using a 3D convolution with a (1,7,7) kernel size, you can control the processing along the depth dimension more flexibly. Later in the network, we change the kernel size to (3,3,3) or another value that combines depth information. This approach allows a smooth transition from spatial to spatio-temporal feature extraction. It allows the same type of layer to handle both 2D and 3D feature extraction, which can make the code more flexible and easier to extend or modify.

```
Define ResNet:
Initialize layers:
    conv1 = Conv3d(in_channels=3, out_channels=64, kernel_size=(1, 7, 7),
   stride=(1, 2, 2), padding=(0, 3, 3), bias=False)
    bn1 = SynchronizedBatchNorm3d(64, eps=1e-05, momentum=0.1, affine=True,
     track_running_stats=True)
    relu = ReLU(inplace=True)
    maxpool = MaxPool3d(kernel_size=(1, 3, 3), stride=(1, 2, 2), padding
   =(0, 1, 1), dilation=1, ceil_mode=False)
    layer1 = Sequential(
        BasicBlock(64, 64),
        BasicBlock(64, 64)
    )
    layer2 = Sequential(
        BasicBlock(64, 128),
        BasicBlock(128, 128)
    )
    corr1 = Get_Correlation(128)
    layer3 = Sequential(
        BasicBlock(128, 256),
        BasicBlock(256, 256)
```

```
    )
    corr2 = Get_Correlation(256)
    layer4 = Sequential(
        BasicBlock(256, 512),
        BasicBlock(512, 512)
    )
    corr3 = Get_Correlation(512)
    avgpool = AvgPool2d(kernel_size=7, stride=1, padding=0)
    fc = Identity()

Define BasicBlock(in_channels, out_channels):
Initialize layers:
    conv1 = Conv3d(in_channels, out_channels, kernel_size=(1, 3, 3), stride
   =(1, 1, 1), padding=(0, 1, 1), bias=False)
    bn1 = SynchronizedBatchNorm3d(out_channels, eps=1e-05, momentum=0.1,
   affine=True, track_running_stats=True)
    relu = ReLU(inplace=True)
    conv2 = Conv3d(out_channels, out_channels, kernel_size=(1, 3, 3),
   stride=(1, 1, 1), padding=(0, 1, 1), bias=False)
    bn2 = SynchronizedBatchNorm3d(out_channels, eps=1e-05, momentum=0.1,
   affine=True, track_running_stats=True)
Return Sequential(conv1, bn1, relu, conv2, bn2)

Define Get_Correlation(in_channels):
Initialize layers:
    down_conv = Conv3d(in_channels, 8, kernel_size=(1, 1, 1), stride=(1, 1,
     1), bias=False)
    down_conv2 = Conv3d(in_channels, in_channels, kernel_size=(1, 1, 1),
   stride=(1, 1, 1), bias=False)
    spatial_aggregation1 = Conv3d(8, 8, kernel_size=(9, 3, 3), stride=(1,
   1, 1), padding=(4, 1, 1), groups=8)
    spatial_aggregation2 = Conv3d(8, 8, kernel_size=(9, 3, 3), stride=(1,
   1, 1), padding=(4, 2, 2), dilation=(1, 2, 2), groups=8)
    spatial_aggregation3 = Conv3d(8, 8, kernel_size=(9, 3, 3), stride=(1,
   1, 1), padding=(4, 3, 3), dilation=(1, 3, 3), groups=8)
    conv_back = Conv3d(8, in_channels, kernel_size=(1, 1, 1), stride=(1, 1,
     1), bias=False)
Return CorrelationModule(down_conv, down_conv2, spatial_aggregation1,
   spatial_aggregation2, spatial_aggregation3, conv_back)
```

TemporalConv

This section processes the sequence of features extracted from the ResNet and correlation layers over time. It further abstracts temporal features to prepare the data for the recurrent layers. The temporal_conv consists of a series of 1D convolutions and max-pooling layers to further abstract short temporal features. Conv1d Layers apply 1D convolution operations across the temporal dimension, capturing temporal patterns and dependencies. Batch normalization and ReLU activation functions follow each convolution operation to normalize the output and introduce non-linearity. MaxPool down samples the temporal dimension, reducing the sequence length while retaining important features.

The combination of these convolutional layers, batch normalization, and ReLU activation functions in the Temporal_Conv architecture helps the model to learn useful temporal features from the input video frames. These features are then used by the recurrent layers to generate accurate transcriptions of the input sign language sequences. The pseudocode to develop for temporal

convolution is presented below

```
    # Reshape for TemporalConv
x = layers.Reshape((-1, x.shape[-2] * x.shape[-1]))(x)

# Temporal Convolution Layer 1
x = layers.Conv1D(
    filters=1024,
    kernel_size=5,
    strides=1,
    padding="same",
    use_bias=False,
    name="temporal_conv_1"
)(x)
x = layers.BatchNormalization(name="temporal_conv_1_bn")(x)
x = layers.ReLU(name="temporal_conv_1_relu")(x)
x = layers.MaxPooling1D(
    pool_size=2,
    strides=2,
    padding="same",
    name="temporal_conv_1_maxpool"
)(x)

# Temporal Convolution Layer 2
x = layers.Conv1D(
    filters=1024,
    kernel_size=5,
    strides=1,
    padding="same",
    use_bias=False,
    name="temporal_conv_2"
)(x)
x = layers.BatchNormalization(name="temporal_conv_2_bn")(x)
x = layers.ReLU(name="temporal_conv_2_relu")(x)
x = layers.MaxPooling1D(
    pool_size=2,
    strides=2,
    padding="same",
    name="temporal_conv_2_maxpool"
)(x)
```

The overall architecture integrates spatial, spatio-temporal, and temporal feature extraction and modeling to effectively handle continuous sign language recognition. By combining 3D convolutional layers, correlation layers, and temporal convolutions, the model can learn complex spatio-temporal patterns and dependencies crucial for the accurate recognition and transcription of sign language sequences.

3.5.2 Temporal Modeling

In continuous sign language recognition, temporal modeling is crucial because it captures the sequential nature of the sign language data. Each sign is not only defined by its characteristics but also by its context within a sequence of signs. Therefore, understanding the temporal dependencies between signs is essential for accurate recognition. To achieve this, the BiLSTM (Bidirectional Long Short-Term Memory) layer is used for temporal modeling.

BiLSTMLayer

The BiLSTM layer captures long-range dependencies in the temporal sequence using bidirectional LSTM units. This layer is instrumental in understanding the sequential nature of sign language, where each sign is influenced by the preceding and following signs. The bidirectional nature of this layer enables the model to have a broader context, considering both past and future information in the sequence.

Bidirectional LSTM Layer: The BiLSTM layer comprises two LSTM layers, each with 512 units per direction, resulting in a total of 1024 units. By processing the input sequence in both forward and backward directions, the BiLSTM can capture dependencies from both past and future contexts. This is particularly important in continuous sign language recognition, where the meaning of a sign can be influenced by signs that come before and after it.

Dropout is introduced in the BiLSTM layer to prevent overfitting. In this context, dropout randomly sets a fraction of the input units to zero at each update during training time, which helps make the model more robust and reduces the risk of overfitting. For continuous sign language recognition, preventing overfitting is crucial due to the variability and complexity of sign language data. Effective dropout ensures the model generalizes well to unseen sequences. The pseudocode for BILSTM is presented below.

```
    # BiLSTM Layer
x = layers.Bidirectional(
    layers.LSTM(
        units=512,
        return_sequences=True,
        dropout=0.3,
        recurrent_dropout=0.3,
        name="bilstm"
    ),
    name="bilstm_bidirectional"
)(x)
```

Overall, the BiLSTM layer in continuous Ethiopia sign language recognition provides a powerful mechanism to capture temporal dependencies, enabling the model to understand and interpret the sequence of signs accurately.

3.5.3 Classifier

In continuous sign language recognition, after the model has extracted features and captured temporal dependencies through the preceding layers, it is essential to translate these processed features into final predictions. This is where the classifier comes into play. The classifier, specifically the final fully connected layer, is responsible for mapping the features obtained from the BiLSTM layer to the predicted sign language classes.

The classifier in the continuous sign language recognition model plays a crucial role in translating the processed features into final predictions. Specifically, the final fully connected layer, often implemented as a 'NormLinear' layer, is responsible for mapping the features obtained from the BiLSTM layer to the predicted sign language classes. This transformation is essential as it reduces the dimensionality of the output from the temporal model to the number of classes in the sign language recognition task. The 'NormLinear' layer likely incorporates normalization techniques to ensure the outputs are well-scaled, enhancing the training stability and convergence of the model. Normalization is particularly important in handling varying scales of input data, which is crucial for achieving better performance.

The output of the BiLSTM layer consists of processed features that encapsulate both past and future temporal dependencies of the sign language sequence. These features remain in a sequence format, preserving the temporal context of the data. The classifier layer (NormLinear) takes this sequential output and aggregates it, typically through averaging or taking the final time step's output, to produce a fixed-size vector. This vector is then passed through the fully connected layer to generate the final class scores. By combining the temporal modeling capabilities of the BiLSTM layer with the classification power of the fully connected layer, the model can effectively translate sequential sign language data into accurate class predictions. The normalization in the 'NormLinear' layer helps maintain stable training and improves the overall performance of the classifier.

```
    # Final Classifier
x = layers.Dense(
    units=num_classes,
    activation="softmax",
    name="classifier"
)(x)
```

Decoding:

The decoding process comes after the neural network has processed the input sequence through various layers, including convolutional, recurrent, or fully connected layers, and produced an output sequence. This output sequence is then decoded into human-readable text or symbols using techniques like CTC (Connectionist Temporal Classification) decoding,

In the proposed model for continuous Ethiopia Sign Language (CESL) recognition, we leverage CTC as the cornerstone of our classification strategy. Connectionist Temporal Classification (CTC) is a powerful technique commonly used in sequence labeling tasks, particularly in the

context of speech and gesture recognition. At its core, CTC addresses the challenge of aligning input sequences with target labels of variable lengths, without requiring explicit alignment information during training.

The CTC (Connectionist Temporal Classification) layer is a specialized layer used in sequence-to-sequence learning tasks where the input and output sequences have variable lengths, and there is no one-to-one correspondence between the input and output sequence elements. This layer is commonly used in sign recognition tasks, where the input sequence is a stream of images and the output sequence is a sequence of words.

CTC achieves this by introducing a special ”blank” symbol along with the target symbols, allowing the model to output multiple repeated symbols and align them with different parts of the input sequence. The model then generates a probability distribution over all possible alignments between the input sequence and the target labels, effectively learning to map input sequences to target sequences flexibly and robustly.

Decoding Methods in the `Decode` Class

The `Decode` class provides two methods for decoding the output of a neural network trained for sequence-to-sequence tasks, particularly with Connectionist Temporal Classification (CTC) loss.

Decoding Methods: We provide two decoding methods: BeamSearch and MaxDecode.

BeamSearch: This method performs beam search decoding on the input neural network output. It takes the softmax output of the neural network (nn_output), along with the length of each sequence (vid_lgt). It uses the CTC beam decoder to perform beam search decoding and returns the decoded results.

MaxDecode: This method performs maximum likelihood decoding on the input neural network output. It simply selects the index with the maximum probability at each time step and constructs the output sequence. It returns the decoded results.

BeamSearch Method

Purpose: This method performs beam search decoding on the neural network’s output. Beam search is a heuristic search algorithm that explores multiple possible sequences to find the most likely one. It’s more sophisticated and accurate than simply taking the most likely class at each time step.

Parameters:

- `nn_output`: The output of the neural network, which should be passed through a softmax layer. This tensor has a shape of (B, T, N), where B is the batch size, T is the sequence length, and N is the number of classes.

- `vid_lgt`: The actual lengths of the sequences in the batch.
- `probs` (optional): If set to `True`, indicates that `nn_output` is already in the form of probabilities.

How it Works:

1. **Softmax Conversion**: If `probs` is `False`, the method first converts `nn_output` to probabilities using a softmax function.
2. **Decoding**: The method then calls `self.ctc_decoder.decode(nn_output, vid_lgt)`, which uses the CTC beam decoder to perform beam search decoding.
3. **Processing Results**: The beam decoder returns several outputs, including the best sequences (`beam_result`), their scores (`beam_scores`), and the lengths of these sequences (`out_seq_len`).
4. **Post-Processing**: The method processes the top beam result for each sequence, removes duplicates, and converts gloss ids to their corresponding glosses using `i2g_dict`.

When to Use: Use `BeamSearch` when you need more accurate decoding of the network output and can afford the computational complexity. Beam search is particularly useful in tasks like speech recognition, where the exact sequence of output is crucial.

MaxDecode Method

Purpose: This method performs maximum likelihood decoding, which is simpler and faster than beam search. It selects the class with the highest probability at each time step.

Parameters:

- `nn_output`: The output of the neural network, typically after a softmax layer. This tensor has a shape of (B, T, N).
- `vid_lgt`: The actual lengths of the sequences in the batch.

How it Works:

1. **Argmax Operation**: The method first computes the index with the highest probability for each time step using `torch.argmax(nn_output, axis=2)`.
2. **Grouping and Filtering**: It then groups consecutive identical predictions to handle CTC's output, which can contain repeated characters and blank tokens.
3. **Removing Blanks**: The method filters out the blank tokens from the grouped results.
4. **Post-Processing**: Finally, it converts gloss ids to their corresponding glosses using `i2g_dict`.

When to Use: Use `MaxDecode` when you need a quick and simple decoding method. It's less computationally intensive compared to beam search but might be less accurate, particularly for sequences where context and dependencies between elements are important.

Detailed Breakdown of Each Method

BeamSearch

```
def BeamSearch(self, nn_output, vid_lgt, probs=False):
    if not probs:
        nn_output = nn_output.softmax(-1).cpu()
    vid_lgt = vid_lgt.cpu()
    beam_result, beam_scores, timesteps, out_seq_len = self.ctc_decoder.
   decode(nn_output, vid_lgt)
    ret_list = []
    for batch_idx in range(len(nn_output)):
        first_result = beam_result[batch_idx][0][:out_seq_len[batch_idx
   ][0]]
        if len(first_result) != 0:
            first_result = torch.stack([x[0] for x in groupby(first_result)
   ])
        ret_list.append([(self.i2g_dict[int(gloss_id)], idx) for idx,
   gloss_id in
                         enumerate(first_result)])
    return ret_list
```

Listing 3.1: BeamSearch Method

1. **Softmax Conversion**: `nn_output.softmax(-1).cpu()` converts the output to probabilities if `probs` is `False`.
2. **Decode**: `self.ctc_decoder.decode(nn_output, vid_lgt)` performs beam search decoding.
3. **Result Processing**: The method extracts the top beam result, handles duplicates, and maps ids to glosses.

MaxDecode

```
def MaxDecode(self, nn_output, vid_lgt):
    index_list = torch.argmax(nn_output, axis=2)
    batchsize, lgt = index_list.shape
    ret_list = []
    for batch_idx in range(batchsize):
        group_result = [x[0] for x in groupby(index_list[batch_idx][:
   vid_lgt[batch_idx]])]
        filtered = [*filter(lambda x: x != self.blank_id, group_result)]
        if len(filtered) > 0:
            max_result = torch.stack(filtered)
            max_result = [x[0] for x in groupby(max_result)]
        else:
            max_result = filtered
        ret_list.append([(self.i2g_dict[int(gloss_id)], idx) for idx,
   gloss_id in
                         enumerate(max_result)])
    return ret_list
```

Listing 3.2: MaxDecode Method

1. **Argmax Operation**: `torch.argmax(nn_output, axis=2)` gets the index with the highest probability for each time step.
2. **Grouping and Filtering**: `group_result` and `filtered` handle duplicate predictions and remove blank tokens.
3. **Result Processing**: The method converts ids to glosses using `i2g_dict`.

For our end-to-end continuous Ethiopian Sign Language recognition system, we have chosen to use the `BeamSearch` method for decoding the neural network outputs. The primary reason for this choice is the method's ability to provide more accurate and contextually appropriate sequences compared to simpler decoding methods like `MaxDecode`. BeamSearch explores multiple potential sequences simultaneously, allowing it to maintain and evaluate a set of the most promising candidate sequences at each step.

This results in a higher likelihood of correctly interpreting complex sign language gestures, which often have dependencies and contextual nuances that simple maximum likelihood decoding might miss. By leveraging the CTC beam decoder, we ensure that our system can handle the intricacies of continuous sign language recognition, providing a robust and reliable translation of sign language gestures into text. This increased accuracy is crucial for creating a system that is both effective and practical for real-world applications in Ethiopian Sign Language interpretation.

3.6 Evaluation

In evaluating continuous Ethiopia Sign Language recognition, we employ the Word Error Rate (WER) as our primary metric. WER quantifies the disparity between the predicted sign language sequence and the reference (ground truth) sequence. It calculates the minimum number of operations required—substitution, insertion, and deletion—to transform the predicted sequence into the reference sequence, or vice versa.

Word Error Rate (WER) is a commonly used evaluation metric in the field of speech and sign language recognition. It provides a quantitative measure of the accuracy of a recognition system by comparing the predicted output with the ground truth reference.

WER is defined as the minimum number of edit operations required to transform the predicted sequence into the reference sequence (or vice versa). These edit operations typically include substitution, insertion, and deletion.

Let N be the total number of words in the reference sequence, S be the total number of substitution errors, I be the total number of insertion errors, and D be the total number of deletion errors.

The formula to calculate WER is given by[11], [18], [20]:

$$WER = \frac{S + I + D}{N} \times 100\%$$

Where:

- S represents the number of words that were substituted in the predicted sequence compared to the reference sequence.
- I represents the number of words that were inserted in the predicted sequence compared to the reference sequence.
- D represents the number of words that were deleted in the predicted sequence compared to the reference sequence.

Substitution occurs when a word in the predicted sequence differs from the corresponding word in the reference sequence. This could happen due to recognition errors where a different sign is recognized.

Insertion occurs when additional words are present in the predicted sequence compared to the reference sequence. This could happen if the recognition system mistakenly adds extra signs.

Deletion occurs when words present in the reference sequence are missing from the predicted sequence. This could happen if the recognition system fails to recognize certain signs.

Interpretation of Word Error Rate (WER)

A lower WER indicates better performance, as it signifies fewer errors in the predicted sequence compared to the reference sequence. Conversely, a higher WER suggests poorer performance, indicating more discrepancies between the predicted and reference sequences.

Chapter Four

Experimentation Results and Discussion

4.1 Overview

The Experimentation Result and Discussion chapter delves into the empirical evaluation of the developed End-to-End Continuous Ethiopia Sign Language Recognition model. This chapter comprises several key sections, each contributing to the comprehensive analysis and interpretation of the experimental findings. Firstly, the Dataset section provides insights into the dataset used for experimentation, detailing its composition and characteristics, which form the basis for the subsequent analysis. The Experimentation section outlines the experimental setup, including parameters, tools, and methodologies employed in conducting the evaluations. Within this section, the Experimental Setup subsection delineates the specific configurations and protocols followed to ensure rigorous experimentation. Subsequently, the Experimental Result subsection presents the findings obtained from the conducted experiments, including performance in Word Error Rate(WER), loss, and any notable observations. Following the presentation of results, the Discussion section critically analyzes the outcomes, contextualizing them within the broader scope of the research objectives and prior literature. Here, potential implications, limitations, and future research directions are explored, fostering a deeper understanding of the model's performance and its implications for Sign Language Recognition technology. By integrating empirical evidence with insightful discourse, this chapter provides valuable insights into the efficacy and potential avenues for improvement of the developed model, ultimately contributing to the advancement of Sign Language Recognition research.

4.2 Dataset

Due to the absence of an existing dataset for continuous Ethiopia sign language, we took the initiative to create our dataset to facilitate the training and assessment of the model we propose. To accomplish this task, data were meticulously compiled from two educational institutions, namely Bahir Dar's Yeakatit 23 Elementary School and Fasilo Secondary School. A group of 22 signers actively contributed to the construction of this dataset. Each participant performed a set of 30 unique sign sentences, repeated twice for consistency. Consequently, as shown in table 3.3 our dataset comprises a total of 1320 videos showcasing these 30 unique sign sentences.

To capture the upper body of signers effectively, we employed a Canon Mark IV camera, strategically positioning it at various distances. This approach was necessary due to the significant height differences among signers. By adjusting the camera's distance from the subjects, we ensured optimal framing of the upper body, as sign language predominantly involves gestures that occur above the upper body region. Camera settings were meticulously configured to record at full HD resolution (1920x1080 pixels) and a frame rate of 25 frames per second, guaranteeing the smooth and accurate capture of signer movements. Each participant in the study performed each sentence twice, contributing to a comprehensive dataset for analysis and model development. The selected 30 sentences are shown in the table below

In our data collection process, we prioritized sentences related to family introductions and other related sentences to Family, recognizing their significance as highlighted by experts specializing in addressing the needs of individuals with hearing impairments. These introductory sentences were identified as crucial areas where people with hearing loss encounter challenges in communication. As mentioned earlier, our dataset was acquired from a group of 22 signers, ensuring a diverse representation of signing styles and expressions. Subsequently, the amassed data was meticulously divided into distinct sets for training, validation, and testing purposes.

4.3 Model Architecture and Hyperparameters

In this section, we provide a detailed description of the hyperparameters used in the development of our end-to-end continuous Ethiopian Sign Language recognition model. These hyperparameters, crucial in shaping the model's structure, capacity, functionality, and optimization methodology, are explored in depth. A thorough understanding and precise adjustment of these hyperparameters is deemed essential, as they profoundly impact the model's effectiveness and adaptability across diverse contexts.

We adopt ResNet18 for feature extraction due to its ability to efficiently capture spatial features from video frames through its deep residual learning framework, which helps in handling complex sign language gestures. For temporal modeling, we use Bidirectional LSTM (BILSTM) because it effectively captures the temporal dependencies and contextual information in continuous sign language sequences, allowing the model to understand the order and context of signs within sentences. This combination ensures robust spatial-temporal feature learning, essential for accurate continuous Ethiopian Sign Language Recognition.

We present the description of each hyperparameter, explaining its role and impact on the model. These hyperparameters encompass the configuration of the 3D convolutional layers, 1D convolution to extract spatio-temporal features, and for temporal modeling, the BiLSTM layers, and various settings related to the optimizer and training procedure. Additionally, we outline the learning rate used in the optimizer and the loss function employed for model training.

Firstly, the model includes a ResNet backbone for 3D convolutional layers, which captures spatio-temporal dependencies in the input video data. The ResNet architecture is chosen for its proven effectiveness in feature extraction. It begins with a 3D convolutional layer applying 64 filters with a kernel size of (1, 7, 7) and stride of (1, 2, 2). This initial layer is followed by batch normalization and a ReLU activation function.

The ResNet backbone consists of four main blocks, each composed of multiple 3D convolutional layers:

- **Layer 1**: Two BasicBlocks with 64 filters.
- **Layer 2**: Two BasicBlocks with 128 filters, where the first block uses a stride of (1, 2, 2) for downsampling.
- **Layer 3**: Two BasicBlocks with 256 filters, where the first block uses a stride of (1, 2, 2) for downsampling.
- **Layer 4**: Two BasicBlocks with 512 filters, where the first block uses a stride of (1, 2, 2) for downsampling.

Each BasicBlock consists of two 3D convolutional layers with batch normalization and ReLU activation. Downsampling is performed using 3D convolution with stride (1, 2, 2).

The temporal convolutional module (`Conv1d`) includes:

- A sequence of `Conv1d` layers with 512 and 1024 filters, kernel sizes of 5, and batch normalization.
- MaxPooling layers with a kernel size of 2 and stride of 2 to downsample the temporal features.
- ReLU activation functions after each convolutional layer.

To capture temporal dependencies in the sign language sequences, the model incorporates a bidirectional LSTM (BILSTM) layer:

- **BILSTM Layer**: Consists of 2 LSTM layers with 512 units each, bidirectional, with a dropout rate of 0.3. This allows the model to capture long-term dependencies in both forward and backward directions.

The classifier component includes a fully connected layer (`NormLinear`) for producing the final output classes.

To prevent overfitting, a dropout rate of 0.3 is applied during training.

The model is optimized using the Adam optimizer with the following settings:

- **Learning Rate**: Set to 0.0001, controlling the step size during model training.
- **Weight Decay**: Set to 0.0001, acting as a regularization technique to prevent overfitting.

The model is trained using the Connectionist Temporal Classification (CTC) loss function, which is specifically designed for sequence-based tasks like sign language recognition. This loss function enables the model to learn alignments between input sequences and target labels.

4.3.1 Model Configuration Summary

Our model architecture follows the approach of state-of-the-art methods[43]. It includes ResNet18 as the 2D CNN, a 1D CNN layer, and incorporates a two-layer BiLSTM with a hidden size of 1024 for extensive temporal modeling. Subsequently, a fully connected layer is employed for predicting sentence outcomes. A summary of the model configurations which are used in our experimentation is listed.

- **Dataset**: CESLR
- **Epochs**: 25
- **Batch Size**: 4
- **Test Batch Size**: 4
- **Workers**: 10
- **Device**: 0,1
- **Optimizer**: Adam
- **Base Learning Rate**: 0.0001
- **Weight Decay**: 0.0001
- **Loss Function**: CTC
- **Model Architecture**: ResNet-18 backbone with 3D convolutions, Temporal Convolutions, and BILSTM layers.

By adjusting these hyperparameters, such as the number of layers, units, kernel(filters) size, and learning rate, we have customized the model according to our specific requirements and dataset characteristics.

4.4 Experimentation

In this section, a comprehensive breakdown is provided, delving into the intricate details of the experiment setup, and Dataset details utilized during the development phase of our continuous Ethiopia sign language recognition model.

4.4.1 Experimental Setup

To Train our Continuous Ethiopia sign language recognition model, all experiments were conducted on RTX800 NVIDIA with an Intel Core i5-6200U CPU running at a primary speed of 3.6GHz (with the ability to increase to 3.8GHz), 120GB of RAM, and 48GB of GPU installed Ubuntu operating system.

PyTorch was selected as the foundational framework because of its extensive support for libraries designed specifically for handling image datasets. With PyTorch, there's a more adaptable and user-friendly programming interface, enabling the creation and modification of intricate neural network structures suited for image processing tasks. Its dynamic computational graphs facilitate efficient memory usage, particularly beneficial when dealing with image sequences of varying lengths. Additionally, PyTorch's extensive ecosystem includes specialized libraries like torch-vision, which further boosts its efficacy in addressing the distinctive complexities of working with image datasets.

This paper presents different ways of dividing the data set for identifying the best method for recognizing sign language. Mukushev et al [44] and Huang et al[28] propose various splits to get baseline performance. Guo et al propose two splits; signer-independent and unseen sentence splits and Mukushev et al add one more split, which is an age-independent split. In this study, an attempt is made to investigate the effect of our dataset by using the first two splits.

Experiment I: Signer independence is one of the main challenges that must be addressed for the real-life value of SLR models. In the signer-independent setup, the system is tested on data from signers it has not seen during training. This is important because signers vary in appearance, age, and signing style, Consistent performance across different signers is required. For implemented architectures to generalize well for unseen signers, training, validation, and testing splits should include different signers. To accomplish this, sentences from 16 signers are utilized for training, with the remaining 6 signers divided into validation and testing sets (3 signers in each).

This training helps the system learn to recognize signs accurately from new signers it has not encountered before. Signer independence is crucial for developing practical sign language recognition technology for the deaf and hard-of-hearing community. As discussed above, the dataset contains variations in fluency between different signers. Also, some signs are performed differently by various participants, which brings additional challenges. To this end, the training set has 1020 samples, the validation set has 150 samples, and the testing set has 150 samples

Experiment II: The unseen sentence split separates the dataset into training and testing groups, ensuring that the sentences arragement in the testing group are different from those in the training group. This means the model learns from some sentences and then tries out completely new ones it hasn't seen before. sentences were composed in a way that there are multiple instances of the same words used in several sentences surrounded by different words and in a different order.

Out of 30 sentences, 25 are used for training and validation, and the remaining 5 are used for testing. This split is crucial for checking and improving the model's capability to handle new content, not seen during training. It helps to evaluate how well sign language recognition systems can handle different situations beyond what they have learned. To this end, the testing set includes signs used in unseen sentences. Each sign varies concerning the order in a sentence it appears in and the context of the sentence.

4.4.2 Experimental Result and Discussions

We conducted two primary experiments to assess the effectiveness of our End-to-End Continuous Ethiopia Sign Language recognition model. The first experiment utilized a Signer-independent setup, while the second employed an Unseen Sentence configuration. We used a total of 1320 Sentence videos for experimentation.

Table 4.1: Result on CESLR on BiLSTM

Dataset	WER	
	Validation	Test
Experiment I:	4.09%	8.82%
Experiment II:	1.54%	58%

Experiment I: In this experiment, we focus on evaluating our proposed recognition model in a signer-independent setup. This setup involves training and testing sets that consist of identical sentences but are signed by different individuals, ensuring there is no overlap of signers between the two sets. According to Table 4.1, our BiLSTM model achieved an impressive Word Error Rate (WER) of 4.09% during the development phase and 8.82% during the testing phase for the signer-independent split. Additionally, as shown in Table 4.2, our BiGRU model achieved a WER of 5.81% in the development phase and 16.77% in the testing phase. These results highlight the effectiveness and robustness of our models in recognizing sign language across different signers.

Table 4.2: Result on CESLR on BiGRU

Dataset	WER	
	Validation	Test
Experiment I:	5.81%	16.77%
Experiment II:	1.79%	47.04%

Experiment II: This experiment aims to evaluate the performance of the recognition model on an unseen sentence setup. As shown in Table 4.1, during the development stage, the BiLSTM model achieved a word error rate (WER) of 1.54%. However, in the testing phase, the WER increased significantly to 58% for the unseen sentence split. Similarly, the BiGRU model as shown in table 4.2 achieved a WER of 1.79% in the development phase and 47.04% during testing. Despite the increase in WER during the testing phase, the BiGRU WER of 47.04% suggests it has a relatively better ability to generalize to unseen sentence structures compared to the BiLSTM 58%. These findings highlight the potential of both models in continuous Ethiopian sign language recognition, with the BiGRU showing promise for further improvement and optimization to handle new and unseen sentence combinations more effectively

A loss curve is a graphical representation that shows the change in loss (error) of a model over training epochs. It is used to monitor the training process and to understand the performance of a model in both the training and validation phases. The loss function quantifies how well or poorly the model's predictions match the true labels, with lower loss values indicating better performance. The x-axis typically represents the number of epochs, while the y-axis represents the loss. A well-behaved loss curve shows a decrease in loss over time, indicating that the model is learning.

Figure 4.1 depicts the training and validation loss curves for an end-to-end continuous Ethiopian sign language recognition on BiLSTM. The experiment was conducted using a signer-independent split, meaning the training and validation sets contain data from different signers to ensure that the model can generalize to new, unseen signers.

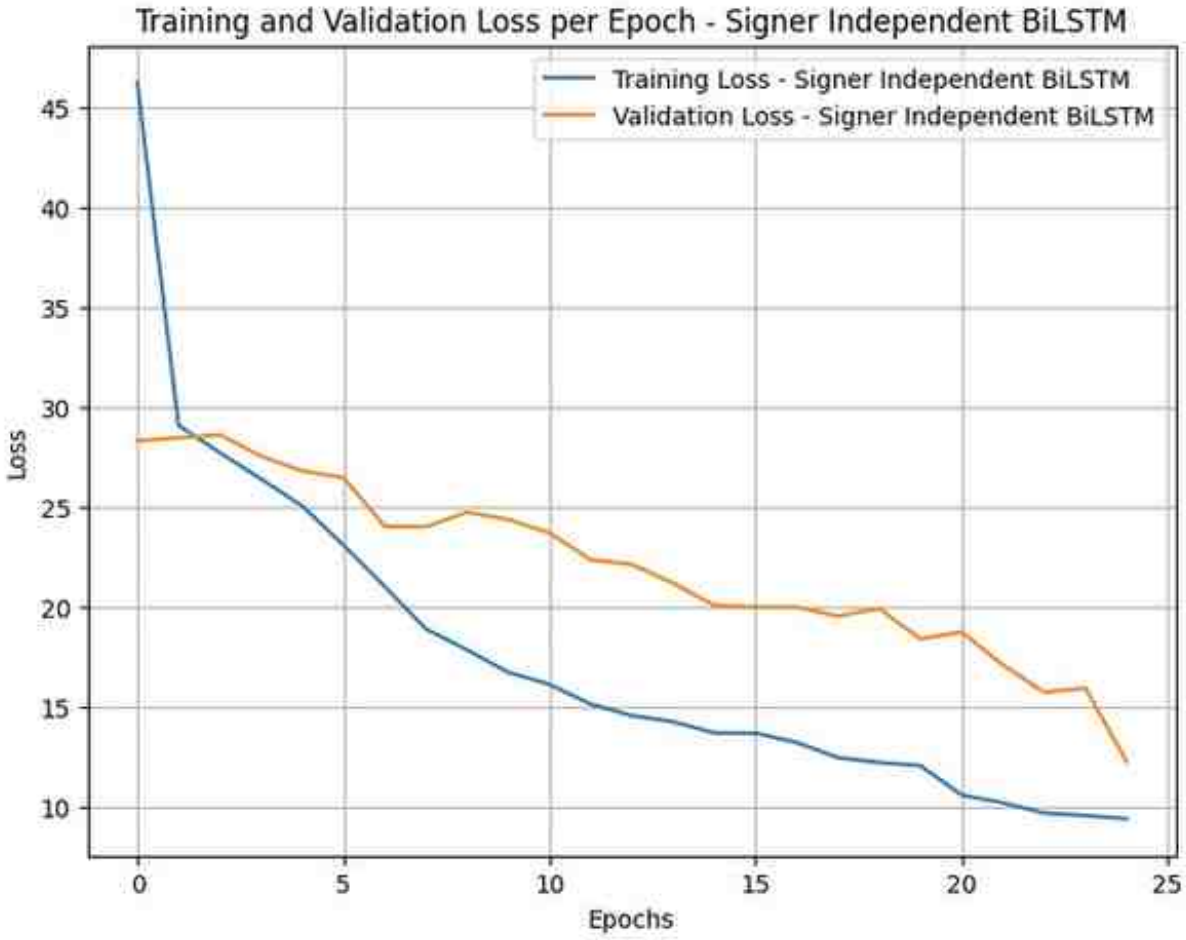


Figure 4.1: Loss of Signer independent Experiment BiLSTM

As figure 4.1 displays the training and validation loss per epoch for a Signer Independent spilt on the Bidirectional Long Short-Term Memory (BiLSTM) model. The x-axis represents the number of epochs, ranging from 0 to 25, while the y-axis represents the loss values.

Initially, as figure 4.1 the training loss, indicated by the blue line, starts at a high value above 45 and shows a sharp decline in the first few epochs, reaching below 30 by the 2nd epoch. It then continues to decrease steadily throughout the remaining epochs, with minor fluctuations, ending below 10 at the 25th epoch. This consistent downward trend in training loss suggests that the model is learning and fitting the training data progressively better over time.

The validation loss, represented by the orange line, starts just below 30 and initially shows a slight increase before starting to decline after the 3rd epoch. It then exhibits a generally downward trend with more variability compared to the training loss. There are periods where the validation loss plateaus or slightly increases, indicating the model's performance on unseen data may fluctuate. However, the overall trend shows a reduction in validation loss, finishing just above 15 by the 25th epoch.

The gap between training and validation loss narrows as the epochs progress, indicating the model's improving generalization capability. This convergence between training and validation loss curves suggests the model is learning well without significant overfitting. However, the presence of fluctuations in the validation loss indicates that there might still be room for



Figure 4.2: Loss of Signer independent Experiment BiGRU

improvements in model generalization and stability.

Figure 4.2 depicts the training and validation loss per epoch for a Signer Independent split on the BiGRU model. The x-axis represents the number of epochs, while the y-axis shows the loss values.

As shown in figure 4.2 the beginning of the training, the training loss starts at a high value of approximately 45 and rapidly decreases. This initial sharp decline indicates that the model is quickly learning from the training data. The validation loss starts lower than the training loss, around 30, and also decreases but more gradually.

As the epochs progress, both training and validation losses continue to decline, with some fluctuations. Around the 10th epoch, the training loss stabilizes below 20 and steadily decreases, reaching a value close to 10 by the 25th epoch. The validation loss, however, shows more variability. It initially decreases but experiences several peaks and troughs, particularly around the 10th, 15th, and 20th epochs, indicating that the model's performance on the validation data is not as stable as on the training data.

Despite these fluctuations, the overall trend for both losses is downward, suggesting that the model is improving its ability to generalize to unseen data. The gap between training and validation loss narrows slightly towards the end, which can be a sign of the model becoming better at generalization, though the persistent variability in the validation loss suggests potential room for

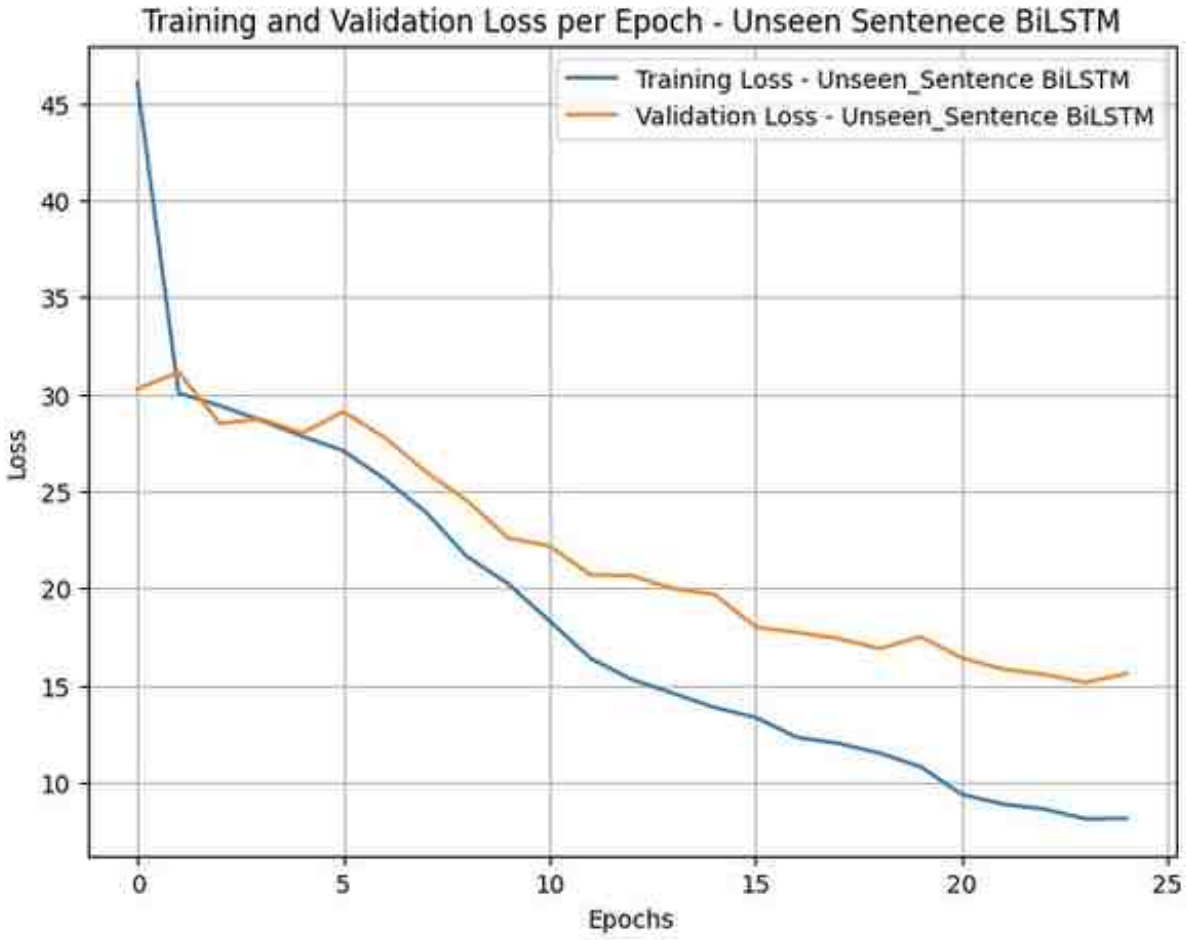


Figure 4.3: Loss of Unseen Sentence Experiment BiLSTM

further optimization or that the model might benefit from regularization techniques to enhance stability.

In summary, the graph shows a clear improvement in both training and validation losses over time, with the training loss decreasing steadily and the validation loss showing more variability but an overall downward trend. The fluctuations in validation loss indicate that while the model is learning, it may benefit from additional fine-tuning, such as regularization or adjusting the learning rate, to achieve more stable performance on unseen data.

Figure 4.3 depicts the training and validation loss over epochs for a BiLSTM model trained on an unseen sentence dataset.

Initially, at epoch 0, the training loss is extremely high, indicating that the model's predictions are far from the actual values. This is expected since the model has not yet learned from the data. The training loss, depicted by the blue line, plummets sharply within the first few epochs. This rapid decrease suggests that the model quickly learns from the training data, significantly reducing the error in its predictions.

In contrast, the validation loss, as shown in figure 4.3, starts around 30 and also drops swiftly in the initial epochs, although it does not decrease as sharply as the training loss. The early decrease in validation loss indicates that the model is also learning to generalize from the training data to unseen data.

However, as training progresses, differences between the training and validation loss behaviors become apparent. While the training loss continues to decline smoothly, reaching a value close to 10 by epoch 25, the validation loss exhibits more variability. This variability is characterized by small peaks and troughs, indicating periods where the model's performance on the validation set temporarily worsens before improving again.

Around epoch 10, the validation loss curve shows more noticeable fluctuations, suggesting the model might be overfitting—learning patterns specific to the training data that do not generalize well to the validation data. Despite these fluctuations, the overall trend of the validation loss is downward, demonstrating that the model is gradually improving its ability to generalize, albeit not as consistently as it improves on the training data.

By epoch 25, the validation loss stabilizes around 15, whereas the training loss is lower, around 10. This gap between the training and validation loss at the later epochs suggests some overfitting, where the model performs better on the training data than on unseen validation data.

Overall, figure 4.3 demonstrates a successful training process for the BiLSTM model on Unseen Split, with both the training and validation losses showing significant reductions over the 25 epochs. The initial sharp decline in losses indicates that the model quickly grasps the underlying patterns in the data. While the validation loss exhibits some variability, it still trends downward, reflecting the model's ability to generalize to unseen data. By the end of the training period, both losses are substantially lower than their starting points, suggesting that the model has learned effectively. Despite minor fluctuations, the model's performance indicates a positive learning trajectory and promising potential for further refinement and optimization.

Figure 4.4 provides a comprehensive visualization of the training and validation loss for a Bidirectional Gated Recurrent Unit (BiGRU) model on Unseen sentence split over 25 epochs.

As shown in figure 4.4 as the training process, the model exhibits high initial loss values, starting at approximately 48 for the training loss and around 35 for the validation loss. This initial high loss is typical as the model parameters are randomly initialized and not yet optimized. During the early epochs (from 0 to 5), both loss curves show a sharp decline. The training loss drops rapidly, indicating that the model quickly learns the basic patterns in the training data. The validation loss also decreases significantly during this phase, reflecting improved performance on the unseen validation data, though it remains higher than the training loss.

From epoch 5 onwards, the training loss continues to decrease in a more gradual, consistent manner. This steady decline suggests that the model is effectively learning and refining its parameters to minimize error. By the 25th epoch, the training loss reaches a value close to 5, indicating a substantial reduction in error from the initial state.

The validation loss, as shown in figure 4.4, follows a similar downward trend but with more fluctuations. These oscillations in the validation loss, especially noticeable around epoch 15, could suggest the model is experiencing some instability or challenges in generalizing across different validation sets. However, the general trend is downward, and by the 25th epoch, the validation loss stabilizes just below 20. This stabilization suggests the model is becoming more consistent in its performance on unseen data.

The gap between the training and validation loss narrows as the number of epochs increases, indicating the model is not overfitting significantly. Overfitting would be evidenced by a large

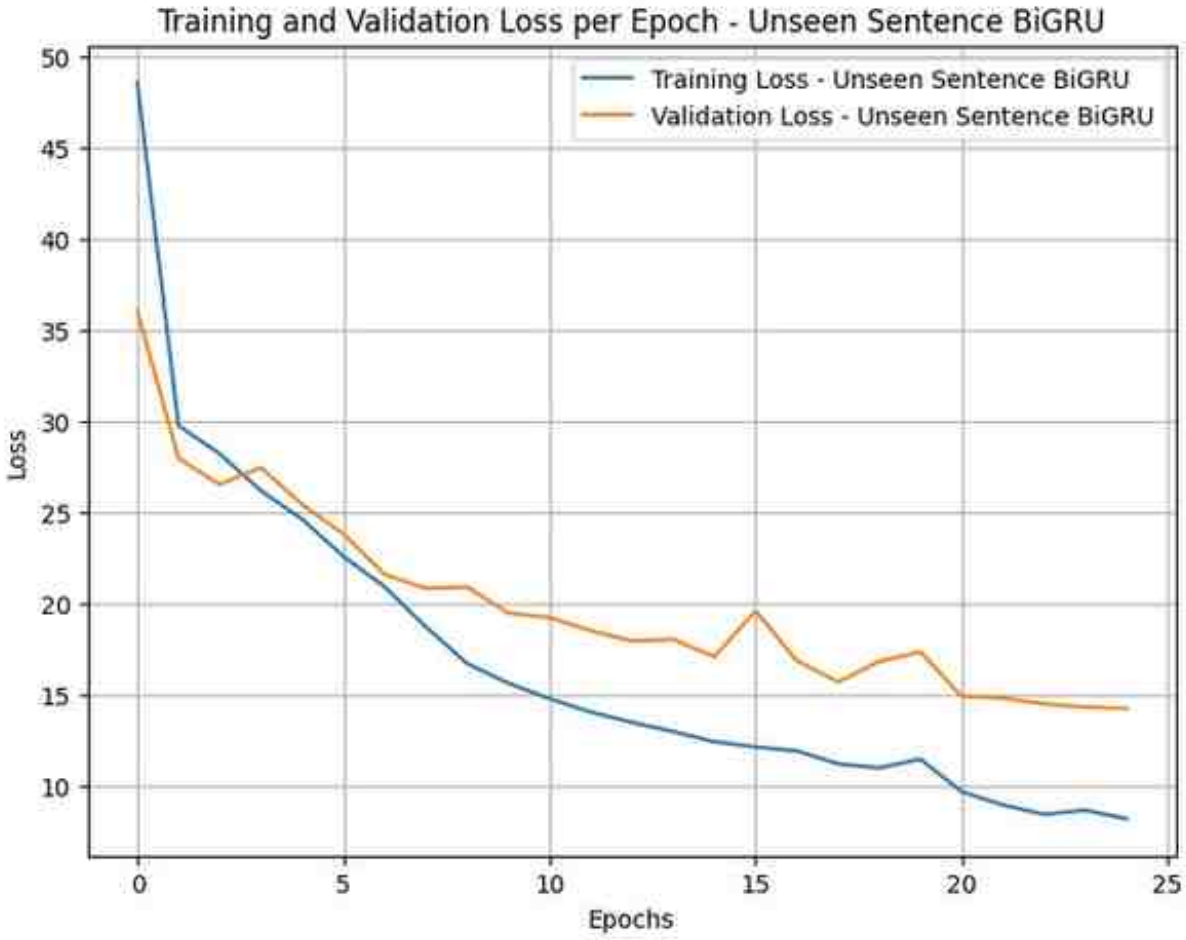


Figure 4.4: Loss of Unseen Sentence Experiment BiGRU

gap where the training loss continues to decrease while the validation loss starts to increase or remains high. Instead, the convergence of both loss curves towards lower values suggests that the model is learning to generalize well from the training data to the validation data.

In conclusion, figure 4.4 demonstrates a successful training process for the BiGRU model on unseen sentences. The significant reduction in both training and validation losses over 25 epochs highlights the model's effectiveness in learning and adapting, while the narrowing gap between the two loss curves suggests robust generalization capabilities. The occasional fluctuations in the validation loss indicate areas for potential further tuning to achieve even more consistent performance.

Table 4.3: Loss of CESL on BiLSTM

	Loss	
Dataset	training	validation
Experiment I:	6.99	7.09
Experiment II:	6.24	15.82

Table 4.3 and table 4.4 provided includes the loss values for training and validation for two different dataset splits: signer independent split (Experiment I) and unseen sentence split (Ex-

periment II) on Continuous Ethiopian Sign Language (CESL) recognition using BiLSTM and BiGRU models under different experimental conditions.

In Experiment I, when we compare the performance of the BiLSTM and BiGRU models on the signer-independent split of the dataset. The signer independent split tests the models' ability to generalize to signers that were not seen during training.

Table 4.4: Loss of CESL on BiGRU

	Loss	
Dataset	training	validation
Experiment I:	5.58	17.74
Experiment II:	6.19	14.30

The BiLSTM models, which use memory cells to retain long-range dependencies and benefit from a bidirectional structure, excel in understanding the full context of Ethiopian sign language sequences. While more complex and slower to train, BiLSTM shows a more stable performance with a smaller gap between training and validation losses. In contrast, BiGRU, with a simpler structure and fewer parameters, is faster to train but does not capture long-range dependencies as effectively, leading to poor generalization to unseen signers.

The analysis concludes that the BiLSTM model is preferred for signer-independent continuous Ethiopian Sign Language recognition. Its superior generalization capabilities and stable performance make it more robust in handling the complexities of sign language data compared to the BiGRU model, which, despite better training performance, struggles with unseen signers.

In Experiment II, when we evaluate the performance of BiLSTM and BiGRU models on an unseen sentence split of our dataset, focusing on their ability to generalize to new sentence structures. Both models exhibited similar training losses (6.24 for BiLSTM and 6.19 for BiGRU), indicating comparable proficiency in fitting the training data. However, the BiGRU model demonstrated a lower validation loss (14.30) compared to the BiLSTM model (15.82), suggesting that BiGRU generalizes slightly better to unseen sentences.

The architectural differences between the models highlight their respective strengths and weaknesses. BiLSTM models, with their memory cells and bidirectional nature, are adept at capturing long-range dependencies and context, but their complexity can lead to slower training and a higher risk of over-fitting. Conversely, BiGRU models, being simpler with fewer parameters, are more computationally efficient and train faster while still effectively capturing dependencies through their gating mechanisms. This efficiency can be particularly advantageous when handling large datasets or limited computational resources.

The BiGRU model's slightly better generalization to new sentence structures, combined with its efficiency and stable performance, makes it the preferred choice for continuous Ethiopian Sign Language recognition in this context.

Overall, the BiLSTM model shows better generalization on the signer-independent split compared to the BiGRU model, as indicated by the much smaller gap between training and validation loss. On the unseen sentence split, both models show a significant increase in validation loss, suggesting difficulty in generalizing to unseen sentences. However, BiGRU has a relatively smaller increase in validation loss compared to BiLSTM, indicating slightly better performance.

Both models exhibit better performance (lower validation loss) on the signer-independent split (Experiment I) compared to the unseen sentence split (Experiment II). This suggests that the task of recognizing unseen sentences is inherently more challenging than recognizing signs from unseen signers.

The substantial increase in validation loss for the unseen sentence split highlights a potential area for model improvement, such as incorporating more diverse sentence structures in the training set or exploring data augmentation techniques to better capture sentence variability.

Table 4.5: WER of CESLR without Correlation Network

	Loss	
Dataset	training	validation
Experiment I:	8.02%	12.5%
Experiment II:	3.02%	62.5%

Impact of Correlation Network on Continuous Sign Language Recognition: A Comparative Analysis of Word Error Rates

The integration of the correlation network into the continuous sign language recognition (CSLR) model has demonstrated significant improvements in word error rate (WER) across both experimental setups, highlighting its effectiveness in enhancing recognition performance.

Experiment I: In the sign-independent split, the baseline model without a Correlation Network achieved a WER of 8.02% on the development set and 12.5% on the testing set. With the incorporation of the Correlation Network, the WER was significantly reduced to 4.09% on the development set and 8.82% on the testing set. This substantial improvement, especially the near-halving of the WER on both sets, indicates that Correlation Network effectively captures the motion trajectories of body parts, which are crucial for distinguishing between different signs. The enhanced temporal and spatial modeling capabilities provided by Correlation Network lead to more accurate recognition, particularly in varied signing conditions.

Experiment II: For the unseen sentence split, the baseline model without Correlation Network exhibited a WER of 3.02% on the development set but faced a drastic increase to 62.5% on the testing set. This significant discrepancy highlights the model's struggle to generalize to completely new sentences. When Correlation Network was integrated, the WER was reduced to 1.79% on the development set and 47.04% on the testing set. While the improvement on the testing set remains substantial, it still underscores the challenge of generalizing to entirely unseen sentences. Nonetheless, the nearly 25% absolute reduction in WER on the testing set demonstrates that Correlation Network substantially enhances the model's ability to handle new sentence structures by better capturing relevant temporal dynamics and spatial relationships.

The comparative results demonstrate the efficacy of the correlation network in reducing WER across different experimental conditions. In the sign-independent split, the reduction in WER suggests that Correlation Network significantly improves the model's robustness and accuracy in recognizing signs from a diverse set of examples. In the unseen sentence split, the notable decrease in WER, despite the inherent difficulty, indicates that Correlation Network enhances the model's ability to generalize better to new sentences by leveraging the motion trajectories effectively.

In conclusion, the addition of Correlation Network to the CSLR model provides a strong advantage in capturing essential body movements, leading to more precise and reliable sign language recognition. The consistent reduction in WER across both experiments underscores the critical role of correlation networks in advancing the performance of continuous sign language recognition systems.

Comparative Analysis of BiLSTM and BiGRU Models in Continuous Ethiopian Sign Language Recognition: A Study on Signer Independence and Unseen Sentence Structures

The performance differences between the BiLSTM and BiGRU models in Experiment I and Experiment II can be attributed to their architectural strengths and the unique challenges presented by each experimental dataset setup. This comparison highlights how each model leverages its architecture to address the specific demands of continuous Ethiopian Sign Language recognition.

In Experiment I, which focuses on signer independence, the BiLSTM model outperforms the BiGRU model. This is likely due to the BiLSTM's ability to capture long-range dependencies through its memory cells and utilize bidirectional context, providing a comprehensive understanding of sign sequences that vary in style and fluency among different signers. The complexity and larger number of parameters in BiLSTM, while generally a challenge, prove beneficial here by enabling the model to adapt to the nuanced variations in signing styles, leading to better generalization to unseen signers.

Our dataset includes 22 different signers, each with unique signing styles, speeds, and distinctive individual techniques. So BiLSTM models are beneficial for recognizing signs across different signers. Additionally, the larger number of signer in the dataset provide enough data for the BiLSTM model to generalize effectively across signers

Conversely, in Experiment II, which tests the models' ability to generalize to unseen sentence structures, the BiGRU model demonstrates superior performance. The BiGRU's simpler gating mechanism and fewer parameters facilitate efficient training and robust generalization, particularly in handling new and varied sentence structures. The BiGRU's efficient control over the flow of information allows it to adapt quickly to new patterns, which is crucial when the model is faced with sentence compositions it has not encountered during training. This parameter efficiency likely helps prevent overfitting, making the BiGRU more adept at generalizing from the training data to new, unseen sentences.

Overall, these results underscore the importance of selecting a model architecture based on the specific requirements of the task at hand. The BiLSTM strengths in handling signer variability make it more suitable for signer-independent scenarios, while the BiGRU efficiency and robustness to new sentence structures make it better suited for tasks involving unseen sentences.

4.5 Error Analysis

Error analysis is a crucial step in understanding the performance of any machine learning model, especially in complex tasks such as sign language recognition. In this study, we focus on the error analysis of a continuous Ethiopian sign language recognition system. Our goal is to thoroughly



Figure 4.5: Evaluation of Signer-independent Split

examine the errors made by the model on two distinct dataset splits: signer-independent and unseen-sentence split. By dissecting these errors, we aim to gain insights into the limitations of the current model and pave the way for future improvements in sign language recognition technology.

In signer-independent split evaluation showcased remarkable performance, achieving a commendably low WER of 7.8% in BiLSTM and slightly higher in 18.94% BiGRU. This outcome indicates that the model demonstrates effective generalization across different signers, which is crucial for practical deployment in real-world scenarios where users may vary widely. It suggests that the model can successfully recognize sign language regardless of the signer, highlighting its potential for broader application and usability. The BiLSTM model has lower WER than BiGRU in signer-independent evaluation demonstrating its strong generalization capabilities.

Figure 4.5 depicts the evaluation of signer-independent performance on the BiLSTM model, where the model successfully recognizes all words except for those with similar signing styles but different glosses. This discrepancy suggests that the model may face challenges in distinguishing between signs with subtle variations in signing style. As you see in figure 4.5 we found some deletions like the ground truth is **የኢትዮጵያ ባንዲራ አረንጓዴ ቢጫ ቀይ ነው** but when we test these sentence with the model we get all sentence but it misses **አረንጓዴ**. Overcoming this issue may require the development of techniques that can effectively capture and represent these nuances, such as incorporating additional different signing styles by diverse signers.



Figure 4.6: Evaluation of Unseen Sentence Split on BiGRU model

A WER of 7.8% in signer-independent split is generally considered to be a good result in the field of Sign language recognition. It suggests that the system is capable of accurately recognizing sign words with a high degree of accuracy, even in the presence of diverse signers in the dataset.

Furthermore, we scrutinized the model's performance on the unseen-sentence split, where the test sentences were entirely new and unseen during training. This evaluation sheds light on the model's ability to handle novel signs and sequences effectively. In unseen sentence split evaluation, we get the WER of 58% in the BiLSTM model and we achieve 47.04% low WER in BiGRU compared to the BiLSTM Model. This result underscores the inherent difficulty in capturing the nuances and complexities of continuous Ethiopia sign language recognition in unseen sentence split compared to signer independent split, particularly when confronted with novel linguistic structure. It suggests that the model may struggle to generalize effectively to unseen sign sequences, indicating the need for more robust techniques.

Notably, This discrepancy suggests a discrepancy between the model's predictions and the actual signing sequence, indicating potential limitations in the model's ability to accurately interpret unseen sentences

Figure 4.6 and 4.7 provide further insight into the model's performance on unseen sentences. It reveals that the model tends to identify frequently occurring sign language words, while other words, signed only once in the sentence, remain unrecognized. As shown in figure 4.6 the model tended to omit certain words present in the ground truth sentence while introducing new words that were not part of the original sentence. For example, we encounter substitution of word in the predicted sentence like **ታደርጋለች** in **እህቴ ጽሎት ታደርጋለች** is substituted in **ናት** and the word **ይወዳል** with **እወዳለሁ** were substituted with similar looking words leading to errors in the output. This observation implies that the model may struggle with rare signs, potentially due to limited exposure to sign-word combinations during training. Addressing this limitation on unseen sentence split involves increasing the training data with a more diverse range of sign combinations. This can help the model better generalize to novel expressions and gestures encountered in real-world scenarios.

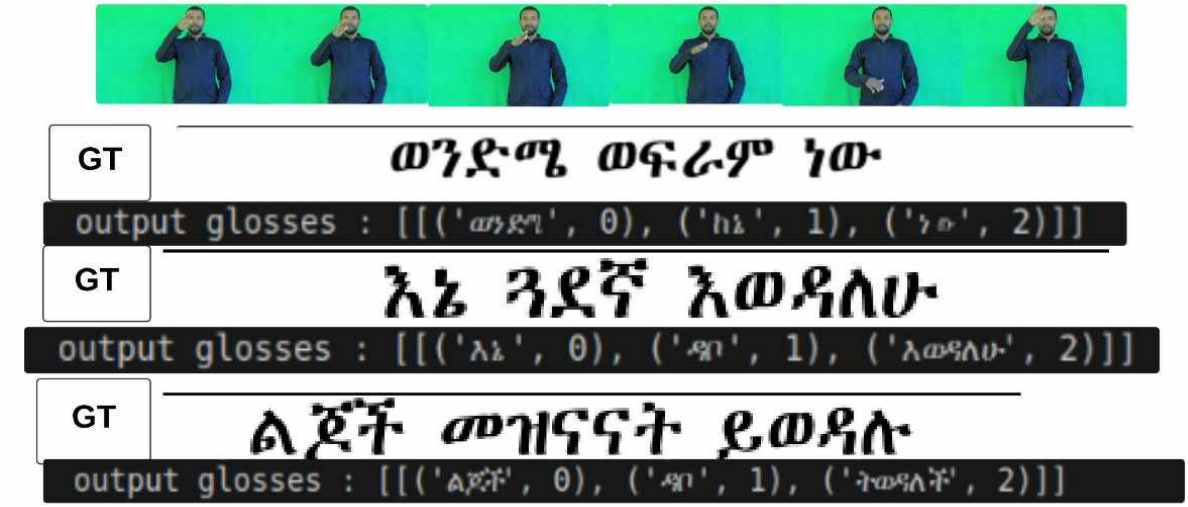


Figure 4.7: Evaluation of Unseen Sentence Split on BiGRU model

Moreover, the analysis revealed that unseen sentences split a striking pattern wherein the initial word of the sentence was consistently recognized correctly, attributed to the ample training data available for these words, encompassing various sequential forms. However, subsequent words in the sentence often faced recognition errors, primarily stemming from a scarcity of diverse training data representing these words' different forms and variations.

These findings underscore the critical importance of dataset design in CESLR research. The significant disparity in performance between the unseen sentence split and the signer-independent split highlights the need for a comprehensive dataset that encompasses diverse linguistic variations and signer characteristics. It is important to ensure that training datasets adequately represent the full spectrum of sign language variations. Looking ahead, future research efforts should focus on developing techniques that enhance the model's adaptability to unseen sign combinations.

In conclusion, the discussed findings highlight both the challenges and successes in CESLR research. While recognizing unseen sign combinations remains a difficult task, the model demonstrates remarkable performance in signer-independent scenarios. By addressing the limitations observed in the evaluation, such as handling low-frequency signs and subtle signing style variations, and by employing comprehensive dataset design, we can pave the way for more effective and practical continuous sign language recognition systems.

Comparison with Previous Studies on Ethiopia Sign Language Recognition

Previous research on Ethiopian Sign Language (EthSL) recognition has mainly targeted on isolated sign recognition, dealing with individual signs at the character, word, and phrase levels. Although these studies have provided valuable insights, they fall short of capturing the full complexity and fluidity of natural sign language conversations, which occur at the continuous (sentence) level. In contrast, the current study focuses on recognizing continuous EthSL, aim-

ing to understand sequences of signs performed by multiple signers, each with their own unique styles and variations.

Previous approaches to Ethiopian Sign Language Recognition have primarily focused on traditional approach. These approach typically break down the task into independent stages such as segmentation, feature extraction, classification, and post-processing, each requiring specialized models and hand-crafted features. In contrast, our research introduces an end-to-end approach for continuous Ethiopian Sign Language Recognition, leveraging advanced deep learning techniques to create a unified model that directly maps raw video input to recognized sign language sentences. This methodology not only simplifies the system architecture and mitigates error propagation but also enhances efficiency and scalability, allowing for the seamless recognition of sign language in continuous input streams. By addressing the unique linguistic challenges and practical needs of the Ethiopian deaf community, our approach significantly advances the field by moving beyond the limitations of isolated sign recognition to provide a more comprehensive and realistic solution for real-world communication.

4.6 Summary

The experiments on continuous Ethiopia sign language recognition unveiled both promising capabilities and notable challenges. In the signer-independent setup, the recognition model showcased remarkable adaptability across different signers, achieving a commendable Word Error Rate (WER) of 4.09% during development and 8.82% during testing on the BiLSTM model. This underscores the model's robustness in accommodating signer variability, a critical aspect for real-world applications requiring broad accessibility. However, the transition to the unseen sentence setup revealed a significant hurdle. Despite a promising WER of 1.79% during development, the model struggled when presented with previously unseen combinations of signs, experiencing a sharp increase in WER to 47.04% during testing. This underscores the intricate nature of sign language recognition, where the model's performance heavily relies on exposure to diverse sign expressions and contextual cues.

Chapter Five

Conclusion and Recommendation

5.1 Overview

The conclusion and recommendation chapter synthesizes the findings from the research on continuous Ethiopian Sign Language Recognition (CESLR) and proposes actionable steps to enhance future work. The study achieved notable success in recognizing continuous Ethiopia sign language in signer-independent and unseen sentence scenarios, demonstrating the model's potential in handling diverse signers.

The recommendations focus on addressing the identified challenges and leveraging the strengths of the current model. Key suggestions include expanding and diversifying the dataset, and incorporating contextual and linguistic features.

5.2 Conclusion

Our End-to-End Continuous Ethiopia Sign Language Recognition model sought to address a critical communication barrier faced by the Deaf and hard-of-hearing community in Ethiopia by developing an advanced system capable of recognizing continuous Ethiopian Sign Language.

Previous Ethiopia sign language recognition systems have predominantly concentrated on identifying isolated signs such as letters, words, and phrases thus struggling to capture the fluidity and naturalness of continuous sign language communication. In contrast, our work in continuous Ethiopia sign language recognition represents a significant leap forward, aiming to understand and interpret continuous sequences of sign language gestures. By focusing on continuous communication, we can capture the richness and nuances of natural sign language conversations, leading to a more comprehensive and intuitive interaction for users and enhancing the accessibility and inclusivity of communication for the deaf and hard-of-hearing community.

This research successfully achieved the main objective of developing an end-to-end continuous sign language recognition model by using deep learning. To achieve this, we developed a novel dataset consisting of 1320 sentences performed by 22 signers, capturing a wide range of signing styles and expressions. The dataset's diversity provided a robust foundation for training a

comprehensive recognition model. Our approach architecture utilized combining 2D and 1D Convolutional Neural Networks (CNNs) to extract spatial and short-temporal features respectively from video frames, and a Bidirectional Long Short-Term Memory (BiLSTM) network to capture temporal dependencies within sign sequences. The Connectionist Temporal Classification (CTC) classifier is employed to align and recognize the continuous sign language sequences without requiring pre-segmented data.

The research on continuous Ethiopian Sign Language Recognition demonstrated promising results in signer-independent scenarios with a Word Error Rate (WER) of 4.09% during development and 8.82% during testing, showcasing the model's robustness in handling variations among different signers. However, when tested on unseen sentence combinations, the model's performance dropped significantly, with a WER of 47.02%. Despite this challenge, it is important to highlight that this is the first work on continuous Ethiopian Sign Language recognition, marking a significant milestone in the field. To build on this pioneering effort, it is recommended to expand and diversify the dataset further, incorporating more varied signing styles and contexts, and to explore advanced data augmentation techniques and model architectures that can enhance the model's adaptability and generalization to novel sign combinations.

The research on continuous Ethiopian Sign Language Recognition (CESLR) has made several significant contributions across various areas, each enhancing the understanding and development of Ethiopia sign language recognition technology.

Novel Dataset Creation: Development of a unique dataset comprising 1320 sentences performed by 22 different signers. This dataset provides a diverse range of signing styles and expressions, crucial for training robust and generalizable sign language recognition models. It sets a foundation for future research in the field and fills a gap where there is no resource available for Continuous Ethiopian Sign Language. The dataset is available on Zenodo for future research: Zenodo Dataset.

In this thesis, we use end-to-end approach for designing and implementing a recognition model that integrates advanced techniques in deep learning for continuous Ethiopia sign language recognition. Our model combines the strengths of 2D and 1D Convolutional Neural Networks (CNNs) alongside a Bidirectional Long Short-Term Memory (BiLSTM) network and a Connectionist Temporal Classification (CTC) classifier. This integrated architecture is specifically crafted to effectively capture both spatial nuances and temporal dependencies inherent in sign language. By adopting an end-to-end methodology, our approach significantly enhances the model's capability to recognize continuous sequences of signs, marking a substantial leap beyond conventional methods that typically handle isolated signs in isolation.

Recognizing that sign language varies in speed and word combinations from one signer to another, To test the architecture two distinct models were developed: one for signer-independent recognition and another for handling unseen sentence data. The performance of the model was rigorously evaluated using a signer-independent, unseen sentence dataset. The primary metric used was the Word Error Rate (WER), which provided a clear measure of the model's accuracy and effectiveness. The results indicated that the model could generalize to new data, although some challenges, such as overfitting, were noted on unseen sentence split. These findings highlighted areas where further optimization and improvement are needed to enhance the model's performance.

We use a Correlation Network that enhances continuous Ethiopia sign language recognition by

capturing and leveraging body motion trajectories across video frames, significantly reducing word error rates in both sign-independent and unseen sentence splits.

Finally, the study compared its results with previous research on Amharic Sign Language (ASL) recognition. Unlike earlier studies that focused on recognizing individual characters, words, and phrases, this research addressed continuous sentence-level recognition. This distinction is crucial for real-time communication and a deeper understanding of language. The integrated approach using CNNs, BLSTM, and CTC Loss Layer set a new benchmark in the field, demonstrating significant progress in the complexity and applicability of sign language recognition models.

5.3 Recommendation

Based on the findings and conclusions of this study, several recommendations are proposed for future research and development in the field of continuous Ethiopian Sign Language recognition.

Fine-tuning for specific domains involves exploring tailored techniques to enhance the model's effectiveness in specialized areas such as medical, legal, or educational sign language recognition. This process entails adapting the model to domain-specific sign language datasets or integrating specialized knowledge during training to optimize its performance for practical applications, including educational settings. By focusing on educational domains, the model can be refined to assist students by accurately interpreting and translating educational content conveyed through sign language.

When capturing our sign language videos, we used a uniform green background. For future research, we recommend using more complex backgrounds or recording in real time to increase dataset robustness. We used a single Canon IV camera for its high-quality image capture capabilities. To further improve model performance, researchers should consider using a variety of camera devices, such as mobile phones and laptops, to collect diverse data sources.

So, it is recommended to enhance dataset diversity. Future work should focus on expanding the dataset to include a broader range of topics, environments, and signer demographics. Increasing the diversity and size of the dataset will improve the model's ability to generalize and perform well across different contexts.

Engaging with the Ethiopian deaf community and collaborating with linguists and educators is recommended. Such collaborations can provide deeper insights into the nuances of Ethiopian Sign Language, ensuring that the models and applications developed are culturally and linguistically appropriate. This engagement will help meet the actual needs of the users.

Finally establishing benchmark datasets and standardized evaluation protocols for the Ethiopian Sign Language recognition model is important. This will facilitate comparison across different studies and models, driving consistent improvements and innovations in the field. Standardized benchmarks will help in measuring progress and setting clear goals for future research.

By implementing these recommendations, future research can build upon the foundation laid by this study. These efforts will advance the state-of-the-art in continuous Ethiopian Sign Language recognition, making significant contributions to the accessibility and inclusivity of technology for the deaf community.

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